



Lean Six Sigma Green Belt Certification

LEARNER GUIDE

SGS ACADEMY



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COURSE ADMINISTRATION

FOREWORD

The course is owned by and certificated as SGS United Kingdom Ltd. and is provided internationally as SGS Certification and Business Enhancement (CBE). The SGS policy and objectives with respect to the course are given below.

POLICY STATEMENT AND OBJECTIVES

The purpose of this course is to equip learners with a general overview and understanding of the requirements of Lean Six Sigma Green Belt Certification.

COURSE AIM

The objective of this course is to equip learners with the knowledge and skills necessary to actively contribute to improvement projects, enhance quality, and boost efficiency within their area of responsibility. By the end of the course, learners will have a solid understanding of Lean Six Sigma principles, be proficient in applying the DMAIC methodology, and possess the tools and techniques to drive continuous improvement.

Learners will need to demonstrate acceptable performance in these areas to complete the course successfully.

PRIOR KNOWLEDGE

Before starting this course, learners are expected to have the following prior knowledge:

- Basic understanding of process improvement concepts.
- Familiarity with organizational processes and metrics.

COURSE BRIEF

1. LEARNER INTRODUCTIONS

At the start of the course, learners will be asked to introduce themselves. This introduction should include information on the individual's job function, organisation, the organisation's product or service, the organisation's certification details, the individual's knowledge and understanding of the management system's standards and their expectations upon completing the course.

2. PARTICIPATIVE LEARNING

This course is presented using techniques that have been designed to make training an enjoyable as well as a beneficial experience. The approach is based on scientific evidence as to how the brain works and people learn. A better understanding of the standard, is guaranteed.

3. SUCCESS CRITERIA

Learners will be graded based on:

- full attendance of the course;
- acceptable performance in the learning objectives.

4. CONTINUOUS ASSESSMENT

Not applicable for the purpose of this course.

5. LEARNING OBJECTIVES

The learning objectives describe what the learners will be able to do by the end of the course and what knowledge they will take from the course. On completion, successful learners will:

- Understand the principles and methodologies of Lean Six Sigma.
- Apply the DMAIC framework for process improvement.
- Utilize statistical analysis tools for data-driven decision making.
- Apply problem-solving techniques and root cause analysis.
- Identify and implement process improvements using Lean principles.
- Collaborate effectively in improvement projects and contribute to a continuous improvement culture.

6. EXAMINATION

The examination is open-book and consist of 80 multiple-choice questions with 2 hrs exam duration.

Phase	Number of expected questions
Introduction to Lean Six Sigma	10-12
Define	14-15
Measure	14-15
Analyze	14-15
Improve	10-12
Control	10-12

7. COURSE CERTIFICATION

The evaluation criteria for the Lean Six Sigma Green Belt (LSSGB) certification involves completing the required training (100% attendance) and examination. Candidates are required to achieve a minimum score of 60% to pass the examination and obtain the certification.

8. REMINDER

The use of mobile phones, iPads, iPhones, Tablets, pagers etc. during the course are not permitted.

9. CONTINUOUS IMPROVEMENT

Learners are given a Course Evaluation Form at the start of the course for completion and submission at the end of the course. This provides SGS CBE with important customer feedback for the continuous improvement of the course.

10. APPEALS AND COMPLAINTS

Learners may appeal or make a complaint about any aspect of the course. Appeals and complaints should be addressed, in writing, to the local SGS Office.

COURSE TIMETABLE

DAY 1		
08:45- 09:30	COURSE REGISTRATION (Only applicable to in-person workshops)	
09:30-10:00	COURSE INTRODUCTION Objectives; activities; team-work; administration; learner Introductions	
10.00-10:45	INTRODUCTION TO LEAN SIX SIGMA	
10:45-11:00	BREAK	
11:00-13:15	DEFINE	Introduction to Define Phase Generate project ideas and identify CTQs Select project Finalize project charter and as-is Process map (cont...)
13:15-14.00	LUNCH	
14.00 -14.30	DEFINE	Finalize project charter and as-is Process map
14:30-16:45	MEASURE	Introduction to Measure Phase Finalize CTQ and set performance standards Perform measurement system analysis
16:45-17:00	BREAK	
17:00-18:00	MEASURE	Basic Statistics (cont...)

DAY 2		
9:30-11:00	MEASURE	Revision Basic Statistics SPC for continuous data
11:00-11:15	BREAK	
11:15-1:15	MEASURE	Introduction to process stability and capability Establish Baseline (Continuous Y) – Sigma Level
13:15-14:00	LUNCH	
14:00-16:00	MEASURE	Establish Baseline (Continuous Y) – Pp & Ppk Box Plot
16:00-16:15	BREAK	
16:15- 18:00	MEASURE	SPC for discrete data Establish Baseline (Discrete Y)

DAY 3		
9:30-11:00	ANALYZE	Introduction to Analyze Phase List all potential causes Identify critical causes – Data analysis methods
11:00-11:15	BREAK	
11:15-13:15	ANALYZE	Identify critical causes – Process flowchart-based methods
13:15-14:00	LUNCH	
14:00 -16:00	ANALYZE	Determine and validate the root cause- Hypothesis testing
16:00-16:15	BREAK	
16:15-18:15	ANALYZE	Determine and validate the root cause- Hypothesis testing

DAY 4		
9:30 -10:00	ANALYZE	Determine and validate the root cause- 5 Why
10:00-10:45	IMPROVE	Introduction to Improve Phase Generate and evaluate solutions (cont...)
10:45-11:00	BREAK	
11:00-12:15	IMPROVE*	Generate and evaluate solutions Select and optimize solutions Pilot implementation and validation
12:15-13:30	CONTROL*	Introduction to Control Phase Implement Control Systems for Critical Xs Validate Controls Documentation & Closure
13:30-14.30	LUNCH	
14:30 -16:00	COURSE REVIEW AND SUMMARY	Revision Q&A Exam Instructions
16:00-18:00	EXAMINATION	
18.00	END OF COURSE	

*The time allotted to the Improve and Control phase might seem lesser since common topics like process stability and capability are already covered in depth in the Measure phase.

INTRODUCTION TO LEAN SIX SIGMA



Taiichi Ohno



Shigeo Shingo

- Lean originated in the manufacturing industry and traces its roots back to the **Toyota Production System (TPS)** developed by **Toyota** in the 1940s-1950s.
- TPS was primarily influenced by the works of **Taiichi Ohno** and Shigeo Shingo, who aimed to eliminate waste and optimize production processes.
- The principles of lean were further popularized by the book "**The Machine That Changed the World**" by James P. Womack, Daniel T. Jones, and Daniel Roos in 1990. Since then, lean principles and practices have been widely adopted across various industries and sectors, including healthcare, services, and software development, to improve efficiency, reduce waste, and enhance customer value.
- Lean is characterized by its focus on **continuous improvement, respect for people, and the elimination of non-value-added activities in processes**.

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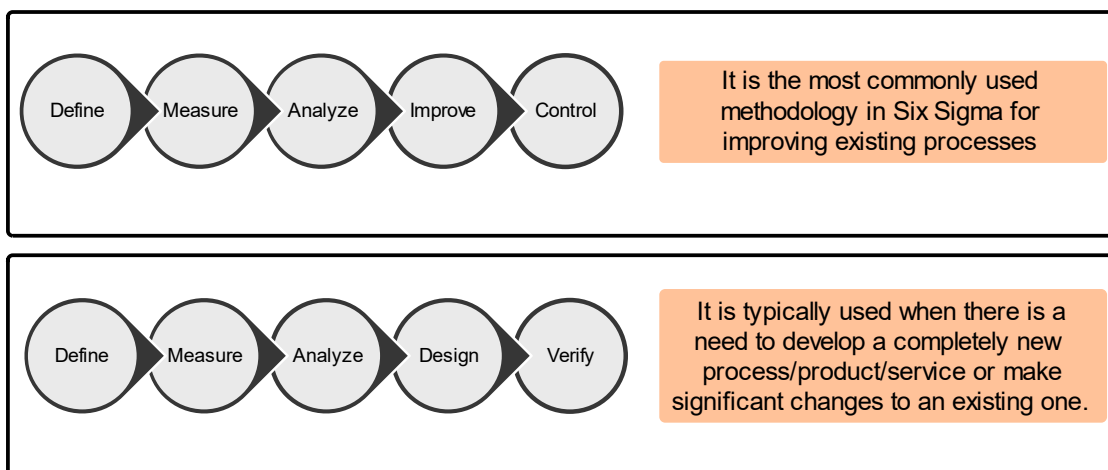
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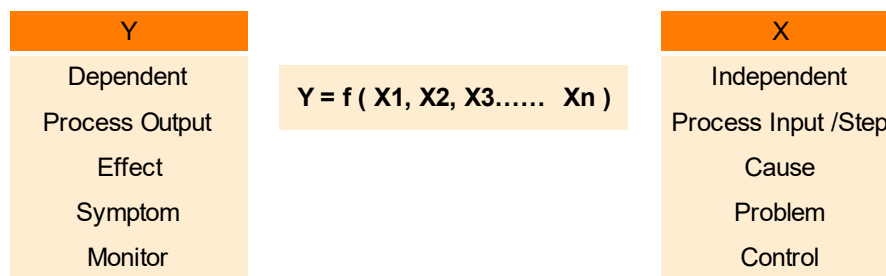
- Six Sigma was developed by **Motorola** in the 1980s as a quality management methodology to improve their manufacturing processes. The term "Six Sigma" refers to a statistical measure of process variation, aiming to reduce defects and errors to an extremely low level.
- **Bill Smith**, an engineer at Motorola, is credited with the development and implementation of the Six Sigma approach. He is also referred to as the '**Father of Six Sigma**'.
- The methodology gained popularity for its **data-driven approach, rigorous problem-solving techniques, and focus on achieving measurable results**.
- It has become a comprehensive business improvement methodology, integrating statistical tools, project management principles, and a structured problem-solving approach.
- Today, Six Sigma continues to be widely implemented by organizations seeking to improve their processes, reduce variation, enhance customer satisfaction, and achieve operational excellence.

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- Lean Six Sigma is a combination of two powerful methodologies, Lean and Six Sigma, that originated independently but eventually merged to create a comprehensive approach for process improvement.
- The integration of Lean and Six Sigma occurred in the 1990s when organizations realized the complementary nature of these methodologies. The combination of Lean's waste reduction and process flow improvement with Six Sigma's data-driven problem-solving and quality management techniques created a comprehensive approach for achieving operational excellence.
- Since its inception, Lean Six Sigma has been widely adopted by organizations across various industries, including manufacturing, healthcare, finance, and services. It has become a comprehensive framework for driving process improvement, waste elimination, and bottom-line results.
- Lean Six Sigma is often supported by a structured DMAIC (Define, Measure, Analyze, Improve, Control) framework, which guides organizations through the stages of problem-solving and process improvement projects.





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DEFINE	MEASURE	ANALYZE	IMPROVE	CONTROL
1.Generate project ideas and identify CTQs 2.Select project 3.Finalize project charter and as-is Process map	4.Finalize CTQ and set performance standards 5.Perform measurement system analysis 6.Establish baseline	7.List all potential causes(Xs) 8.Identify critical causes (Xs) 9.Determine and validate the root cause(s)	10.Generate and evaluate solutions 11.Select and optimize solutions 12.Pilot implementation and validation	13.Implement Control Systems for Critical Xs 14.Validate Controls 15.Documentation & Closure

CTQ: Critical to Quality

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DEFINE PHASE



Aim of the session

- Provide a clear understanding of the Define phase in the DMAIC methodology.
- Define the objectives, focus, and key deliverables of the Define phase.
- Establish the importance of problem identification and project selection.
- Explain the process of Critical-to-Quality (CTQ) identification and its significance in project success.
- Introduce the concept of the project charter and its role in guiding the project.
- Highlight the importance of creating an accurate as-is process map.

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By the end of this session, you will be able to:

- Understand the importance of the Define phase in the DMAIC methodology.
- Clearly define the problem or opportunity that your project aims to address.
- Identify and prioritize the CTQ parameters that are crucial for customer satisfaction and business success.
- Create a comprehensive project charter that outlines the project's purpose, objectives, scope, and stakeholders.
- Develop an as-is process map to visualize the current state of the process and identify areas for improvement.

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The Define phase is the first stage in the DMAIC methodology . It sets the foundation for a successful improvement project by clarifying its objectives, focusing on key areas of improvement, and delivering essential project documentation .

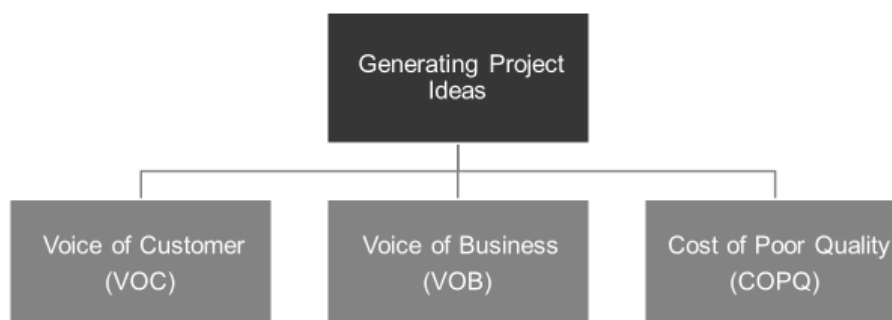
Objective: The primary objective of the Define phase is to clearly define the problem or opportunity that the project aims to address. It involves understanding the project's scope, goals, and expected outcomes, as well as aligning stakeholders on the importance of the project.

Steps:

1. Generate Project Ideas and Identify CTQs
2. Select Project
3. Finalize Project Charter and As-Is Process Map

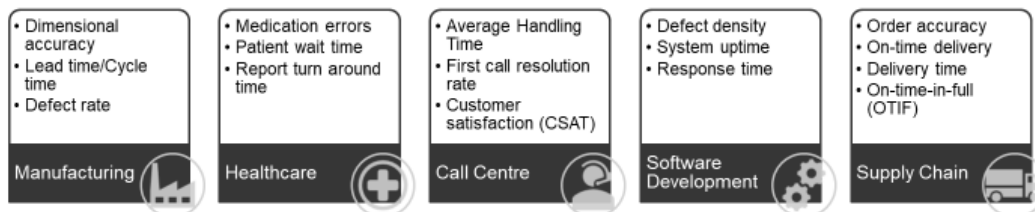
Main Deliverable : Approved Project Charter

Other Deliverables : High level process maps, SIPOC.



CTQ is a key **measurable characteristics** of a product, process, or service that are critical to meeting customer requirements and achieving high quality. CTQs are the specific parameters or metrics that need to be controlled within predefined limits to ensure customer satisfaction and process efficiency.

CTQs are derived from the VOC, VOB and COPQ analysis. By focusing on CTQs, organizations can prioritize improvement efforts, set measurable targets, and monitor performance against those targets. Below are some examples of CTQ metrics across different industries.



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VOC focuses on understanding the **needs, expectations, and preferences of customers**, whether they are **internal or external** to the organization. It involves gathering feedback, conducting surveys, interviews, or observing customer behaviour to gain insights into what they value and what improvements they desire.

Example : In a software development company, conducting user surveys and interviews to understand the challenges faced by customers when using their product would be a part of VOC analysis. This would help identify areas for improvement and guide the development of new features based on customer needs.

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There are several sources for gathering Voice of the Customer (VOC) information.

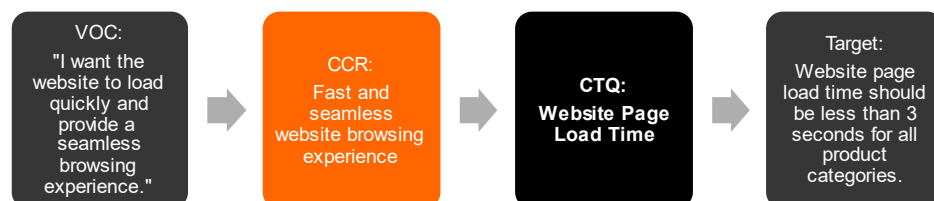
- Surveys
- Interviews
- Focus Groups
- Observations
- Social Media and Online Reviews
- Customer Feedback Channels
- Sales and Support Teams

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Voice of the Customer (VOC) can be converted into a Critical Customer Requirement (CCR) and further translated into a Critical-to-Quality (CTQ) parameter.



By converting the VOC into a CCR and then into a specific CTQ parameter, we can now measure and monitor the website's performance based on the customer's expectation of a fast and seamless browsing experience. This CTQ parameter provides a clear target for improvement efforts and enables us to assess whether the website meets the defined customer requirement.

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VOB analysis plays a vital role in project identification. The VOB incorporates the Balanced Scorecard framework, enabling a holistic view of performance across **financial, customer, internal processes, and learning and growth perspectives**. By defining specific KPIs within each perspective, projects that align with the overall business strategy can be selected and prioritized.

Example: In a pharmaceutical company, through analysis of internal processes, it was discovered that documentation errors in the manufacturing and quality control procedures are causing delays, rework, and potential compliance issues. Addressing these documentation errors becomes a priority project to enhance efficiency, reduce costs associated with rework, and ensure regulatory compliance.

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Some common sources that contribute to capturing the VOB:

- Inputs from leadership and management
- Business strategy and plans
- Financial reports and analysis
- Market research and competitive analysis
- Customer feedback and complaints
- Operational performance metrics
- Industry regulations and standards

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If the Voice of the Business (VOB) statement is "Our on-time order fulfilment is only 60%" we can convert it into a Critical Business Requirement (CBR) and further refine it into a Critical to Quality (CTQ) metric.



The VOB statement emphasizes the need to improve on-time order fulfilment, which currently stands at 60%. The Critical Business Requirement (CBR) sets a specific target. The CTQ metric focuses on measuring the on-time order fulfilment rate on a monthly basis.

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The Cost of Poor Quality (COPQ) analysis helps identify areas where **quality issues are causing financial losses, waste, or inefficiencies**.

- COPQ includes the direct costs associated with **rework, scrap, customer complaints, warranty claims**, and the indirect costs of **lost business opportunities, damaged reputation, or decreased customer loyalty**.
- By quantifying the financial impact of poor quality, organizations can prioritize improvement projects that target high COPQ areas, leading to significant cost savings and enhanced profitability.

Example : For a battery manufacturing company a large number of batteries are experiencing premature failures, resulting in customer dissatisfaction and warranty claims. The company recognizes that this not only incurs direct costs associated with replacing faulty batteries but also indirect costs due to customer dissatisfaction, potential damage to brand reputation, and lost sales opportunities.

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Common sources of data and information used in conducting a COPQ analysis:

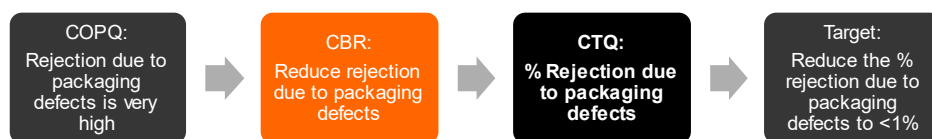
- Internal Quality Data
- Rework and Scrap Data
- Field Failures and Warranty Data
- Supplier Quality Data
- Customer Feedback and Complaints
- Audit and Compliance Findings
- Employee Productivity Data

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In a food and beverage company, it is identified that rejections due to packaging defects account for a significant loss.



By converting the COPQ analysis into a CTQ metric focused on the rejection rate due to packaging defects, the company establishes a clear quality objective and metric to address packaging-related issues. This approach enables them to proactively manage quality, reduce costs associated with rejected products, and deliver high-quality products to customers.

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CAN A PAIN AREA REFLECT ACROSS VOC, VOB AND COPQ ANALYSIS?

Suppose a company has a chronic pain area regarding lengthy lead times for order fulfilment. This pain point can be captured and analysed through VOC, VOB, and COPQ perspectives.

VOC	VOB	COPQ
By collecting and analysing customer feedback, surveys, and complaints, the company identifies the pain area related to lead time.	In this case, the company may have a VOB statement highlighting the importance of efficient operations and customer satisfaction, which impacts business performance and profitability.	Lengthy lead times can contribute to COPQ through various costs, such as increased inventory carrying costs, expedited shipping fees, missed delivery deadlines, customer order cancellations, and potential loss of business due to customer dissatisfaction.

ACTIVITY 1 : CTQ IDENTIFICATION EXERCISE



ACTIVITY 1: CTQ IDENTIFICATION EXERCISE

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VOC/VOB/COPQ	CTQ
We experience a high rate of customer returns due to incorrect product specs	
Customers complain about long wait times on the phone	
We incur substantial rework costs due to product defects	
We incur additional costs for compensating guests due to service failures	
Our assembly line is experiencing high rework rates due to quality defects	
Our production costs have increased significantly due to machine breakdowns	
Clients express dissatisfaction with the time taken to process loan applications	
We face high customer churn due to billing errors and incorrect charges	
Our website experiences frequent downtime during peak shopping season	
We incur high costs due to product recalls caused by contamination	

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OUTPUT

Identified CTQs. For example, for the first question in the table the appropriate CTQs can be percentage of products meeting specification, customer return rate.

TIME ALLOWED

- Preparation: 10 min
- Feedback: 10 min



An operational definition of CTQ is a **clear and precise definition** that describes how the CTQ metric will be measured or assessed in a specific and quantifiable manner.

- An operational definition typically includes the following elements: **the specific parameter being measured, the measurement method or technique, the unit of measurement, and any predetermined standards or criteria for acceptability.**
- While the operational definition should be established during the Define phase, it may be further refined and validated during the Measure phase based on the insights gained from data analysis and process understanding.

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In a pharmaceutical company, a project was initiated to reduce the changeover time for their compression machine, which is used to manufacture tablets. During the Define phase of the project, it was discovered that the internal definition of changeover time referred only to the cleaning time between two batches. The team realized the need for a more comprehensive definition of changeover time and decided to adopt the "good-to-good" logic.

This new definition meant that the changeover time was defined as the time from the last good tablet of the previous batch to the first good tablet of the next batch. This encompassed not only the cleaning time but also other activities, such as equipment setup, calibration, pre-runs, etc. It provided a more accurate representation of the time required for a successful transition from one batch to another, considering all relevant factors that could impact tablet quality and production efficiency.

However, this shift in the operational definition instigated the need for fresh data collection since the existing data was based on the previous, limited definition of changeover time.

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In a home finance company, a project was initiated to reduce the time taken for processing loan applications, specifically focusing on the duration from the applicant's successful CIBIL score run to the login of the file. During the Define phase of the project, it became evident that there were different interpretations of the stop time among the various departments involved, namely Sales, Credit, and CPA (Central Processing Agency).

Sales considered the stop time as the submission of the file to the CPA. However, for the CPA team, the stop time was contingent upon receiving complete and accurate files. On the other hand, the Credit team defined the stop time as the entry of the scanned file into the system. These differing definitions of the stop led to confusion and misalignment among the departments involved in the loan application process.

To overcome this challenge, it was crucial to establish a common and precise operational definition. After thorough discussions and understanding the process flow, the project team decided to use the scanning of the file as the definitive close time. This decision ensured that all departments aligned their activities and measurements to this specific event.

- **DMAIC Suitability:** DMAIC is suitable for projects where **process improvement, data-driven analysis, and structured problem-solving** are key objectives. DMAIC is effective for tackling **complex, chronic, and recurring problems** that require a systematic approach.
- **Type of Problem:** The cause of the **problem should be unknown or not clearly understood**. There should be no predetermined or overtly evident ideal resolution. If you possess prior knowledge of the solution, then proceed directly to resolving it!
- **Duration:** Green Belt projects are generally expected to be completed within a few months, typically ranging from **3 to 6 months**.
- **Complexity:** Green Belt projects should typically involve problems of moderate complexity.
- **Impact:** Green Belt projects should have a meaningful impact on the organization, whether it is in terms of financial savings, process efficiency improvements, customer satisfaction enhancements, or other key performance indicators.

- Kaizen is a continuous improvement approach that originated in Japan. It involves making **small, incremental changes to improve processes, products, or services over time**. The word "Kaizen" itself means **"change for the better" or "continuous improvement."**
- Kaizen encourages a mindset of continuous improvement, where **individuals at all levels of the organization** are **empowered to suggest and implement small improvements** in their work processes.
- Kaizen events typically have a smaller scope and focus on specific processes, work areas, or improvement opportunities within the organization.
- The focus is on identifying and implementing small improvements on an ongoing basis unlike DMAIC project which are more structured and episodic.
- Kaizen events often target areas where **quick improvements** can be made with relatively low effort and resources. Projects that can deliver immediate results and demonstrate the effectiveness of the Kaizen approach are typically preferred.

The A3 methodology provides a structured approach to problem-solving and decision-making by capturing all relevant information on a single sheet of paper. It gets its name from the international paper size A3, which is approximately equivalent to 11 x 17 inches.

An A3 typically includes:

1. Problem Statement
2. Current State Analysis
3. Desired State
4. Analysis and Countermeasures
5. Implementation Plan
6. Follow-up and Results :

Title: _____	
Plan	Do, Check Act
1. Background	5. Countermeasures
2. Current Condition	
3. Future Goal	
4. Root Cause Analysis	
	6. Implementation Plan
	7. Follow-up Action

Title: _____	
Plan	Do, Check Act
1. Background	5. Countermeasures
2. Current Condition	
3. Future Goal	
4. Root Cause Analysis	
	6. Implementation Plan
	7. Follow-up Action



- A3 can be a useful tool within the Kaizen approach to document and guide the problem-solving process.
- The structured framework of A3 helps teams analyze the current state, identify root causes, develop countermeasures, and track progress towards the desired future state.
- By utilizing A3 as part of the Kaizen process, teams can ensure that improvements are based on data, involve systematic problem-solving, and are effectively communicated to stakeholders.

STEP 3: FINALIZE PROJECT CHARTER AND AS-IS PROCESS MAP



- The project charter serves as a formal document that outlines the project's purpose, scope, objectives, and key stakeholders.
- It serves as a written agreement between management and the team, establishing clear expectations for the project.
- The development of a project charter is an iterative process, creating a dynamic document that may evolve throughout the different phases of DMAIC.
- Additionally, an as-is process map is created to document the current state of the process, providing a visual representation of how the process works and the areas for improvement

EXAMPLE CASE STUDY 3

Marvel Motor relies on Sparkling Glass Ltd for the supply of a critical glass component used in their vehicles. This particular component (windshield), undergoes manual visual inspection at Sparkling Glass Ltd's facility before being shipped to Marvel Motors. In the past quarter, quality issues with the windshields have resulted in frequent product recalls and customer complaints for Sparkling Glass Ltd.

The common reasons for rejections were cracks or chips, scratches, distortion, warping, improper logo or markings, etc. Size or dimensional deviations were rarely observed. While there are multiple reasons for the defects observed during the visual inspection of windshields, including the glass manufacturing process, handling procedures, environmental conditions, and quality control measures, the actual causes are not clearly understood.

The poor quality of windshields has resulted in increased rework, replacement, and potential penalties for Sparkling Glass Ltd. due to non-compliance with agreed-upon quality standards. In the last quarter, they incurred approximately \$150,000 in financial losses due to rejected shipments and customer compensation.

Defective windshields that reach Marvel Motors result in disruption of the production process, line stoppages, additional quality checks, and rework. If the quality problems persist, Marvel Motors may explore alternative suppliers, resulting in a potential revenue loss of \$1 million per year for Sparkling Glass Ltd.

Marvel Motors expects a high quality level of 99% for the supplied windshields to meet their stringent safety and performance standards. However, Marvel Motors has observed through their own quality checks that the actual quality level of the received glass components from Sparkling Glass Ltd. is at 91%. This discrepancy in quality levels has raised concerns and strained the relationship between the two companies .

Recognizing the importance of addressing the quality issues, top executives at Sparkling Glass Ltd. have made it a priority to take up a comprehensive improvement project to reduce defects, and achieve an accepted quality level of 99% within the next 6 months.

What are the possible CTQ's for this project? Finalize one CTQ and formulate the operational definition. Is the project suitable for DMAIC?

1. Sparkling Glass Ltd. manufactures glass sheets using specialized glass manufacturing processes, such as float glass or tempered glass production.
2. The molten glass is shaped, cooled, and cut into large glass sheets suitable for further processing. The large glass sheets are cut into smaller pieces according to the required dimensions for windshields .
3. The cut glass pieces are cleaned to remove any dust, dirt, or contaminants .
4. Interlayers are applied between multiple layers of glass to create a laminated structure for enhanced strength and safety.
5. The glass layers and interlayers are bonded together using adhesive materials and subjected to controlled pressure and heat.
6. Visual inspection of the laminated glass components is conducted to identify any defects, such as cracks, scratches, or impurities.
7. Inspectors assess the compliance of the windshields with specifications and quality standards.
8. Approved windshields are carefully packaged and labelled .
9. Packaged windshields are dispatched to Marvel Motors.



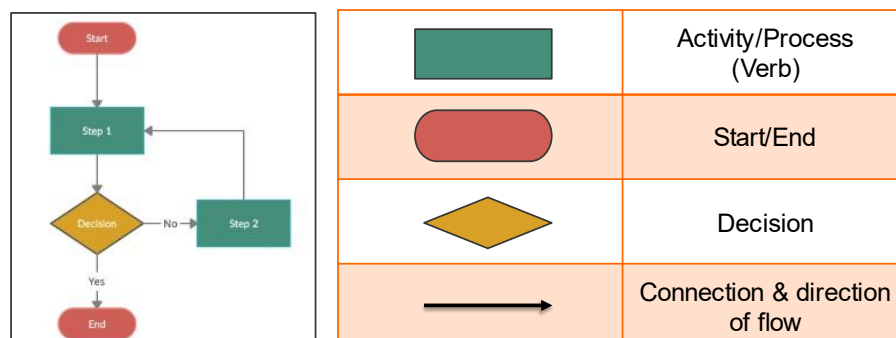
- A flowchart serves as a **visual representation of individual process steps arranged in sequential order**.
- It is a versatile tool that can be customized to suit a wide range of purposes and can be utilized to depict various processes, such as manufacturing processes, administrative or service processes, or project plans.
- Flowcharts are widely used as a common tool for process analysis and are recognized as **one of the seven fundamental quality tools**.

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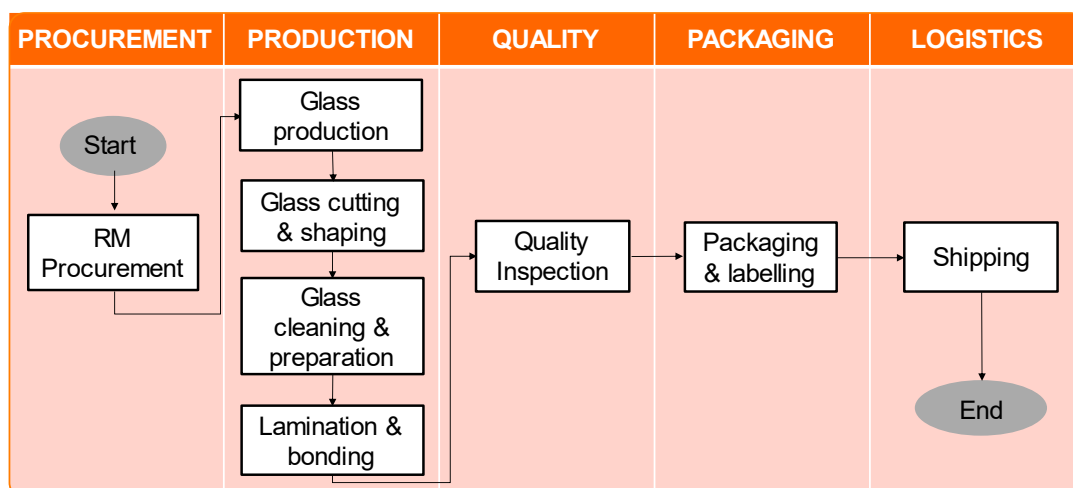
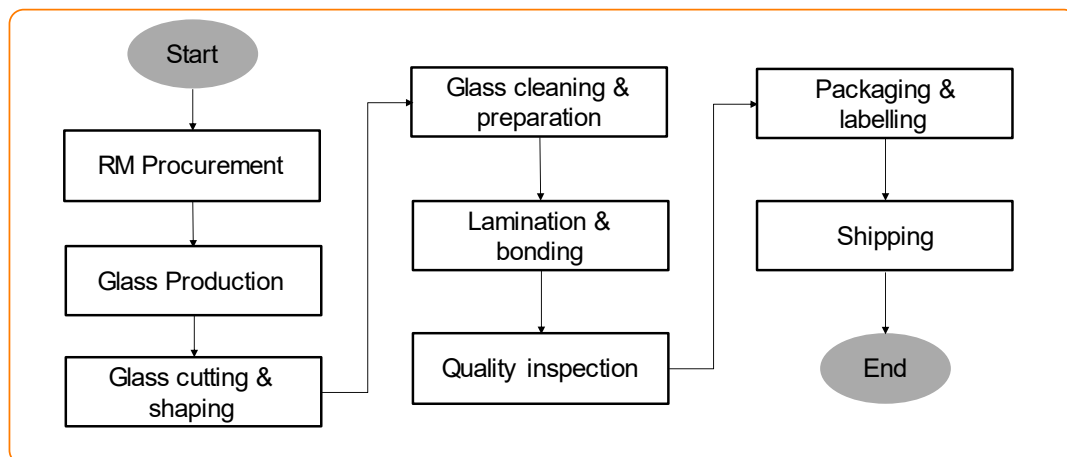


Process maps typically use a set of standardized shapes to represent different elements of a process. The basic shapes commonly used in process maps include:



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SIPOC stands for Suppliers, Inputs, Process, Outputs, and Customers. SIPOC provides a high-level overview of a process, highlighting its key elements and their relationships.

1. **Suppliers:** Suppliers are the individuals, departments, or organizations that provide the necessary inputs to the process. They can be internal or external entities.
2. **Inputs:** Inputs refer to the materials, data, or resources required to initiate or support the process. Inputs are received from the suppliers and are used as raw materials or information to carry out the process.
3. **Process:** The process describes the series of activities, steps, or tasks that transform the inputs into desired outputs. It outlines how the work is done and the specific actions taken.
4. **Outputs:** Outputs are the results or deliverables generated by the process. They can be products, services, reports, or any other tangible or intangible outcomes that fulfil a specific purpose or meet customer requirements.
5. **Customers:** Customers are the recipients or end-users of the outputs generated by the process.

While both tools provide valuable insights into a process, there are specific advantages to using SIPOC over a process map:

- SIPOC's big-picture view helps in **identifying the scope of the process and its boundaries**.
- It facilitates better **communication and alignment among stakeholders**.
- SIPOC is useful for **identifying gaps or areas of improvement** in a process. By clearly outlining the inputs, outputs, and suppliers, it becomes easier to spot missing elements or potential bottlenecks.
- It is useful in situations where a **rapid overview of a process** is required. It helps in scoping projects, identifying critical process steps, and understanding the interfaces between different process components.

SIPOC EXAMPLE

SUPPLIER	INPUT	PROCESS	OUTPUT	CUSTOMER
RM suppliers	procurement specifications	RM procurement	Procured RM meeting quality standards	Sparkling Glass Ltd (Production team)
RM procurement department	Procured RM (silica sand, soda ash, limestone)	Glass production	Manufactured glass sheets	Production (Glass cutting and shaping team)
Production (Glass cutting and shaping team)	Glass sheets, cutting and shaping equipment, cutting patterns/templates	Cutting & shaping	Cut and shaped glass pieces	Production (Cleaning & preparation team)
Production (Cleaning & preparation team)	Cut and shaped glass pieces, cleaning solutions, cleaning equipment	Cleaning & preparation	Cleaned and prepared glass pieces	Production (Lamination & bonding team)
Production (Lamination & bonding team)	Cleaned and prepared glass pieces, interlayers, adhesive materials	Lamination & bonding	Laminated glass components	Quality inspection department (Sparkling Glass)
Quality inspection department	Laminated glass components, quality inspection criteria, inspection equipment	Quality inspection	Accepted and rejected windshields	Packaging and labelling department
Packaging and labelling department	Quality Pass shields, packaging materials, labels	Packaging & labelling	Packaged and labelled windshields	Logistics team
Logistics team	Packaged and labelled windshields	Shipping & delivery	Shipped and delivered windshields	Marvel Motors

ACTIVITY 2 : STAKEHOLDER IDENTIFICATION EXERCISE



ACTIVITY 2: IDENTIFY STAKEHOLDERS

SGS ACADEMY

Stakeholder	Responsibility	Candidate
Sponsor	Provides strategic guidance, resources, and support for the improvement project.	
Team Leader (TL)	Leads the improvement project, coordinates team activities, and ensures project objectives are achieved.	
Team Members	Support the TL in the execution of the project, providing input, insights, and assistance as needed. actively contribute to the team's goals and objectives, working collaboratively towards achieving desired outcomes .	
Process Owners	Process owners are individuals responsible for the overall management and performance of a specific process within an organization. They have a deeper understanding of the process and are accountable for its design, execution, and continuous improvement.	
Subject Matter Experts	SMEs provide guidance, advice, and insights based on their expertise to support decision-making and problem -solving. SMEs may or may not have direct accountability for the management or performance of a process.	
Other Stakeholders	Support	

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OUTPUT

Identified designations/positions in Sparkling Glass Ltd. who will be suitable candidates for the project with respect to example case study 3. For example, a good candidate of Sponsor can be the COO at Sparkling Glass Ltd.

TIME ALLOWED

- Preparation: NA (Case study already discussed)
- Feedback: 5-10 min

Problem Statement	The problem statement clearly defines the issue or challenge that the project aims to address.
Business Case	The business case outlines the justification for undertaking the project, including the expected benefits and potential return on investment.
Goal Statement	The goal statement (SMART) sets a specific, measurable, attainable, relevant, and time-bound objective that the project aims to achieve.
Scope	The scope identifies the boundaries of the project, defining what is included (in scope) and what is excluded (out of scope).
Team Members	The team members are the individuals who will be involved in executing the project, bringing their expertise and skills to accomplish the project objectives.
Timelines	The project charter may include five milestones based on the DMAIC methodology, representing the key stages of the project's timeline and progress.

Project Code, Project Title	
Internal project nomenclature, Appropriate short title for the project	
Business Case	Problem Statement / Opportunity Statement
Why is it important for us to work on this project? What advantages does it bring to the company? What would be the outcome if we choose not to proceed with this project?	What specific problem or challenge is being experienced? Provide quantifiable information about the extent, location, timing, and magnitude of the pain
Goal Statement	Scope
The goal of this project should be described using the SMART format, which stands for Specific, Measurable, Achievable, Relevant, and Time-bound.	What is in-scope and what is out of scope?
Team Members	Project Timeline
Mention the names of the team sponsor, team leader, team members, subject matter experts, customers, process owners, and other stakeholders	What is the expected timeline for this project. Start and end dates and dates for key milestones (DMAIC)
Approval	
Project Sponsor: _____ Date: _____	

ACTIVITY 3 : PROJECT CHARTER CREATION EXERCISE



ACTIVITY 3: FINALIZE THE CHARTER FOR EXAMPLE CASE STUDY 3

SGS ACADEMY

Project Code, Project Title	
Internal project nomenclature, Appropriate short title for the project	
Business Case	Problem Statement / Opportunity Statement
Goal Statement	Scope
Team Members	Project Timeline
Approval	
Project Sponsor: _____ Date: _____	

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OUTPUT

- Fill in all the 6 sections of the charter template with inputs from example case study 3. For example, the problem statement can be;

Problem Statement: The current visual inspection process for windshields conducted by Sparkling Glass Ltd. is resulting in a high rate of rejection, with a quality level of only 91%, causing financial losses, strained relationships with Marvel Motors, and compromised brand reputation for both Sparkling Glass Ltd. and Marvel Motors.

TIME ALLOWED

- Preparation: NA (Case study already discussed)
- Feedback: 15-20 min

DEFINE PHASE TOLLGATE CHECKLIST

- Is the problem or opportunity clearly defined and documented?
- Has the primary CTQ been finalized for the project?
- Has the project been chosen in alignment with organizational goals?
- Does a financial business case exist?
- Are the in-scope and out-of-scope items clearly defined, identifying the specific boundaries and limitations of the project?
- Is the SMART goal statement established?
- Is the project team formed, including the necessary resources with relevant skills and expertise?
- Has the project charter been created, outlining the project's purpose, objectives, deliverables, timelines, and key stakeholders?
- Have all key stakeholders been identified and their expectations understood?
- Has the initial project plan been developed, outlining the major activities, milestones, and timelines for each phase of the improvement project?
- Is there a communication plan in place to ensure effective communication among team members and stakeholders?
- Have all necessary approvals and sign-offs been obtained from relevant stakeholders?

Have the required resources, such as budget, personnel, and equipment, been allocated to the project?

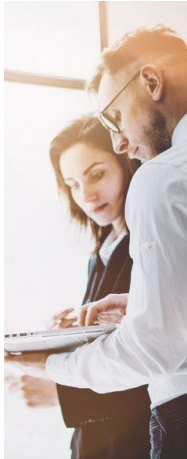
- The Define phase is the first stage of the DMAIC (Define, Measure, Analyze, Improve, Control) methodology .
- It aims to clearly define the problem or opportunity that the project will address.
- Key objectives of the Define phase include understanding the project scope, goals, and expected outcomes .
- The phase focuses on generating project ideas, identifying Critical-to-Quality (CTQ) factors, and selecting the most suitable project.
- A project charter is created, which outlines the project's purpose, objectives, scope, deliverables, stakeholders, and timeline .
- An As-Is Process Map is developed to document the current state of the process.
- The Define phase sets the foundation for the project by establishing a clear problem statement, project scope, and common understanding among team members and stakeholders .

MEASURE PHASE

OBJECTIVES

MEASURE PHASE



Aim of the session

- Provide a clear understanding of the Measure phase in the DMAIC methodology .
- Define the objectives, focus, and key deliverables of the Measure phase .
- Establish the importance of measurement system analysis.
- Learn how to perform attribute agreement analysis to assess the level of agreement among different evaluators or measurement methods for categorical data .
- Gain a foundational understanding of basic statistical concepts to analyse process stability and capability .
- Learn how to establish a baseline for process performance to compare against future improvements

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MEASURE PHASE OBJECTIVES



By the end of this session, you will be able to:

- Understand the significance of the Measure phase in the DMAIC methodology and its role in gathering accurate data for process evaluation .
- Set performance standards or targets based on the identified CTQ metric to establish clear expectations for process improvement .
- Conduct a Measurement System Analysis to assess the reliability and validity of the measurement system used to collect data .
- Establish a baseline by collecting data to quantify the current performance of the process and identify areas for improvement .
- Use various statistical techniques to analyse process capability and measure the extent to which the process meets customer requirements .

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MEASURE PHASE

The Measure phase is the second stage in the DMAIC methodology . It focuses on establishing a robust measurement system, defining performance standards, and collecting baseline data to quantify the current state of the process.

Objective: The primary objective of the Measure phase is to establish a reliable measurement system and gather accurate data to assess the process's current performance.

Steps:

4. Finalize CTQ and Set Performance Standards
5. Perform Measurement System Analysis (MSA)
6. Establish Baseline

Main Deliverable : Baseline performance

Other Deliverables : Established measurement system, Updated project documentation

In this step, the Critical-to-Quality (CTQ) metrics identified in the Define phase are further refined, and specific performance standards or targets are established . These standards provide the desired level of performance that the process should achieve.

CTQ	Data Type	Operational Definition	LSL	USL	Target



- MSA (Measurement System Analysis) evaluates the **reliability, accuracy, and precision** of a measurement system.
- It assesses the quality of measurement instruments, techniques, and procedures to ensure accurate and consistent data collection.
- MSA techniques, such as Gage R&R and Attribute Agreement Analysis, identify sources of variation and measure system capabilities.
- MSA helps organizations make reliable decisions by ensuring the validity and reliability of the data used.
- It is typically conducted during the Measure phase of DMAIC to validate the measurement system before data collection.
- **MSA should be a routine activity for organizations, even if they are not specifically engaged in a DMAIC project, to continuously monitor and improve the quality of their measurement systems.**

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A measurement system refers to the combination of measurement instruments, techniques, procedures, and personnel involved in collecting data. It can vary depending on the context and the nature of the process being measured. Below are some examples of measurement systems:

Measuring Instruments	Thermometers, scales, rulers, callipers, gauges, spectrophotometers, and pH meters
Testing Equipment	Tensile testing machines, hardness testers, moisture analysers, and vibration analysers
Software Systems	Computer-based systems used for data collection, analysis, and reporting.
Inspection Tools	Magnifying glasses, microscopes, X-ray machines, and ultrasonic testing equipment
Data Collection Forms	Checklists, rating scales, surveys, or questionnaires
Human Observers/ Appraisers	Visual inspection, subjective assessments

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	Logical Validation	Attribute Agreement Analysis (AAA)	Bias, Linearity, Stability, Gauge Repeatability and Reproducibility (R&R)*
Type of data	Numeric, Attribute	Attribute	Numeric
Purpose	Ensures the system captures what it is supposed to capture	Assess the level of agreement among different evaluators	Assess accuracy, variability and consistency

* AAA is relatively simpler and is covered in Lean Six Sigma Green Belt. Bias, Linearity, Stability, Gauge R&R, being more complex, require a deeper understanding of statistical concepts and methods. These techniques are typically covered in the Lean Six Sigma Black Belt curriculum.

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Escalation Desk Response Time Analysis

A B2B financial services company has a dedicated escalation desk email for their financial advisors. The company has set an internal service level agreement (SLA) to respond to any email received in this inbox with suitable action or resolution within 24 hours. To monitor and evaluate their compliance with this SLA, the escalation team records data regarding the time of receiving the email, type of complaint, sending advisor, date, time of response, etc. in an Excel sheet. This data is manually entered and analysed at the end of each month for reporting to the management.

MSA approach: Use sampling techniques (Random, Stratified, Systematic) to select a representative subset of the emails for analysis.

1. Time Validation: Verify the accuracy of the recorded date and time stamps for email receipt and response. Identify any potential outliers or anomalies in the recorded response times.
2. Data Review: Excel sheet needs to be checked for any data entry errors or inconsistencies, such as missing or incorrect information, duplicate entries, etc.
3. SLA compliance calculation: Verify the calculations.

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1. **Define the attribute and categories:** Clearly define the attribute or characteristic that will be assessed and identify the categories. Ensure that the categories are MECE.
2. **Select appraisers/evaluators:** Determine the number of appraisers. Appraisers should be knowledgeable and trained on the attribute and the classification criteria.
3. **Prepare the evaluation set:** Prepare sample set (representative of the population) that will be evaluated by the appraisers.
4. **Determine the standard responses (Optional):** Provide the evaluation sample set to an expert/SME (or a panel) and ask them to evaluate and classify each item based on the established criteria. Record the standard responses for each sample.
5. **Conduct the assessment:** Ensure that each appraiser independently assesses the samples in random order without knowledge of other appraisers ratings. Collect the classifications provided by each appraiser for each sample. Repeat the activity with a new random order. Do not reveal to the appraisers that samples are repeated.

6. **Analyze the data:** Compare the classifications provided by the appraisers using appropriate statistical methods.
7. **Interpret the results:** Analyze the results to determine the level of agreement among the appraisers. Identify areas of disagreement or inconsistency and investigate potential sources of variation.
8. **Take corrective actions:** Based on the analysis and findings, take corrective actions to improve the attribute agreement. This may involve additional training for appraisers, clarification of classification criteria, or addressing sources of variation to enhance consistency and reliability.
9. **Monitor and validate:** Periodically validate the attribute agreement to assess the effectiveness of corrective actions and improvements made.

In Attribute Agreement Analysis (AAA), the following four percentages are calculated:

	Within Appraisers (Repeatability)	It assesses how closely appraisers agree with themselves when evaluating the same samples multiple times.
	Each appraiser vs standard (Appraiser accuracy)	It indicates how closely appraisers assessments align with the standard or reference evaluations.
	Between appraisers (Reproducibility)	It assesses how closely appraisers agree with each other when evaluating the same samples.
	All appraisers vs Standard (Team Accuracy)	It indicates how closely the team's evaluations align with the standard or reference evaluations.

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In Attribute Agreement Analysis (AAA), the following four percentages are calculated:

	Within Appraisers (Repeatability)	It assesses how closely appraisers agree with themselves when evaluating the same samples multiple times.
	Each appraiser vs standard (Appraiser accuracy)	It indicates how closely appraisers assessments align with the standard or reference evaluations.
	Between appraisers (Reproducibility)	It assesses how closely appraisers agree with each other when evaluating the same samples.
	All appraisers vs Standard (Team Accuracy)	It indicates how closely the team's evaluations align with the standard or reference evaluations.

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- The acceptable percentage is often determined by the organization based on their **quality standards, customer requirements, and the criticality of the attribute** being assessed.
- It may also take into consideration any applicable **regulatory or industry-specific guidelines**.
- Commonly used standard for accepting/rejecting the measurement system:

Reject	Caution	Accept
Team accuracy* <70%	70-90%	>90%

*In AAA, the "Team Accuracy" or "All Appraisers vs Standard" percentage is likely to be the smallest among the four percentages. This is because the "Team Accuracy" percentage compares the combined evaluations of all appraisers with the standard or reference evaluations.

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QA manager at Macrosoft, a software company, wants to evaluate the effectiveness of their bug reporting system as they suspect that some team members may not fully understand the criteria for categorizing bugs. To assess this measurement system, the QA manager randomly selects 25 bug reports (BR) from the team's database and assigns them to the 4 QA team members for evaluation. The team members are asked to classify each bug report as either "Critical" or "Non-critical" based on the severity of the bug and its impact on the software's functionality. After completing the initial evaluation, the QA manager takes the same 25 bug reports and distributes them to the 4 team members once again for a second round of evaluation. However, the team members are unaware that they are reviewing the same bug reports as before. Is the performance of the measurement system (QA team members) acceptable?

Data File: Macrosoft

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Stat>Quality
Tools>Attribute
Agreement
Analysis...

Attribute Agreement Analysis

Data are arranged as

☐ Attribute column:

Samples:

Appraisers:

☒ Multiple columns:

'QA1 Trial 1'-'QA4 Trial 2'

(Enter trials for each appraiser together)

Number of appraisers: 4

Number of trials: 2

Appraiser names (optional):

Known standard/attribute: 'Quality Manager' (Optional)

☐ Categories of the attribute data are ordered

Information... Options... Graphs... Results...

Select Help OK Cancel

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Within Appraisers

Assessment Agreement

Appraiser	# Inspected	# Matched	Percent	95% CI
1	25	25	100.00	(88.71, 100.00)
2	25	25	100.00	(88.71, 100.00)
3	25	25	100.00	(88.71, 100.00)
4	25	25	100.00	(88.71, 100.00)

Matched: Appraiser agrees with him/herself across trials.

Between Appraisers

Assessment Agreement

# Inspected	# Matched	Percent	95% CI
25	23	92.00	(73.97, 99.02)

Matched: All appraisers' assessments agree with each other.

Each Appraiser vs Standard

Assessment Agreement

Appraiser	# Inspected	# Matched	Percent	95% CI
1	25	23	92.00	(73.97, 99.02)
2	25	24	96.00	(79.65, 99.90)
3	25	25	100.00	(88.71, 100.00)
4	25	24	96.00	(79.65, 99.90)

Matched: Appraiser's assessment across trials agrees with the known standard.

All Appraisers vs Standard

Assessment Agreement

# Inspected	# Matched	Percent	95% CI
25	23	92.00	(73.97, 99.02)

Matched: All appraisers' assessments agree with the known standard.

Team accuracy > 90% suggests that the team as a whole is performing well in assessing the attribute being measured.

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ACTIVITY 4: PRACTICE CASE STUDY 1

In the case of Marvel Motors and Sparkling Glass Ltd., it was observed that Marvel Motors expects a high quality level of 99% for the windshields supplied by Sparkling Glass, but their own quality checks have revealed a quality level of only 91%. This discrepancy raises concerns regarding the effectiveness and consistency of the visual inspection process conducted by Sparkling Glass Ltd.

QC manager at Sparkling Glass selected a set of windshields to form the evaluation set for the study. Appraisers chosen from Marvel Motors provided the standard responses. The selected windshields were independently evaluated by the inspectors at Sparkling Glass Ltd. as pass (0) or fail (1). The same set of windshields were provided to the inspectors for a second round of evaluation after a gap of few days. Is the visual inspection process of windshields a Sparkling Glass reliable?

Data File: Visual Inspection

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PURPOSE

- Evaluate understanding of AAA
- Evaluate right usage of Minitab for analysis

TASK

- Perform AAA

OUTPUT

- Results and Interpretation

TIME ALLOWED

Preparation: 3-5 min

Feedback: 1-2 min

ACTIVITY 5: PRACTICE CASE STUDY 2



Customer support department at a company called Tech Solutions handles incoming calls from customers seeking technical assistance with the company's products. The Quality Assurance (QA) team within the company regularly audits calls recording of the customer support operators to ensure high-quality service. To evaluate the consistency of the QA team's assessments, a Measurement System Analysis (MSA) study was conducted. The study involved a set of 30 recorded customer calls randomly selected from the call database. These recordings were anonymized to remove any identifiable information.

The two QA team members, Alice and Bob, independently listened to each of the 30 customer call recordings and classified the responses of the customer support operators as either "Satisfactory" or "Unsatisfactory" based on the quality of assistance provided. Is there a high level of agreement between the two team members in categorizing the customer responses?

Data File: Tech Solutions

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PURPOSE

- Evaluate understanding of AAA
- Evaluate right usage of Minitab for analysis

TASK

- Perform AAA

OUTPUT

- Results and Interpretation

TIME ALLOWED

Preparation: 3-5 min

Feedback: 1-2 min

Knowledge of statistics is essential primarily for the Measure, Analyze and Control phases of the DMAIC methodology for several reasons:

1. **Data Analysis :** The Measure phase involves analysing data to establish a baseline, measure process performance, and identify areas for improvement. Statistics provides the necessary tools and techniques to analyse data, such as descriptive statistics, graphical representations, hypothesis testing, and correlation analysis. These techniques help in uncovering patterns, trends, and relationships in the data, enabling meaningful insights and informed decision-making in the analyse phase.
2. **Process Capability Analysis :** Assessing process capability requires statistical knowledge to calculate process capability indices which help in determining if the process is capable of meeting customer specifications and identifying areas where process improvements are needed.
3. **Statistical Process Control (SPC):** SPC involves monitoring and controlling process performance using statistical techniques. Knowledge of statistics is crucial for establishing control charts, setting control limits, interpreting control chart patterns to identify special cause variation and take appropriate actions.

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Discrete/Attribute	Continuous
• Yes/No responses • Count data: e.g. number of defects in a product, the number of customer complaints received, or the number of emails received in a day • Categorical Data: e.g. type of payment method used (credit card, cash, check), the brand of a product (Brand A, Brand B, Brand C) • Likert Scale Responses • Pass/Fail data	• Dimensions: Length, Width, etc. • Temperature, Pressure, Volume • Time: TAT, lead time, ACHT, delivery time, etc. • Speed, Distance • Sales revenue, Profit, Cash • Blood pressure, Blood Sugar, etc.

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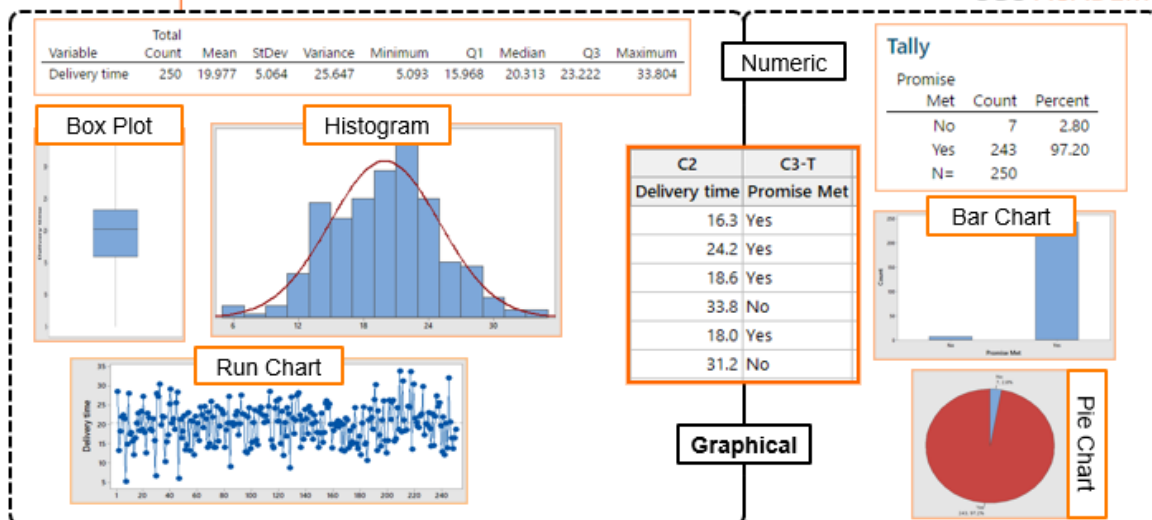
A bustling online food delivery platform called 'Swigatto' has recently introduced an exciting new feature, the "30-Minute Food Delivery Promise." This promise guarantees that customers will receive their orders within a swift 30-minute timeframe, ensuring their meals arrive fresh and piping hot at their doorsteps.

Data File: Swigatto

C2	C3-T
Delivery time	Promise Met
16.3	Yes
24.2	Yes
18.6	Yes
33.8	No
18.0	Yes
31.2	No

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Measures of central tendency (Location)

- **Mean:** The mean or average, represents the arithmetic balance point of a data distribution.
- **Median:** The median corresponds to the mid-point or 50th percentile of a data distribution. The data is arranged in ascending or descending order, and the median is the middle value in the ordered list.
- **Mode:** Mode represents the value that occurs most frequently in a data distribution

Measures of dispersion (Spread)

- **Range:** Range is the difference between the largest observation and the smallest observation in the data set. A smaller range suggests a lesser degree of variability,
- **Interquartile Range (IQR):** IQR is determined by the difference between the 75th percentile and the 25th percentile in a dataset. It provides a measure of the spread of the middle 50% of the data.
- **Variance:** Variance represents the average squared deviation of each data point from the mean.
- **Standard deviation (SD):** SD is the square root of the variance. It provides a more interpretable measure of variation by returning to the original unit of the data. Standard deviation represents the average deviation of the values from the mean. It indicates the typical or average distance between each data point and the mean.

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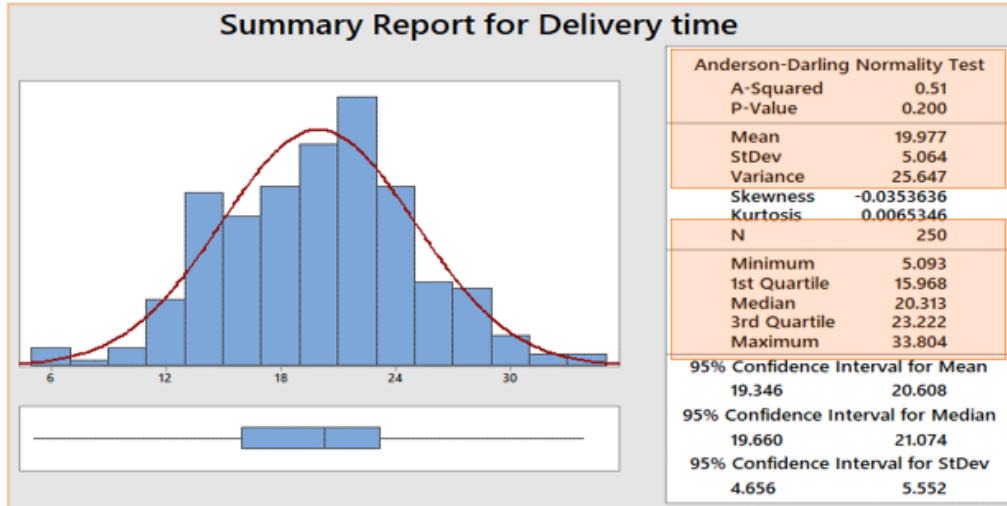
72



Stat>Basic Statistics>
Graphical Summary...

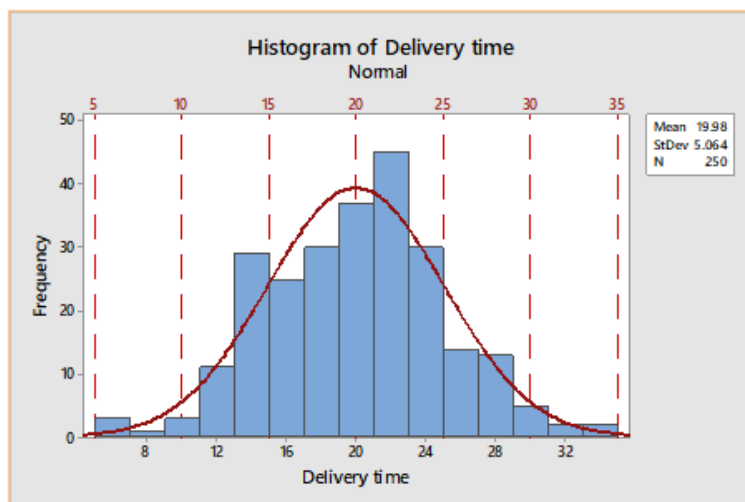
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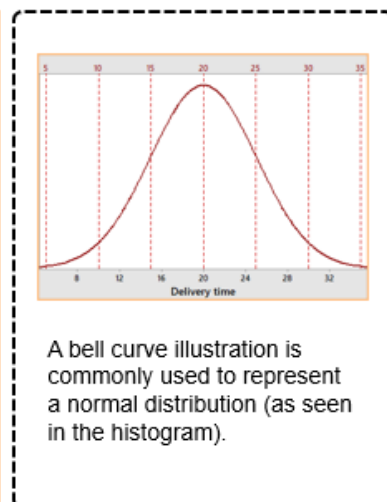


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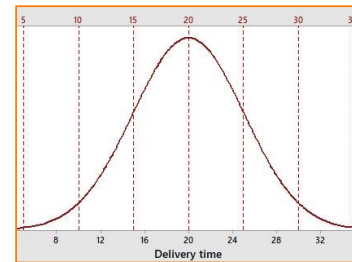


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The normal distribution, also known as the Gaussian distribution or bell curve, is a continuous probability distribution that is widely used in statistics and probability theory.

The key features of the normal distribution are as follows :

1. **Symmetry:** The distribution is symmetrical around its mean.
2. **Bell-shaped:** The distribution has a characteristic bell shape, with the highest point (peak) at the mean.
3. **Parameters:** The normal distribution is fully defined by two parameters: the mean (μ) and the standard deviation (σ). The mean determines the centre of the distribution, while the standard deviation controls the spread or dispersion of the data.



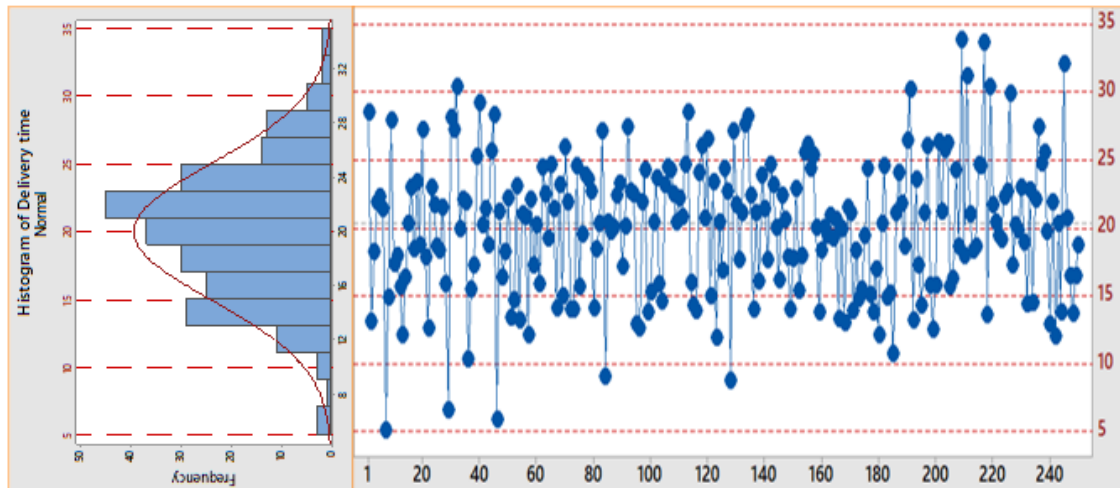
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4. **Central Limit Theorem:** The normal distribution is often encountered in real-world data due to the Central Limit Theorem.
 - According to this theorem, the sum or average of a large number of independent and identically distributed random variables tends to follow a normal distribution, regardless of the shape of the original distribution.
 - The Central Limit Theorem has significant implications in statistics. It allows for the use of normal distribution-based techniques, such as hypothesis testing, confidence intervals, and estimation, even when the underlying population distribution is not normal.
 - By relying on the Central Limit Theorem, statisticians can make reliable inferences about population parameters using sample statistics.
5. **68-95-99.7 rule:** The normal distribution has an interesting property known as the empirical rule or 68-95-99.7 rule. According to this rule, approximately 68.27% of the data falls within one standard deviation of the mean, about 95.46% falls within two standard deviations, and nearly 99.73% falls within three standard deviations.

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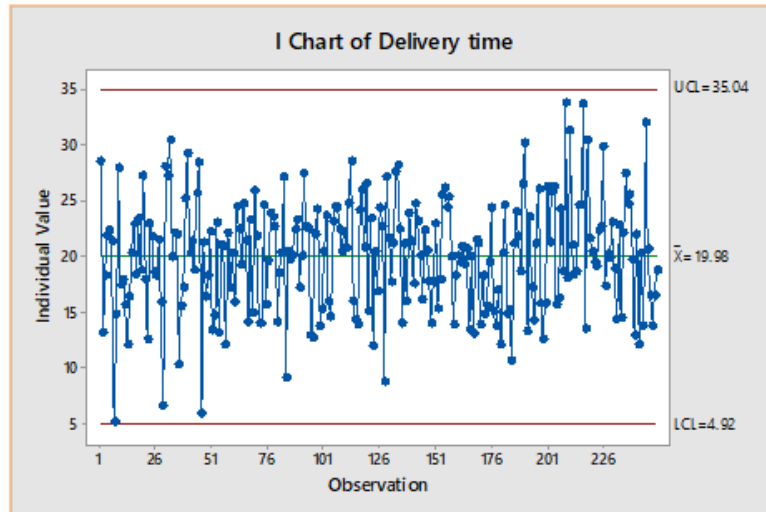
78

- Common cause variation, also known as random variation or inherent variation, refers to the natural variability that is inherent in a process. It represents the typical variability observed in a process when no specific assignable cause is present. The normal distribution is often used to model common cause variation. When data follows a normal distribution, it suggests that the variation observed in the process is primarily due to common causes.
- Special Cause Variation: Special cause variation, also known as assignable cause variation or non-random variation, refers to variation that is caused by specific factors or events that are not part of the typical functioning of the process. Special causes are sporadic or intermittent and can lead to unexpected or unusual outcomes. Special cause variation is characterized by a significant deviation from the normal or expected pattern of variation.
- Special cause variation is considered non-routine and requires investigation and corrective action to identify and eliminate the specific assignable cause.
- Control charts are valuable tools for differentiating between common cause and special cause variation.

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The control limits in control charts are typically set at ± 3 standard deviations from the process mean based on statistical principles and the characteristics of a normal distribution.



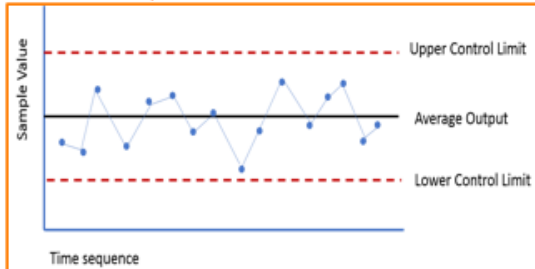
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- SPC is based on the principle that most processes exhibit inherent variability, which can be categorized as common cause variation or special cause variation.
- Control charts are a fundamental tool in SPC. They provide a graphical representation of process data over time.
- They allow practitioners to monitor the process, identify variations, and distinguish between common cause and special cause variation.
- The primary purpose of a control chart is to provide an indication of whether a process is stable (in-control) or unstable (out of control) based on the presence of common cause and special cause variations.

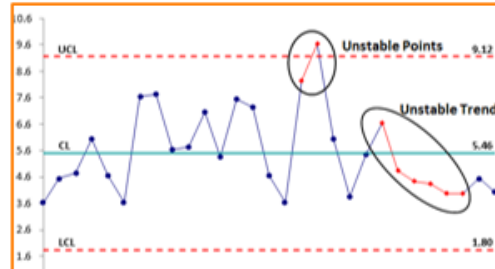
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Process in-control

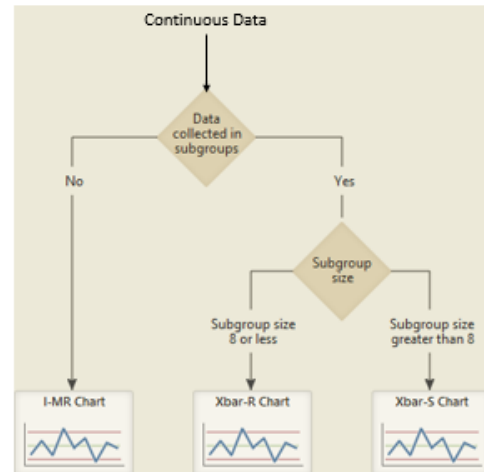
When a process is stable and only exhibits common cause variation (all points within the control limits), proceed with checking the process capability



Process out of control

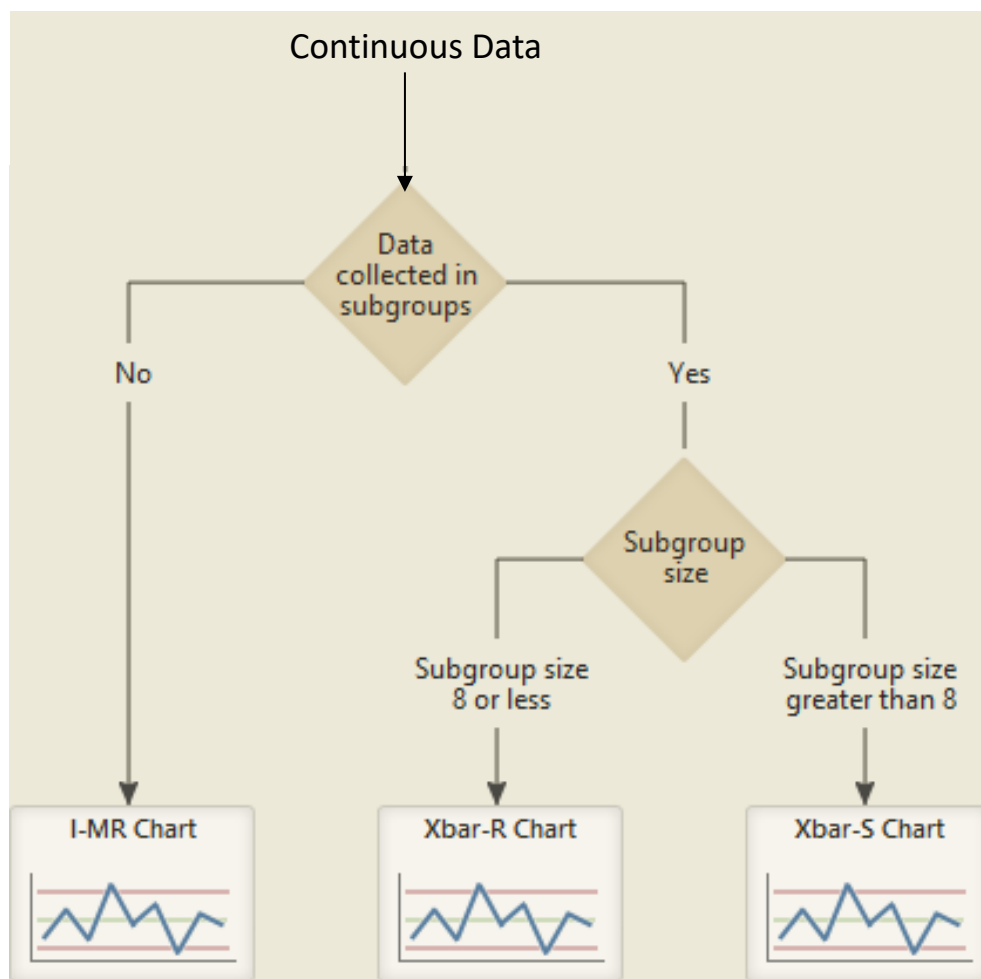
An out of control process is typically identified by specific patterns or signals, such as data points consistently falling outside control limits, non-random patterns, trends, shifts, or cycles along with common cause variation. Such scenario necessitate RCA and CAPA is required to address the identified issues and prevent their recurrence in the future.

- Continuous data control charts, typically consist of two control charts: one for the process mean (central tendency) and another for the process variation (dispersion).
- Charts monitoring the central tendency: Individual (I), Mean ($\bar{X}$)
- Charts monitoring the variation: Moving Range (MR), Range (R), Standard Deviation (s)
- By having two charts, any shifts or changes in either the mean or the variation can be detected separately, providing a more comprehensive view of the process behaviour.



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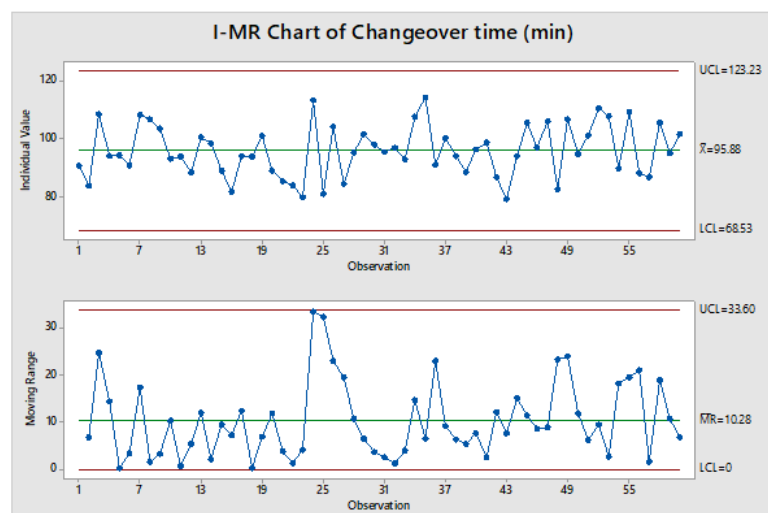
1. A Type-B change-over process is involved when switching from one formulation to another on a blister packaging machine for tablets. Use an appropriate control chart to monitor the change-over time. [Data File: CO Time]
2. Vivida, a food manufacturing company, needs to monitor the acidity level (pH) of their signature tomato-based pasta sauce. The ideal pH range for their sauce is between 3.8 and 4.2, as it provides the desired taste and preservation properties. Six samples of the sauce at regular time intervals are collected and measured during production. Use an appropriate control chart to monitor the process. [Data File: Vivida]
3. A company called Mobicare provides repair services for mobile phone companies. As a manager at Mobicare, you want to monitor the time required for quality checks post the repair process. Quality checks are crucial to ensure that the repaired mobile phones meet the required standards before being returned to customers. To monitor this, you randomly select and record the time required for quality checks for 12 repaired mobile phones every day. [Data File: Mobicare]

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84

Stat > Control Charts > Variable charts for Individuals > I-MR...

The process mean and variation are stable. No points are out of control on either chart.

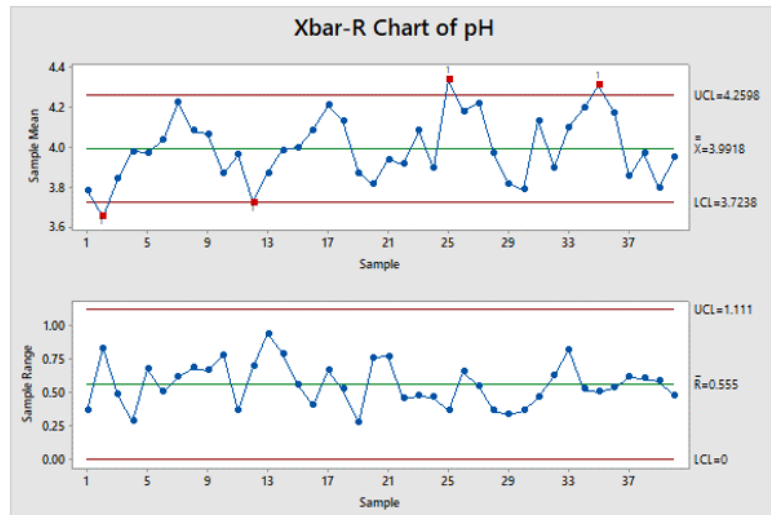


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85

Stat > Control Charts >
Variable charts for
Subgroups > Xbar-R...

The process variation is stable. No subgroups are out of control on the R chart. However, the process mean may not be stable.

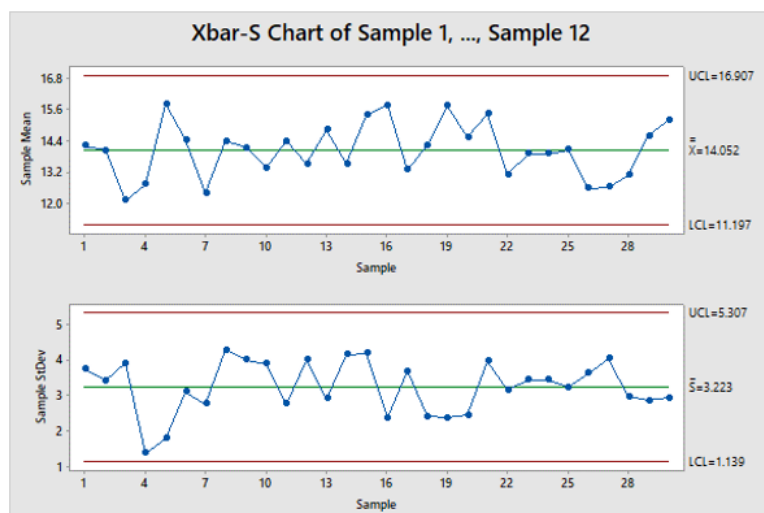


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86

Stat > Control Charts >
Variable charts for
Subgroups > Xbar-S...

The process mean and variation are stable. No subgroups are out of control on either chart.



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- Specification limits, also known as **tolerance limits** or target values, are the predetermined acceptable ranges within which a product or process characteristic must fall to meet customer requirements.
- Specification limits are typically defined based on **customer expectations, regulatory requirements, industry standards, or internal quality standards**.
- They provide a clear indication of the acceptable variation around the desired target value.
- There are two types of specification limits:

Upper Specification Limit (USL): This is the maximum acceptable value for a given characteristic beyond which the product or process is considered nonconforming or unacceptable.

Lower Specification Limit (LSL): This is the minimum acceptable value for a given characteristic, below which the product or process is considered nonconforming or unacceptable.

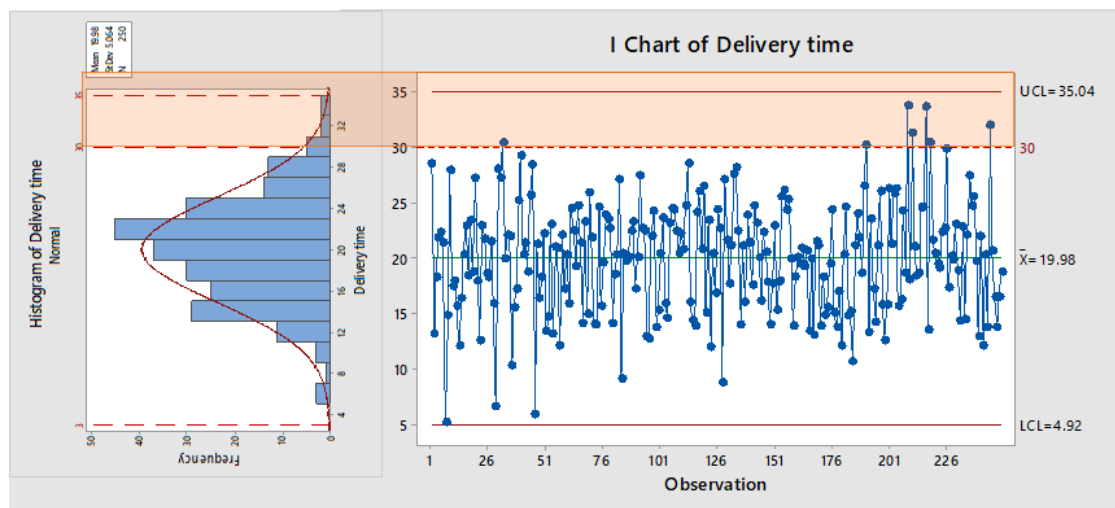
Specification limits can be defined as either a single limit (either LSL or USL) or as both LSL and USL. The choice depends on the specific characteristic or attribute being measured and the requirements of the customer or industry standards.

- **Single Limit (LSL or USL):**
 - LSL only: e.g. the purity of a chemical must not be lower than 98% to meet the desired quality standards.
 - USL only: e.g. In a call centre, the average call handling time has an USL of 5 minutes.
- **Both LSL and USL:**
 - e.g. The specification limits for tensile strength of a metal component may require it to be within a range of 70 MPa to 90 MPa.

- Process Stability refers to the consistency of a process in terms of important process characteristics such as the average value or variation of a parameter. If a process behaves consistently over time, it is considered stable or in control.
- On the other hand, Process Capability measures the ability of a process to meet specified requirements or specifications. Various methods can be employed to assess process capability, including estimating defects per million opportunities (DPMO) or utilizing Capability indices such as Z, Pp, and Ppk.
- It is important to note that process stability and process capability are distinct concepts. However, there is a significant connection between the two. Process capability assessment should only be conducted after establishing process stability.
- If the process is unstable, predicting its capability becomes unreliable. If the process average is shifting over time, an estimation of process capability only reflects the current state and does not guarantee future performance. It is insufficient to demonstrate capability at a single point in time.
- The process must achieve desired capability consistently - and that's where process stability comes into play.

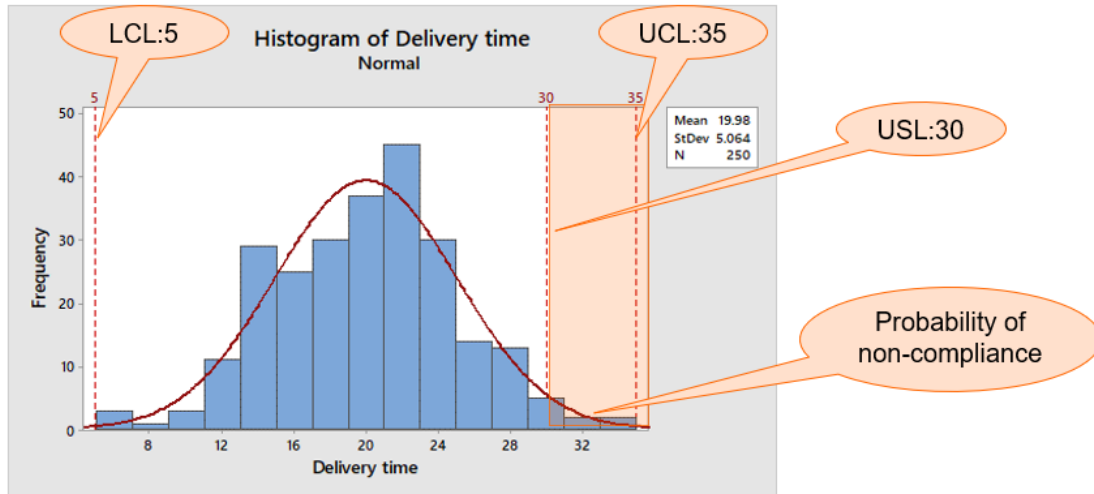
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- Establishing a baseline in the Measure phase of DMAIC is essential for understanding the **current performance of a process and determining its capability**. The baseline provides a reference point against which future improvements can be measured.
- Once the baseline data is collected and the measurement system is validated, process capability analysis can be conducted. It helps determine if the process is capable of consistently meeting the customer requirements or if it needs improvement to achieve the desired level of performance.
- By **establishing a baseline in the Measure phase and subsequently recalculating process capability in Improve and Control phase**, practitioners can evaluate the effectiveness of their improvement efforts, validate the impact of process changes, and ensure that the process is capable of consistently delivering the desired outcomes.

When evaluating the current performance of a process with a continuous data (normally distributed) metric/CTQ and establishing a baseline for future comparison, the following capability metrics can be considered:

Information/Data Available	Preferred Metric
Either LSL or USL, Mean and Standard Deviation	Sigma Level (Z overall)
Both LSL and USL, Mean and Standard Deviation	Pp & Ppk



- Sigma level (Z) is a statistical measure used to quantify the performance and capability of a process.
- The higher the sigma level, the more capable process, the higher the ability of the process to consistently produce output that meets customer requirements.
- Sigma level is a measure of process variation and represents the number of standard deviations between the process mean and the nearest specification limit.

$$Z_{overall} = \frac{USL - Xbar}{S_{overall}}$$

$$Z_{overall} = \frac{Xbar - LSL}{S_{overall}}$$

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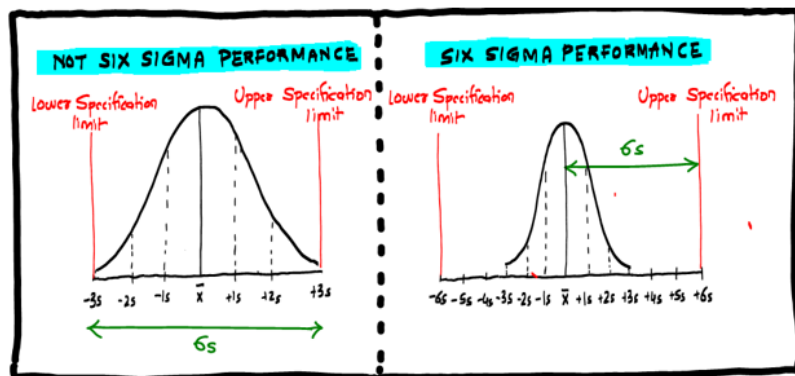
95

Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output
6	0.001	999999.999	3	1349.898	998650.102	0	500000	500000	-3	998650.102	1349.898
5.9	0.002	999999.998	2.9	1865.813	998134.187	-0.1	539827.837	460172.163	-3.1	999032.397	967.603
5.8	0.003	999999.997	2.8	2555.130	997444.870	-0.2	579259.709	420740.291	-3.2	999312.862	687.138
5.7	0.006	999999.994	2.7	3466.974	996533.026	-0.3	617911.422	382088.578	-3.3	999516.576	483.424
5.6	0.011	999999.989	2.6	4661.188	995338.812	-0.4	655421.742	344578.258	-3.4	999663.071	336.929
5.5	0.019	999999.981	2.5	6209.665	993790.335	-0.5	691462.461	308537.539	-3.5	999767.371	232.629
5.4	0.033	999999.967	2.4	8197.536	991802.464	-0.6	725746.882	274253.118	-3.6	999840.891	159.109
5.3	0.058	999999.942	2.3	10724.110	989275.890	-0.7	758036.348	241963.652	-3.7	999892.200	107.800
5.2	0.100	999999.900	2.2	13903.448	986096.552	-0.8	788144.601	211855.399	-3.8	999927.652	72.348
5.1	0.170	999999.830	2.1	17864.421	982135.579	-0.9	815939.875	184060.125	-3.9	999951.904	48.096
5	0.287	999999.713	2	22750.132	977249.868	-1	841344.746	158655.254	-4	999968.329	31.671
4.9	0.479	999999.521	1.9	28716.560	971283.440	-1.1	864333.939	135666.061	-4.1	999979.342	20.658
4.8	0.793	999999.207	1.8	35930.319	964069.681	-1.2	884930.330	115069.670	-4.2	999986.654	13.346
4.7	1.301	999998.699	1.7	44565.463	955434.537	-1.3	903199.515	96800.485	-4.3	999991.460	8.540
4.6	2.112	999997.888	1.6	54799.292	945200.708	-1.4	919243.341	80756.659	-4.4	999994.587	5.413
4.5	3.398	999996.602	1.5	66807.201	933192.799	-1.5	933192.799	66807.201	-4.5	999996.602	3.398
4.4	5.413	999994.587	1.4	80756.659	919243.341	-1.6	945200.708	54799.292	-4.6	999997.888	2.112
4.3	8.540	999991.460	1.3	96800.485	903199.515	-1.7	955434.537	44565.463	-4.7	999998.699	1.301
4.2	13.346	999986.654	1.2	115069.67	884930.330	-1.8	964069.681	35930.319	-4.8	999999.207	0.793
4.1	20.658	999979.342	1.1	135666.061	864333.939	-1.9	971283.440	28716.560	-4.9	999999.521	0.479
4	31.671	999968.329	1	158655.254	841344.746	-2	977249.868	22750.132	-5	999999.713	0.287
3.9	48.096	999951.904	0.9	184060.125	815939.875	-2.1	982135.579	17864.421	-5.1	999999.830	0.170
3.8	72.348	999927.652	0.8	211855.399	788144.601	-2.2	986096.552	13903.448	-5.2	999999.900	0.100
3.7	107.800	999892.200	0.7	241963.652	758036.348	-2.3	989275.890	10724.110	-5.3	999999.942	0.058
3.6	159.109	999840.891	0.6	274253.118	725746.882	-2.4	991802.464	8197.536	-5.4	999999.967	0.033
3.5	232.629	999767.371	0.5	308537.539	691462.461	-2.5	993790.335	6209.665	-5.5	999999.981	0.019
3.4	336.929	999663.071	0.4	344578.258	655421.742	-2.6	995338.812	4661.188	-5.6	999999.989	0.011
3.3	483.424	999516.576	0.3	382088.578	617911.422	-2.7	996533.026	3466.974	-5.7	999999.994	0.006
3.2	687.138	999312.862	0.2	420740.291	579259.709	-2.8	997444.870	2555.130	-5.8	999999.997	0.003
3.1	967.603	999032.397	0.1	460172.163	539827.837	-2.9	998134.187	1865.813	-5.9	999999.998	0.002
									-6	999999.999	0.001

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Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output	Z overall	DPMO	Conforming Output
6	0.001	999999.999	3	1349.898	998650.102	0	500000	500000	-3	998650.102	1349.898
5.9	0.002	999999.998	2.9	1865.813	998134.187	-0.1	539827.837	460172.163	-3.1	999032.397	967.603
5.8	0.003	999999.997	2.8	2555.130	997444.870	-0.2	579259.709	420740.291	-3.2	999312.862	687.138
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5.6	0.011	999999.989	2.6	4661.188	995338.812	-0.4	655421.742	344578.258	-3.4	999663.071	336.929
5.5	0.019	999999.981	2.5	6209.665	993790.335	-0.5	691462.461	308537.539	-3.5	999767.371	232.629
5.4	0.033	999999.967	2.4	8197.536	991802.464	-0.6	725746.882	274253.118	-3.6	999840.891	159.109
5.3	0.058	999999.942	2.3	10724.110	989275.890	-0.7	758036.348	241963.652	-3.7	999892.200	107.800
5.2	0.100	999999.900	2.2	13903.448	986096.552	-0.8	788144.601	211855.399	-3.8	999927.652	72.348
5.1	0.170	999999.830	2.1	17864.421	982135.579	-0.9	815939.875	184060.125	-3.9	999951.904	48.096
5	0.287	999999.713	2	22750.132	977249.868	-1	841344.746	158655.254	-4	999968.329	31.671
4.9	0.479	999999.521	1.9	28716.560	971283.440	-1.1	864333.939	135666.061	-4.1	999979.342	20.658
4.8	0.793	999999.207	1.8	35930.319	964069.681	-1.2	884930.330	115069.670	-4.2	999986.654	13.346
4.7	1.301	999998.699	1.7	44565.463	955434.537	-1.3	903199.515	96800.485	-4.3	999991.460	8.540
4.6	2.112	999997.888	1.6	54799.292	945200.708	-1.4	919243.341	80756.659	-4.4	999994.587	5.413
4.5	3.398	999996.602	1.5	66807.201	933192.799	-1.5	933192.799	66807.201	-4.5	999996.602	3.398
4.4	5.413	999994.587	1.4	80756.659	919243.341	-1.6	945200.708	54799.292	-4.6	999997.888	2.112
4.3	8.540	999991.460	1.3	96800.485	903199.515	-1.7	955434.537	44565.463	-4.7	999998.699	1.301
4.2	13.346	999986.654	1.2	115069.67	884930.330	-1.8	964069.681	35930.319	-4.8	999999.207	0.793
4.1	20.658	999979.342	1.1	135666.061	864333.939	-1.9	971283.440	28716.560	-4.9	999999.521	0.479
4	31.671	999968.329	1	158655.254	841344.746	-2	977249.868	22750.132	-5	999999.713	0.287
3.9	48.096	999951.904	0.9	184060.125	815939.875	-2.1	982135.579	17864.421	-5.1	999999.830	0.170
3.8	72.348	999927.652	0.8	211855.399	788144.601	-2.2	986096.552	13903.448	-5.2	999999.900	0.100
3.7	107.800	999892.200	0.7	241963.652	758036.348	-2.3	989275.890	10724.110	-5.3	999999.942	0.058
3.6	159.109	999840.891	0.6	274253.118	725746.882	-2.4	991802.464	8197.536	-5.4	999999.967	0.033
3.5	232.629	999767.371	0.5	308537.539	691462.461	-2.5	993790.335	6209.665	-5.5	999999.981	0.019
3.4	336.929	999663.071	0.4	344578.258	655421.742	-2.6	995338.812	4661.188	-5.6	999999.989	0.011
3.3	483.424	999516.576	0.3	382088.578	617911.422	-2.7	996533.026	3466.974	-5.7	999999.994	0.006
3.2	687.138	999312.862	0.2	420740.291	579259.709	-2.8	997444.870	2555.130	-5.8	999999.997	0.003
3.1	967.603	999032.397	0.1	460172.163	539827.837	-2.9	998134.187	1865.813	-5.9	999999.998	0.002
									-6	999999.999	0.001



If the difference between the mean and the specification limit is six times the standard deviation, it indicates that the process is performing at a sigma level of six. A sigma level of six is considered excellent performance, as it corresponds to a very low defect rate.

A bustling online food delivery platform called 'Swigatto' has recently introduced an exciting new feature, the "30-Minute Food Delivery Promise." This promise guarantees that customers will receive their orders within a swift 30-minute timeframe, ensuring their meals arrive fresh and piping hot at their doorsteps. What is the ability of a process to consistently meet the 30 min requirement?

Data File: Swigatto

Sigma level is about 2 & the failure rate is 2.39%

Data Type	Continuous
Distribution	Normal
Mean	19.977
Standard Dev (Overall)	5.064
Lower Specification Limit (LSL)	
Upper Specification Limit (USL)	30
Sigma Level (Overall)	1.98
DPMO	23893

Calculator Excel File: ProcessCapabilityCalculator Continuous(Normal)Data

ACTIVITY 6: PRACTICE CASE STUDY 3

The processing time data for a standard task is available for two setups; Office and Work-From-Home (WFH). The expectation is that the processing time should not exceed 30 min.

- Has the new setup (WFH) impacted the processing time?
- To comment on which one is better, would you like to consider variation too?
- Calculate a single measure for process performance that incorporates the central tendency, variation and specification limits?

Data File: WFH

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PURPOSE

- Evaluate understanding of Sigma Level
- Evaluate right usage of Minitab and Excel calculator for analysis

TASK

- Calculate Sigma Level

OUTPUT

- Results and Interpretation

TIME ALLOWED

Preparation: 5 min

Feedback: 5 min

ACTIVITY 7: PRACTICE CASE STUDY 4

Electra has performed a series of pull tests on 50 randomly selected solder joints on their circuit boards. The target minimum strength requirement for each solder joint is 5 Newtons (N). The mean and overall standard deviation of the pull test measurements were found to be 5.2 N and 0.2 N respectively. Calculate Z overall and DPMO.

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PURPOSE

- Evaluate understanding of Sigma Level
- Evaluate right usage of Excel calculator

TASK

- Calculate Sigma Level

OUTPUT

- Results and Interpretation

TIME ALLOWED

Preparation: 2 min

Feedback: 2 min

Pp and Ppk are process capability indices that are used to assess the ability of a process to meet customer specifications. They provide a numerical measure of process performance in relation to the specified tolerances or specification limits.

- Pp: Pp measures the overall process capability by considering the total process variation in relation to the specification width. Generally accepted value of $Pp \geq 1.33$.

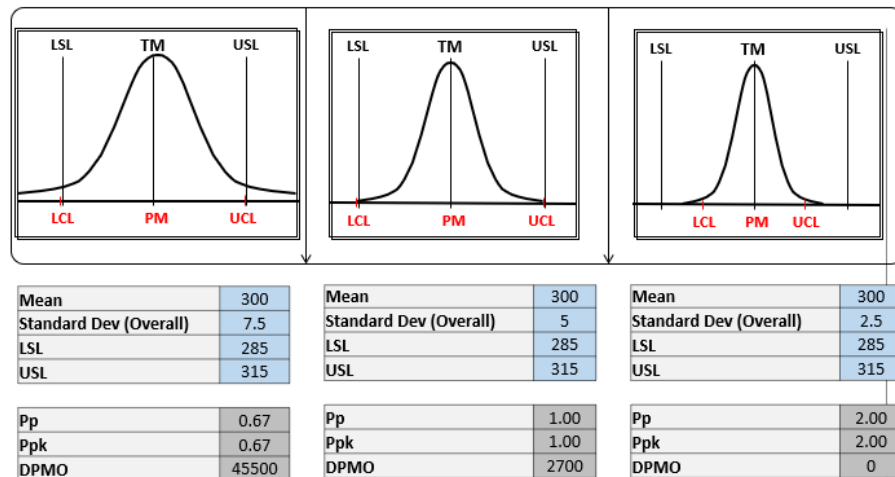
$$Pp = (USL - LSL) / (6 * \text{Standard Deviation})$$

- Ppk: Ppk takes into account both process centring and process spread to assess process capability. Ideal value of $Ppk = Pp$.

$$Ppk = \min[(USL - \text{Mean}) / (3 * \text{Standard Deviation}), (\text{Mean} - LSL) / (3 * \text{Standard Deviation})]$$

- Pp focuses on overall process performance, while Ppk considers both process centering and spread.

OJ, a company which specializes in producing packaged fruit juices, wants to assess the capability of their filling process by measuring fill volume accuracy. They measured the fill volumes of 200 randomly selected bottles of their orange juice product. The target fill volume for each bottle is 300 ± 15 ml.

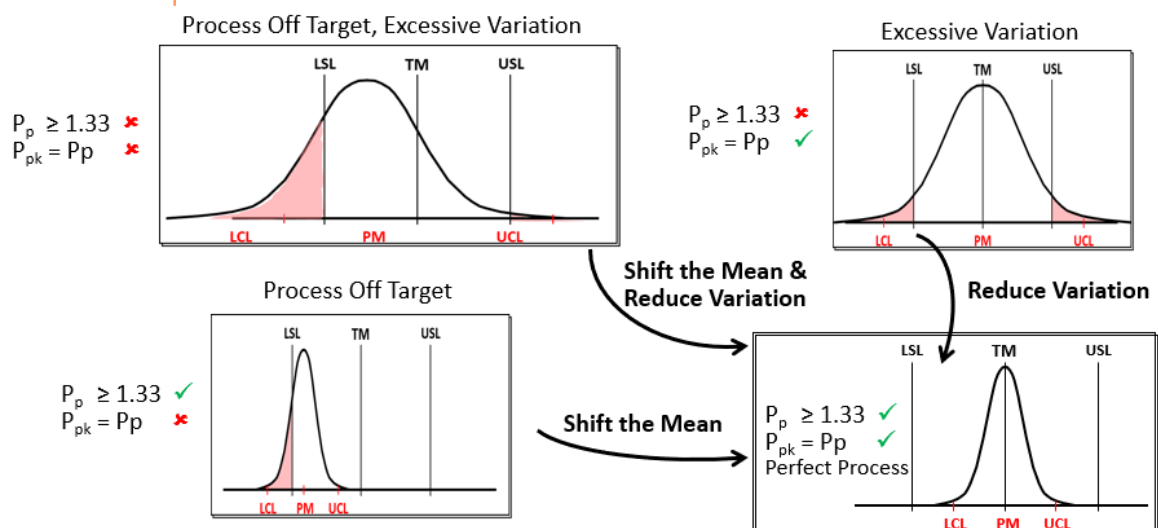


LSL	USL	Target	Process Mean	SD	P_p	P_{pk}
285	315	300	290	2.5		
285	315	300	295	2.5		
285	315	300	300	2.5		
285	315	300	305	2.5		
285	315	300	310	2.5		

- Which parameter(s) affects P_p ?
- Which parameter(s) impacts P_{pk} ?
- $P_{pk} = P_p$ indicates _____
- When is $P_{pk} < P_p$?

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ACTIVITY 8: PRACTICE CASE STUDY 5

Precision Parts has measured the diameters of 100 randomly selected pistons produced by their manufacturing process. The target diameter for each piston is 75 ± 0.1 mm. The mean and overall standard deviation are 75 and 0.03 mm respectively. Calculate the Pp and Ppk values to evaluate the capability of the piston manufacturing process. What will be your recommendation; shift the mean or reduce variation or both?

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PURPOSE

- Evaluate understanding of Pp and Ppk analysis
- Evaluate right usage of Excel calculator

TASK

- Calculate Pp and Ppk

OUTPUT

- Results and recommendation

TIME ALLOWED

Preparation: 2 min

Feedback: 2 min

ACTIVITY 9: PRACTICE CASE STUDY 6



AgroCare produces a liquid fertilizer. One critical parameter in the production of liquid fertilizer is the nitrogen content, which is essential for promoting plant growth. The target content is $5 \pm 0.5\%$. 50 samples of the liquid fertilizer were taken to measure the nitrogen content. The mean was found to be 4.7% and overall standard deviation was 0.2%. Calculate the Pp and Ppk values to evaluate the capability of manufacturing process. What will be your recommendation; shift the mean or reduce variation or both?

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PURPOSE

- Evaluate understanding of Pp and Ppk analysis
- Evaluate right usage of Excel calculator

TASK

- Calculate Pp and Ppk

OUTPUT

- Results and recommendations

TIME ALLOWED

Preparation: 2 min

Feedback: 2 min

ACTIVITY 10: PROVIDE RECOMMENDATIONS



Pp	Ppk	Shift Mean/ Reduce SD/ Both
2.5	2.5	
0.8	0.8	
1.5	0.8	
0.8	0.3	

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PURPOSE

- Evaluate understanding of Pp and Ppk analysis
- Evaluate if correct recommendations are provided based on Pp and Ppk values

TASK

- Provide suitable recommendations based on Pp and Ppk values

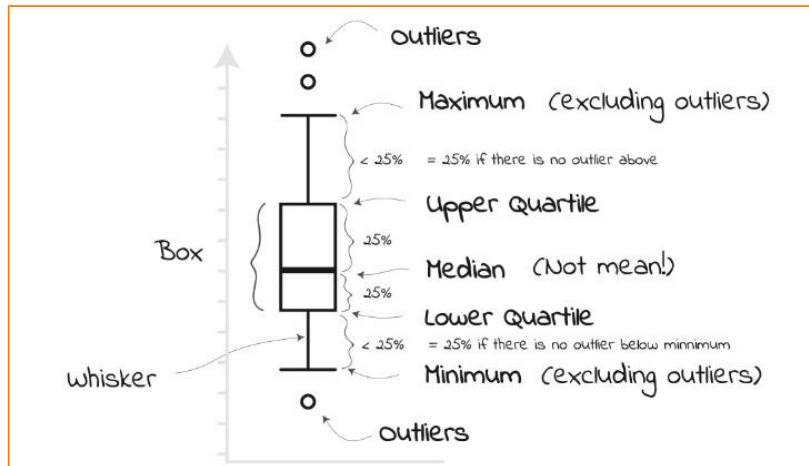
OUTPUT

- Recommendations

TIME ALLOWED

Preparation: 3 min

Feedback: 3 min



A box plot, also known as a box-and-whisker plot, is a graphical representation of the distribution of a dataset. Box plots are useful for visualizing the spread, skewness, and potential outliers in a dataset. They provide a concise summary of the data distribution and can be used to compare multiple datasets.

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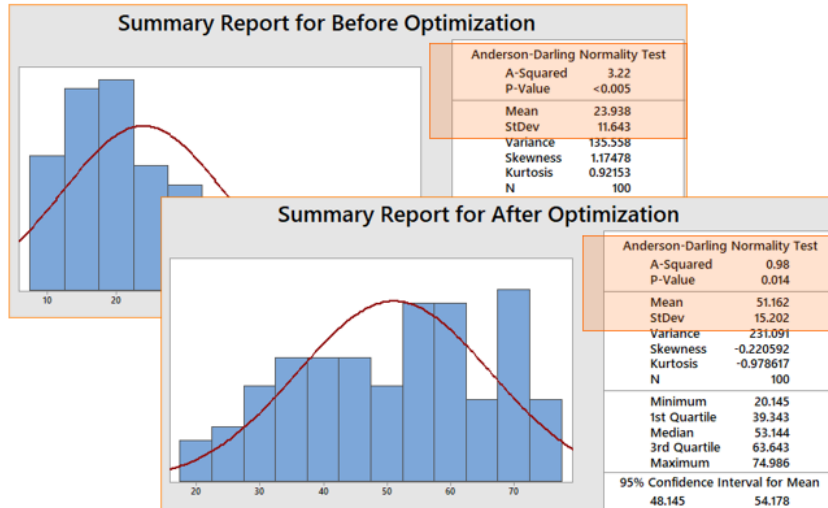
108

A company called Gigatel, has been experiencing issues with the internet connection speed for their customers. To improve the performance of their network they implemented a new optimization technique. This includes network infrastructure upgrades, intelligent traffic management, protocol optimization, employing caching mechanisms and utilizing content delivery networks, etc. The download speed data was collected from 100 random customers before the implementation and after the implementation. Has the performance improved?

Data File: Gigatel

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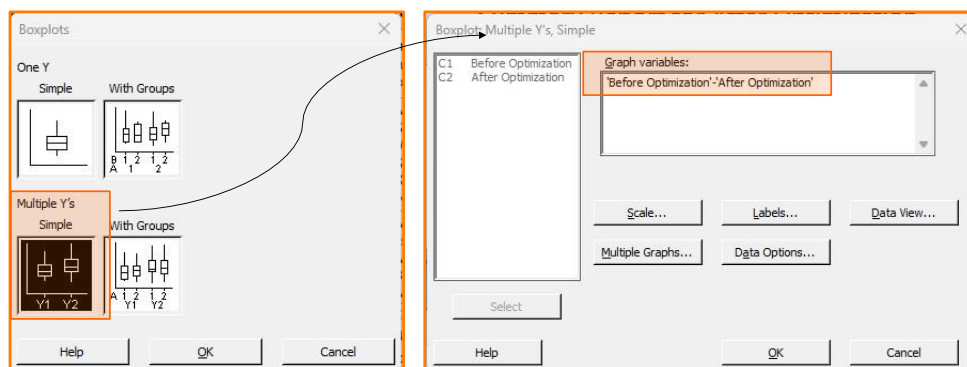
As indicated by the Anderson-Darling Normality test, both the datasets do not follow a normal distribution. When data is not normally distributed, comparing means and standard deviations may not provide an accurate representation of the data or yield meaningful insights.

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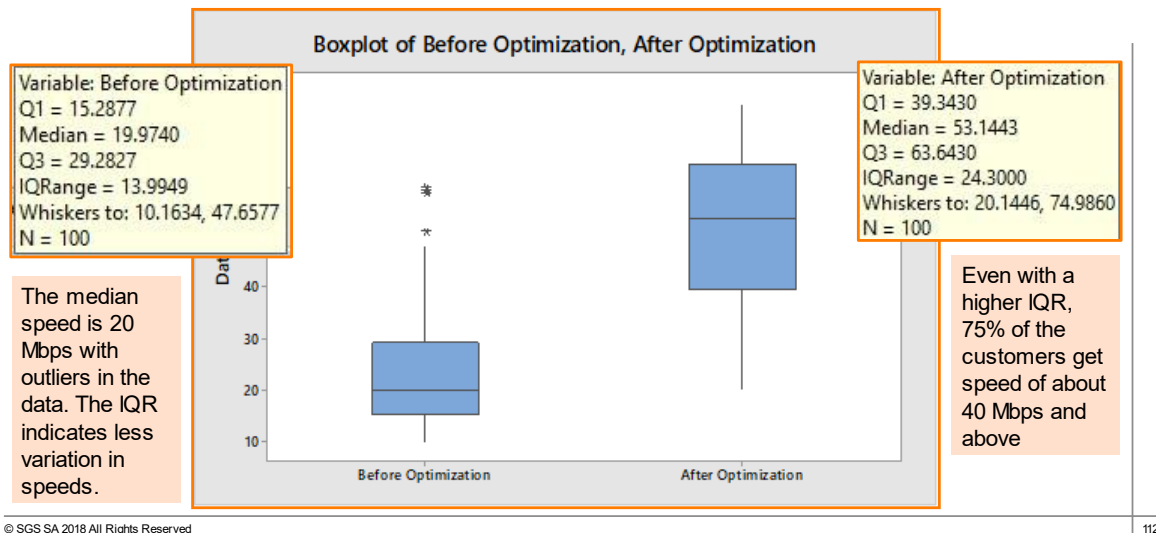
In this scenario, a box plot can be used to visually compare the datasets before and after improvement. The box plot will provide information about the central tendency (median) and spread (interquartile range) of the data, allowing for a quick assessment of the download speed performance.

Graph>
Box-Plot...



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A bustling online food delivery platform called 'Swigatto' has recently introduced an exciting new feature, the "30-Minute Food Delivery Promise." This promise guarantees that customers will receive their orders within a swift 30-minute timeframe, ensuring their meals arrive fresh and piping hot at their doorsteps.

Data File: Swigatto

We were discussing continuous data

C2	C3-T
Delivery time	Promise Met
16.3	Yes
24.2	Yes
18.6	Yes
33.8	No
18.0	Yes
31.2	No

Let's discuss attribute data

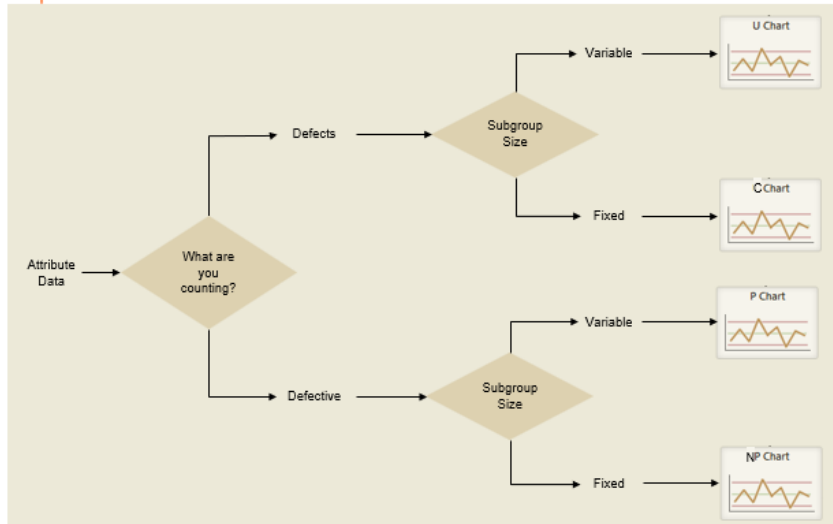
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- Discrete/Attribute data control charts are statistical tools used to monitor and control processes where the data is discrete or categorical in nature.
- Attributes data represents the presence or absence of a certain characteristic or the count of occurrences within a specific category. Examples include pass/fail outcomes, defect counts, defect types, yes/no responses, and categorical ratings.
- Defect: A defect refers to a single nonconformity or error in a product or process. It represents an individual flaw or imperfection that renders an item unacceptable or deviating from the desired specifications. Defects are typically counted on a per-unit basis, where each unit can have zero or more defects.
- Defective: A defective refers to a unit or item that contains one or more defects. Defectives are counted on a binary basis, where each unit is classified as either defective (at least one defect) or non-defective (no defects).

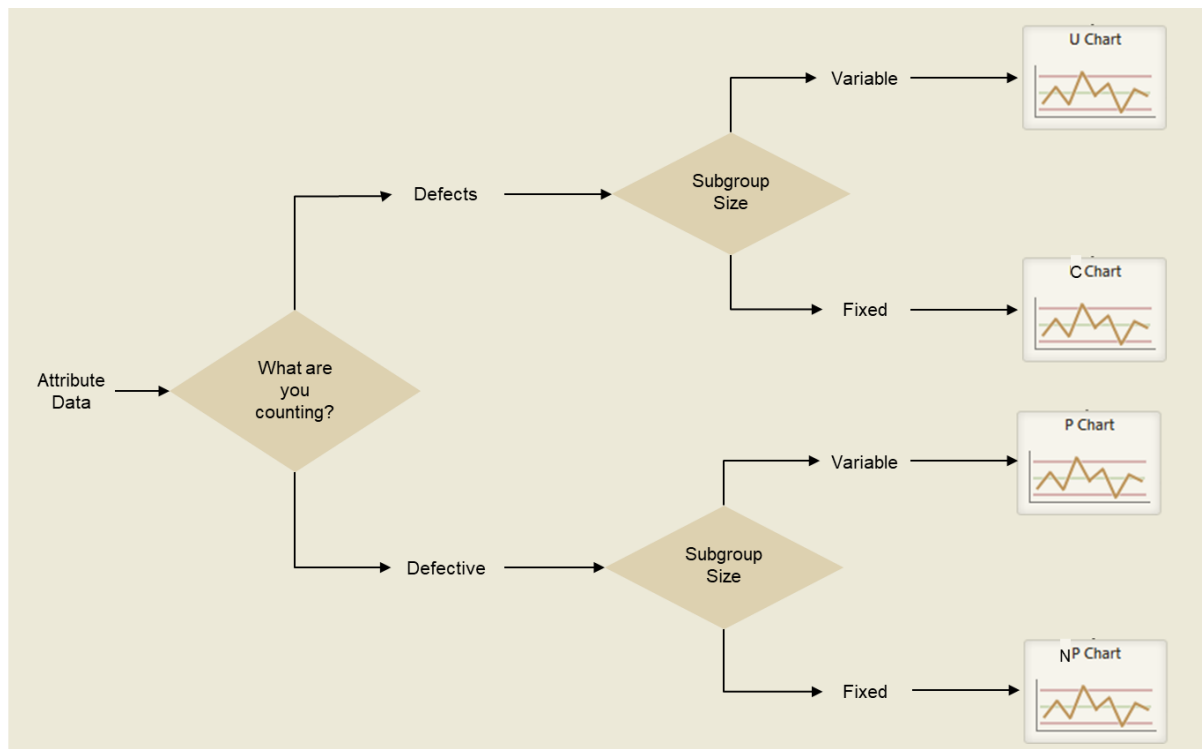
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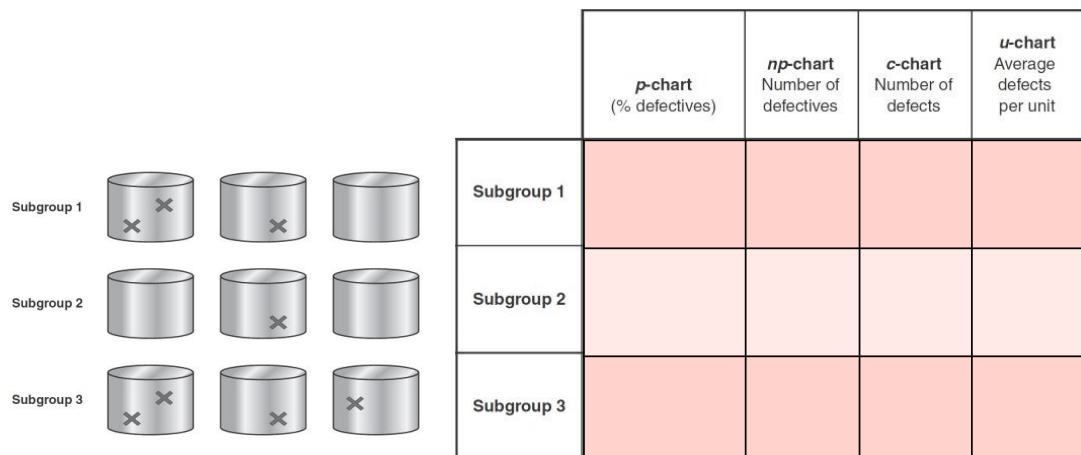
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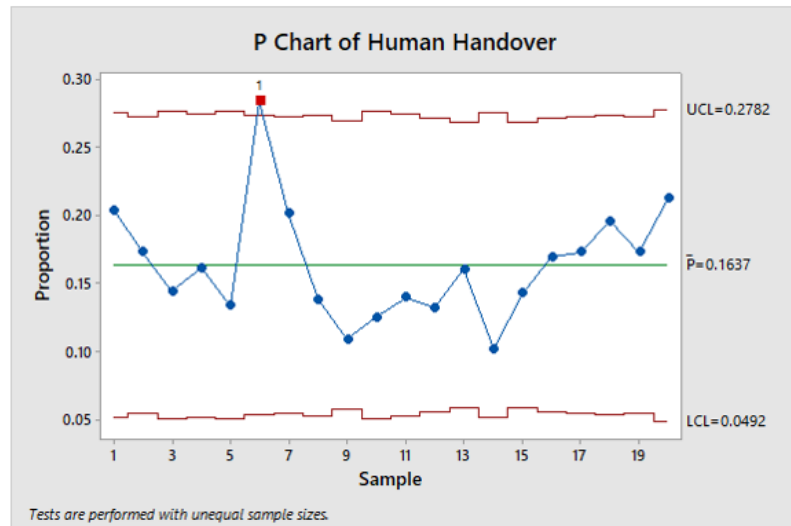
1. In the digital team at SS Bank, a chatbot is designed to handle a wide range of customer inquiries and provide real-time assistance. A defective unit refers to each instance when a customer requests to connect with a human agent during the chatbot interaction. To monitor the effectiveness of the chatbot, data has been collected for the last 20 days. [Data File: SS Bank]
2. At Chokolate, the quality control team is responsible for ensuring that each lot of chocolate bars meets the company's quality standards. To assess the quality of the chocolate bars, destructive testing is conducted, which involves opening and inspecting a random sample of 200 bars from each lot. Data is available for 30 manufacturing lots. [Data File: Chokolate]
3. At Fabtex, fabric rolls are produced and inspected for defects such as holes, stains, or irregular patterns. To ensure quality control, a sample of 100 fabric rolls is randomly selected from each production batch, and the number of defects is recorded for each roll. Data for 40 batches is available. [Data File: Fabtex]
4. At Customer Connect BPO, the call audit team is responsible for evaluating the performance and adherence to quality standards of their customer service representatives. To ensure quality control, 10% percentage of calls are randomly selected for auditing. The auditors assess various aspects, such as adherence to scripts, professionalism, accuracy of information, and customer satisfaction. Number of errors identified for each audited call are recorded. [Data File: Customer Connect]

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Stat > Control Charts >
Attributes Charts> P...

The proportion of defective items may not be stable. 1 subgroup is out of control and should be investigated

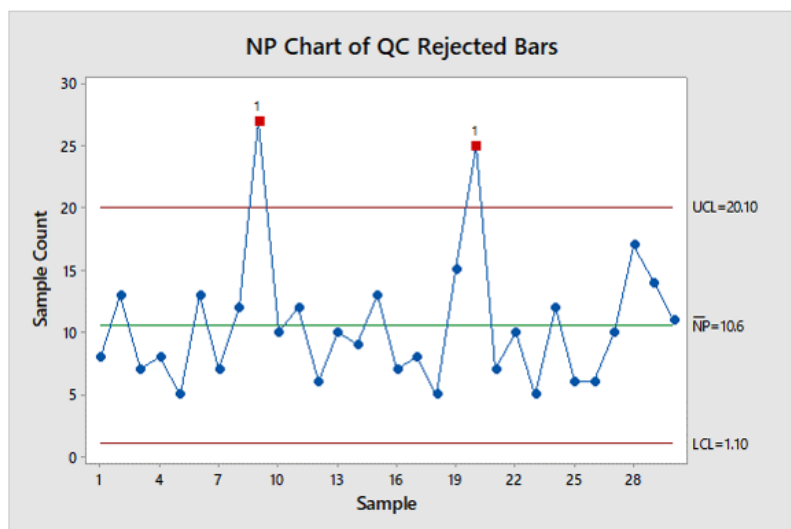


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Stat > Control Charts >
Attributes Charts> NP...

The number of defective items is not stable. 2 subgroups are out of control and should be investigated

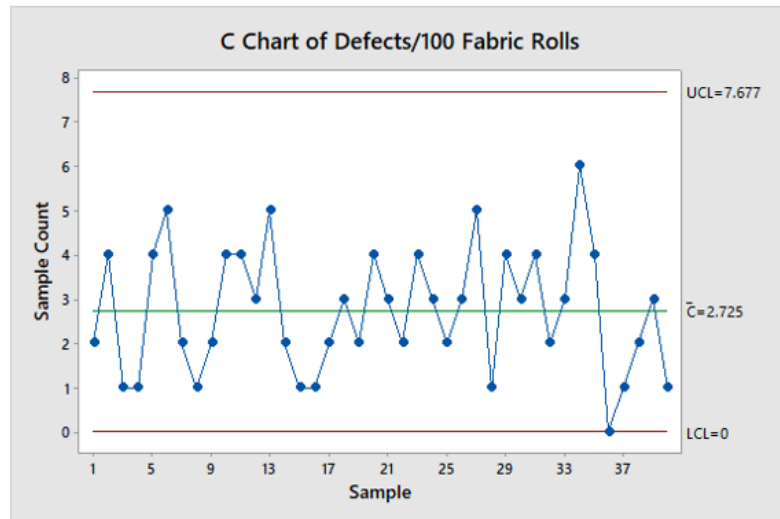


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Stat > Control Charts >
Attributes Charts> C...

The number of defects is stable. No data points are out of control.

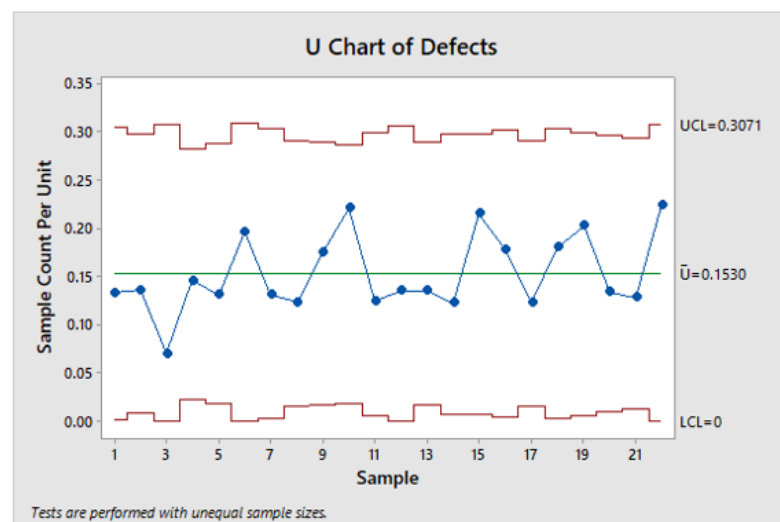


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Stat > Control Charts >
Attributes Charts> U...

The number of defects per unit is stable. No data points are out of control.



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When assessing process capability for discrete data, the following metrics are commonly used:

Information/Data Available	Preferred Metric
Defects, Sample, OFE	Defects Per Million Opportunities (DPMO)
Defects, Sample	Defects Per Unit (DPU) Defects Per Hundred (DPH) Defects Per Million (DPM)
Defectives, Sample	Traditional Yield (%), % Defective First Time Yield (FTY) Rolled Throughput Yield (RTY)

Grap Winery, a prominent wine manufacturer, inspects the quality of its bottled wines to ensure adherence to specific attributes. The attributes checked for potential defects include wine label alignment, cork integrity, bottle seal tightness, wine clarity, and wine aroma. A sample of 500 bottles was randomly selected for inspection, and a total of 8 defects were found. What is DPMO?

Information available :

1. Nature of data collected: Count of defects
2. Sample Size: 500
3. OFE: 5
4. Defects: 8

For every one million opportunities or attributes inspected in the wine production there is a likelihood of 60,000 defects.

OFE Available	
Defects	150
Sample	500
OFE	5
TOFE	2500
DPO	0.06
DPMO	60000

In a software development company called SoftTech, the quality assurance team conducts inspections to identify defects in software code. They focus on a specific type of defect called "syntax errors. The team inspects 200 code files and identifies a total of 350 syntax errors across those files. What is DPU?

Information available :

1. Nature of data collected: Count of defects
2. Sample Size: 200
3. OFE: NA
4. Defects: 350

The DPU value of 1.75 indicates that, on average, there are slightly less than two syntax errors per code file.

OFE Not Available	
Defects	350
Sample	200
OFE	NA
DPU	1.75

DPH (Defects Per Hundred Units) and DPM (Defects Per Million Units) are used when the defect rate is relatively low, and expressing it per hundred units or per million units provides a more manageable and meaningful representation.

- For example, let's consider the previous example of SoftTech. Suppose the quality assurance team inspects 200 code files and identifies 2 instances of solder defects. In this case, the DPU would be $2 / 200 = 0.01$ DPU. Converting this to DPH, we would have $0.01 \text{ DPU} * 100 = 1 \text{ DPH}$, indicating that on average, there is 1 defect per hundred code files.
- By using DPH, SoftTech's quality assurance team can better communicate and track the defect rate, especially when the defect levels are expected to be very low. It allows them to identify even minor deviations from the expected defect rate and take appropriate corrective actions.

- **Traditional Yield:** Traditional Yield represents the proportion of defect-free units or products in a process.
- **Defective Percentage** measures the percentage of units or products that contain one or more defects. $(100\% - \text{Defectives } \%) = \text{Yield } \%$
- **First-Time Yield (FTY):** First-Time Yield measures the percentage of units or products that pass through an entire process without any rework or repairs. While, traditional yield focuses on the quality of the final output, FTY assesses the effectiveness of the entire process in producing defect-free units or products without the need for rework or repairs.
- **Rolled Throughput Yield (RTY):** RTY is a measure of the overall yield or success rate of a multi-step process. RTY is calculated by multiplying the FTYs of each step in the process.

A semiconductor manufacturing company produced 10,000 microchips in a particular production run. During the inspection process, it was found that 200 microchips were defective and could not be reworked or repaired. Calculate defective percentage or yield.

Information available :

1. Nature of data collected: Count of rejected microchips
2. Sample Size: 10000
3. Defectives: 200

A yield of 98% means that out of the total units produced, 98% are defect-free and meet the desired specifications or requirements..

Traditional Yield	
Defectives	200
Sample	10000
Defective (%)	2%
Yield (%)	98%

A financial consulting firm aims to achieve a high rate of first-time right file submissions, which means that the loan application is complete, accurate, and meets all the necessary requirements without the need for any revisions or additional information. The firm handled 100 loan applications for their clients last month. While all the files were accepted and submitted to the bank, it was found that 80 of the loan applications were accepted and submitted to the bank without any issues or the need for revisions.

Information available :

1. Nature of data collected:

Count of files submitted without revisions

1. Sample Size: 100
2. First time right submissions : 80

First Time Right Yield	
First time right units (excluding rework)	80
Total pass units (including rework)	100
Sample	100
Defective (%)	0%
Traditional Yield (%)	100%
FTY (%)	80%
Rework (%)	20%

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Diagnostic firm Medi Lab receives 1000 patient samples for testing every week. These samples undergo multiple steps in the diagnostic process. Each step introduces the potential for errors or defects that may cause rejection of the sample. The FTY of each step is given below. What is RTY?

Step	FTY
Sample Collection	90%
Transportation	89%
Sample Preparation	94%
Sample Analysis	98%

To calculate the Rolled Throughput Yield (RTY), we multiply the yield of each step together.

$$RTY = 0.9 \times 0.89 \times 0.94 \times 0.98 = 0.74 = 74\%$$

A Rolled Throughput Yield (RTY) of 77.4% means that out of the total number of samples that entered the process, approximately 77.4% successfully passed through all the steps without encountering any errors or defects. It represents the overall effectiveness or success rate of the entire process

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ACTIVITY 11: CALCULATE APPROPRIATE METRICS**ACTIVITY 11: CALCULATE APPROPRIATE METRICS**

SGS ACADEMY

1. In an electronics manufacturing company, the quality control team inspects circuit boards for soldering defects. The team inspects 500 circuit boards and identifies a total of 75 instances of solder defects across those boards.
2. A bank receives thousands of credit card applications from customers which go through a series of steps for processing and approval. The steps include Complete application submission (90%), Document verification (95%), Credit check (60%), Income verification (87%), Final approval (97%).
3. Fresh2Door, a leading hyperlocal grocery delivery service, emphasizes the accuracy of order delivery to guarantee customer satisfaction. The attributes checked for potential defects encompass correct item selection, accurate quantity, proper packaging, timely delivery, and complete order fulfilment. A sample of 100 orders was randomly selected for evaluation, and a total of 27 defects were identified.
4. A furniture manufacturing company produces 500 wooden chairs in a particular production batch. During the quality inspection process, it was found that 450 chairs met all the required specifications and were deemed acceptable. However, 50 chairs had defects. Out of 50 chairs, 19 could be corrected, but the remaining 31 chairs had significant structural defects which could not be repaired.

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PURPOSE

- Evaluate if correct metric is identified and calculated for the available scenario

TASK

- Identify and calculate the appropriate metrics based on the information provided

OUTPUT

- Metric and Calculation

TIME ALLOWED

Preparation: 5 min

Feedback: 2 min

MEASURE PHASE TOLLGATE CHECKLIST

- Is the problem statement clearly defined and aligned with the project's goals and objectives?
- Has the Critical-to-Quality (CTQ) been finalized and performance standards set?
- Have logical validation checks been performed to ensure data accuracy and reliability?
- Is the measurement system properly validated through a Measurement System Analysis (MSA)?
- Have data collection plans been established, including what data to collect, how to collect it, and from which sources?
- Has the baseline data been collected and analysed to understand the current state of the process?
- Was a process capability analysis conducted to assess the process's ability to meet customer specifications?
- Is the project documentation up to date, including the project charter, data collection plans, baseline data, and process capability analysis results?
- Have any necessary revisions or adjustments been made based on the findings and insights from the Measure phase?

- The Measure phase is crucial for gathering accurate data, evaluating process capability, and setting the stage for further analysis and improvement in the Analyze and Improve phases of the DMAIC methodology .
- Its primary objective is to gather data and establish a baseline measurement of the current state of the process.
- During this phase, you finalize the Critical-to-Quality (CTQ) factors and set performance standards for measuring process performance.
- It involves performing a Measurement System Analysis (MSA) to assess the accuracy and reliability of the measurement system used.
- Process capability analysis is conducted to determine if the process meets customer specifications and identify areas for improvement .
- Project documentation is updated to include the baseline data, process capability results, and any necessary revisions to the project charter.

[illegible]

ANALYZE PHASE



Aim of the session

- Gain a comprehensive understanding of the Analyze phase in the DMAIC methodology .
- Define the specific objectives and focus of the Analyze phase and understand its key deliverables .
- Emphasize the criticality of identifying and validating the root causes of the identified problem or opportunity .
- Explore the process of listing potential causes (Xs) and selecting critical causes for further analysis.
- Highlight the importance of determining and validating the root cause(s) through rigorous analysis and data investigation .
- Discuss the role of cause-and-effect diagrams, data analysis, and statistical techniques in the Analyze phase

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By the end of this session, you will be able to:

- List all potential causes (Xs) that may contribute to the problem or opportunity.
- Identify the critical causes (Xs) that have the most significant impact on the process.
- Determine and validate the root cause(s) that are responsible for the observed problem or opportunity.
- Conduct in-depth analysis, data investigation, and hypothesis testing to establish cause -and-effect relationships.
- Understand the difference between statistical and practical significance.

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ANALYZE PHASE

The Analyze phase is a crucial stage in the DMAIC methodology where the focus shifts towards understanding the causes of the problem or opportunity identified in the Define phase. This phase involves in-depth analysis of data and processes to determine the root causes.

Objective: The main objective of the Analyze phase is to identify the critical causes (Xs) that contribute to the problem or opportunity and validate the root cause(s). It aims to gain a comprehensive understanding of the underlying factors influencing the Project Y/CTQ.

Steps:

7. List all potential causes (Xs)
8. Identify critical causes (Xs)
9. Determining and validating the root cause(s)

Main Deliverable : Validated Root Causes/Critical Xs

Other Deliverables : Cause-and-effect diagrams, data analysis results

A root cause is the fundamental or underlying reason that triggers a series of cause-and-effect relationships, leading to the manifestation of a problem or issue. It possesses two key characteristics:

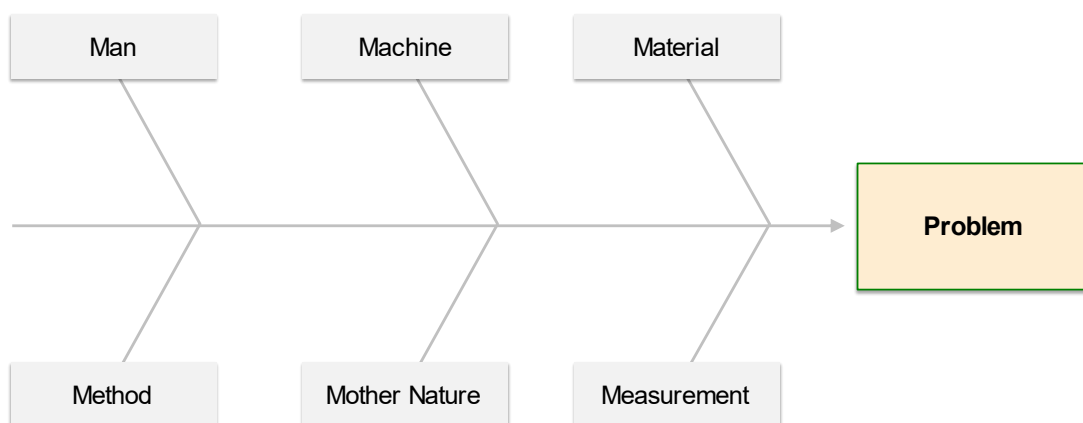
1. **Actionability:** A root cause is actionable, meaning that it can be **addressed or acted upon** to bring about meaningful change. By targeting the root cause, it becomes possible to address the problem at its source rather than simply treating the symptoms.
2. **Preventability :** Once a root cause has been identified and appropriate actions have been taken, it should prevent the **recurrence or occurrence** of the problem. By eliminating or mitigating the root cause, the likelihood of similar issues arising in the future is significantly reduced.

Identifying and addressing the root cause is a crucial step in the Analyze phase of the DMAIC methodology, as it allows for targeted and effective solutions that can bring about long-lasting improvements in the process or system.

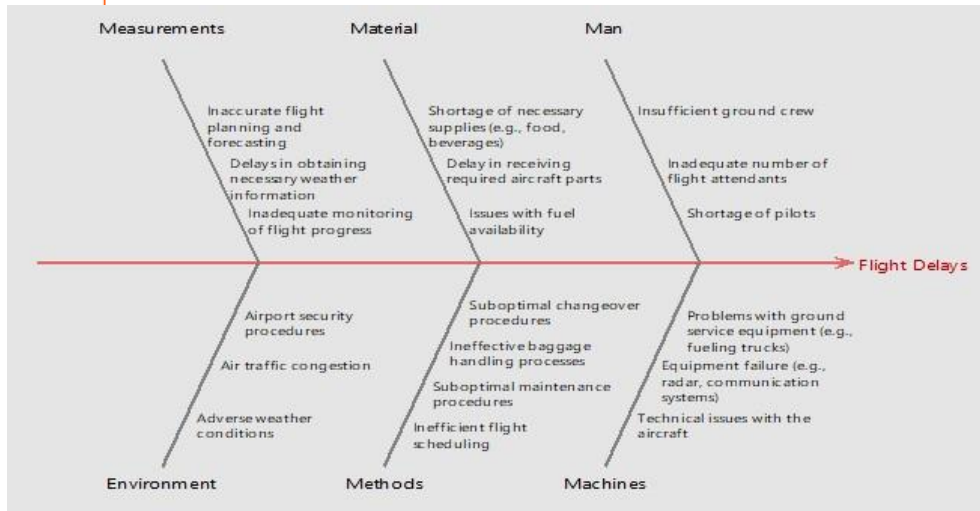
STEP 7: LIST ALL POTENTIAL CAUSES: FISHBONE DIAGRAM

- A fishbone diagram, also known as a cause-and-effect diagram or Ishikawa diagram, was developed by Dr. Kauro Ishikawa and is a visual tool used to identify and categorize potential causes of a problem or an effect.
- The main purpose of a fishbone diagram is to facilitate a structured brainstorming session to identify all possible causes contributing to a particular problem.
- The typical categories used in a fishbone diagram include the 6 M's:
 1. Manpower : People-related factors such as skills, training, and knowledge .
 2. Methods : Processes, procedures, and work methods .
 3. Machines : Equipment, tools, and technology used in the process.
 4. Materials : Raw materials, components, and supplies.
 5. Measurements : Metrics, data, and measurements used to assess performance.
 6. Environment : Physical conditions, surroundings, and external factors.

FISHBONE DIAGRAM USING 6M

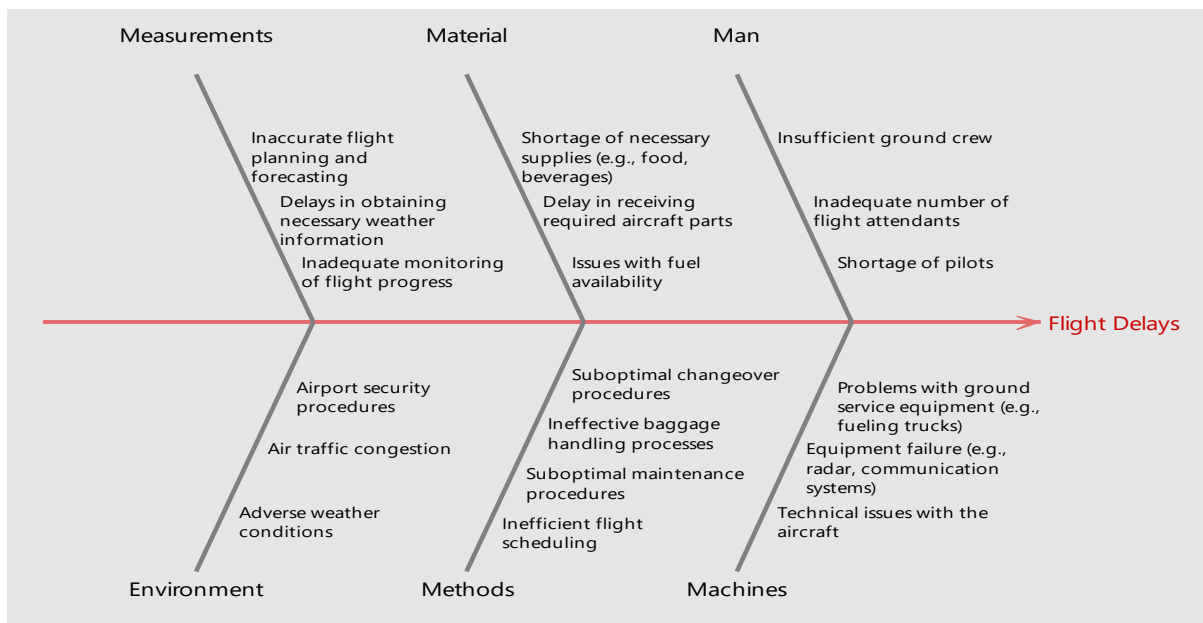


EXAMPLE CASE STUDY 14: FISHBONE FOR FLIGHT DELAYS



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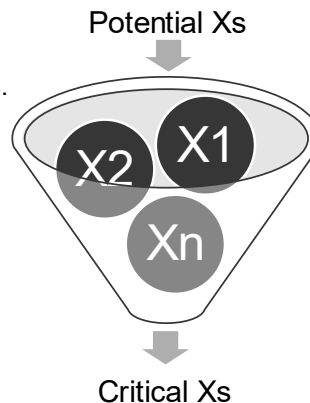
Data Analysis Methods:

Descriptive statistics:

1. Numeric stats (Mean, SD, Median, etc)
2. Graphical: Pareto, Scatter, Box plot, etc.

Inferential statistics (validation)

1. 2 Proportion test
2. Chi-Square test
3. 2 Sample t test
4. One-Way ANOVA
5. Correlation & Linear Regression



Process Flowchart Based Methods:

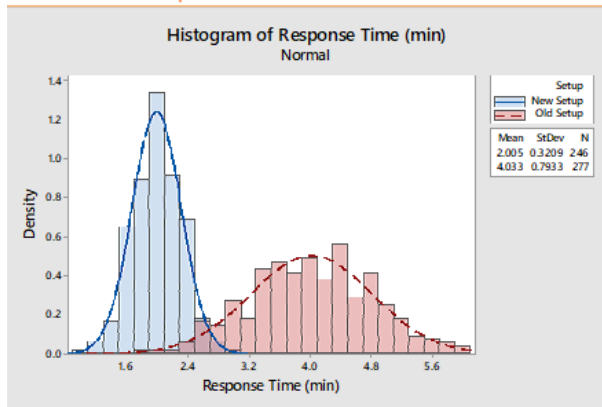
1. Failure Modes and Effects Analysis (FMEA)
2. Value Add/Non - Value Add (VA/NVA) Analysis

In Astollo Hospitals, there have been complaints from both patients and their families regarding poor response times by nursing staff after patients use the call bells in their rooms as part of the nurse call bell system. During the Gemba walks, it is discovered that the root cause of the delayed response is the division of patient beds among the nursing staff.

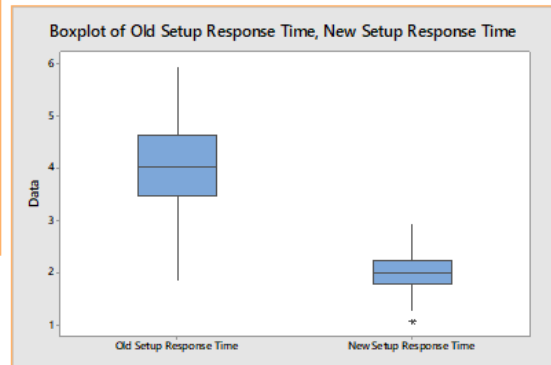
Each nurse is responsible for a specific set of patient beds, and when a call bell is activated, only the nurse assigned to those beds responds. Other nurses, seeing that the call is not for their assigned patients, do not respond to the call even if they are available to assist. This causes delays when the primary nurse is unavailable or occupied. To validate the cause a new configuration was tested. Every primary nurse was linked to a second line nurse. If the primary nurse is unable to respond within a certain time frame, typically 2 minutes, the responsibility automatically falls to the second-line nurse to attend to the call. Also a survey was conducted with both the setups with the question "Are you satisfied with the response on call-bell by the nursing staff?" [Data File: Astollo]

1. Is the setup a critical cause contributing to response time?
2. Will the change impact customer satisfaction?

IS THE SETUP A CRITICAL CAUSE CONTRIBUTING TO RESPONSE TIME?



Variable	Total Count	Mean	StDev
Old Setup Response Time	277	4.0334	0.7933
New Setup Response Time	246	2.0049	0.3209

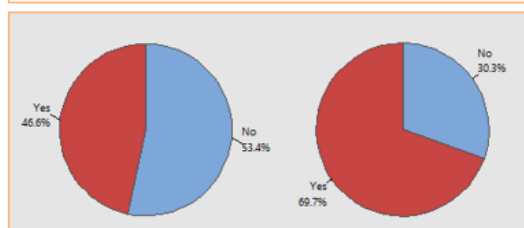


Yes. All the data analysis method show a clear difference between the response times of the two setups.

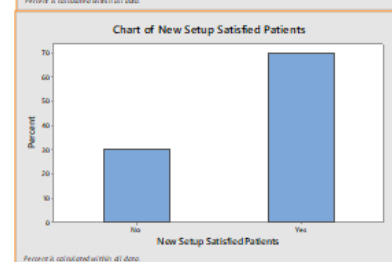
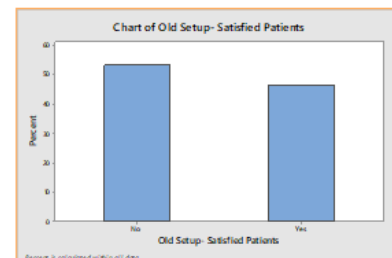
WILL THE CHANGE IMPACT CUSTOMER SATISFACTION?

Tally

Old Setup- Satisfied Patients			New Setup Satisfied Patients		
Patients	Count	Percent	Patients	Count	Percent
No	95	53.37	No	37	30.33
Yes	83	46.63	Yes	85	69.67
N=	178		N=	122	



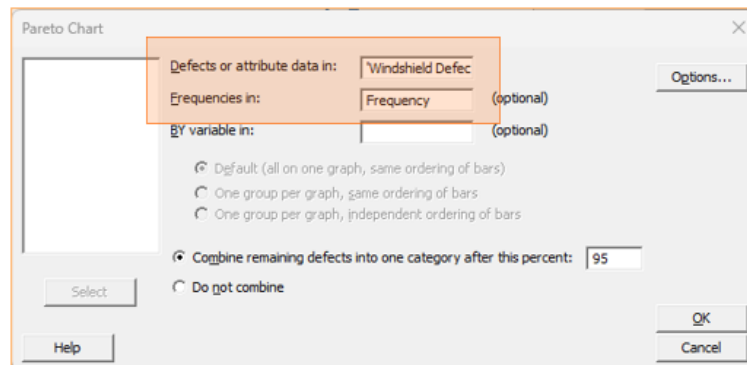
Yes. All the data analysis method show a significant increase in satisfied costumers.



During the analysis of the windshield rejections at Sparkling Glass Ltd, several potential causes of defects were identified along with occurrence (frequency) from historic data. Please carry out appropriate analysis to identify the vital few defects which when fixed will reduce the instances of windshield rejections.

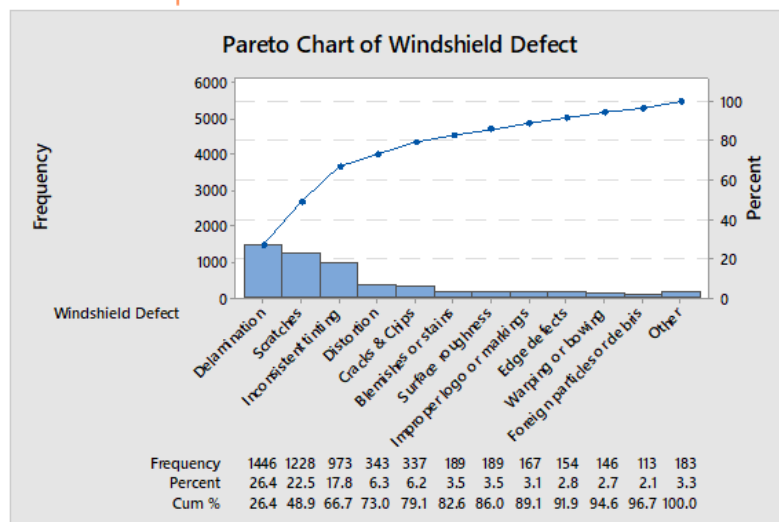
Data File: Glass Defects

Stat > Quality Tools>
Pareto Chart...



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About 80% of the defects are caused by 38% (5/13) of the causes.

ACTIVITY 12: PRACTICE CASE STUDY 7

The start-up bike ride aggregator 'Aloha' has noticed a significant number of instances where customers open the app, enter their destination to book a ride as a pillion rider, but do not proceed with the booking. This lack of conversion from intent to actual booking poses a challenge for the company.

To gather insights and understand the reasons implements a small feedback mechanism via app notifications. The notification leads to a list of reasons or options that the customer can select as the primary reason for not booking the ride. Please carry out appropriate analysis to identify the vital few reasons which when fixed will improve conversion.

Data File: Aloha

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PURPOSE

- Evaluate understanding of Pareto plot
- Evaluate right usage of Minitab for analysis

TASK

- Construct Pareto plot in Minitab and identify the vital few reasons which when fixed will improve conversion.

OUTPUT

- Results and identified vital few reasons

TIME ALLOWED

Preparation: 2 min

Feedback: 2 min

Process map-based analysis in the Analyze phase involves examining the process flowcharts or maps to identify potential causes of problems or areas for improvement.

They help identify various issues, such as:

1. **Bottlenecks** : Identify steps or activities that slow down the process flow or create delays.
2. **Redundancies** : Identify redundant or duplicated activities that can be eliminated to streamline the process.
3. **Handoffs**: Identify handoff points between individuals or departments that may lead to miscommunication or errors.
4. **Decision Points**: Analyze decision-making processes and identify areas where decisions may lead to suboptimal outcomes.
5. **Waste**: Identify non-value-added activities or steps that can be eliminated or reduced.

Process FMEA and VA/NVA analysis complement the process map-based analysis by helping in the identification of critical causes from the potential causes.

- **Process FMEA:** Process Failure Modes and Effects Analysis (FMEA) helps identify potential failure modes, their causes, and their effects on the process. By assessing the severity, occurrence, and detectability of each failure mode, it helps prioritize potential causes based on their potential impact on the process performance and outcomes.
- **VA/NVA Analysis :** Value-Added/Non-Value-Added analysis helps in identifying activities that add value to the process and those that do not. By eliminating or minimizing non-value-added activities, the analysis helps streamline the process and identify critical causes that hinder process effectiveness and efficiency.

ACTIVITY 13: IDENTIFY APPROPRIATE METHOD

1. In a manufacturing facility, there is a high rate of defective products in the final assembly process.
2. In a healthcare clinic, patient waiting times are excessive, leading to dissatisfaction and delays in receiving medical care.
3. A software development team is experiencing delays in delivering projects within the agreed timelines, impacting customer satisfaction.
4. A customer service department is experiencing a high number of customer complaints related to incorrect order processing.
5. A transportation company is experiencing a high rate of late deliveries and damaged goods during transit.
6. An administrative department in a company is struggling with a high volume of paperwork, causing delays in processing.

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PURPOSE

- Evaluate ability to choose between PFMEA and VA/NVA analysis basis problem statement and scenario.

TASK

- Decide appropriate technique between PFMEA and VA/NVA analysis

OUTPUT

- Chosen method

TIME ALLOWED

Preparation: 4 min

Feedback: 4 min

FAILURE MODE AND EFFECTS ANALYSIS (FMEA)

- FMEA is a systematic approach used to identify and evaluate potential failures or risks within a system, process, or product, and to determine their impact and likelihood of occurrence. [FMEA helps in proactively addressing potential failures to prevent them from happening or minimize their impact.](#)
- The history of FMEA dates back to the 1940s when it was developed by the U.S. military for use in reliability engineering. It was initially used in the aerospace and defence industries but has since been widely adopted across various sectors, including automotive, manufacturing, healthcare, and more.
- There are three main types of FMEA:
 1. System FMEA (SFMEA): It analyses potential failures and their effects on the overall system or product.
 2. Design FMEA (DFMEA): It focuses on potential failures and their effects during the design or development phase of a product.
 3. Process FMEA (PFMEA): It assesses potential failures and their effects within a specific process or manufacturing operation.

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USE OF PFMEA IN DMAIC PROJECTS

- The process FMEA used in DMAIC is similar to the one used in the automobile industry in terms of the general methodology and purpose of identifying and mitigating potential failures. However, the specific application and focus may differ based on the industry and context.
- In the automobile industry, process FMEA is commonly used to assess potential failures and risks associated with specific manufacturing processes in order to ensure product quality and safety.
- Process FMEA is helpful in identifying critical causes by systematically examining each process step and evaluating the potential failure modes, their severity, likelihood of occurrence, and detectability.
- By analysing the potential risks and effects, it allows teams to prioritize critical causes that have a high severity rating and a significant impact on the process performance. This helps in directing improvement efforts towards addressing these critical causes to enhance process reliability, efficiency, and quality.

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1. Identify the Process
2. Assemble the Team: Form a cross-functional team consisting of subject matter experts who have knowledge of the process and its potential failure modes.
3. Create a as-is Process Flowchart (if not available)
4. Identify Failure Modes: Brainstorm and identify potential failure modes for each step of the process. Failure modes are the ways in which the process can deviate or fail to meet the desired outcomes.
5. Assess Severity
6. Determine Causes and Occurrence
7. Evaluate Current Controls & Detection: Identify any existing controls or preventive measures in place to detect or prevent the failure modes.
8. Calculate Risk Priority Number (RPN): Calculate the RPN by multiplying the severity, occurrence, and detection ratings for each failure mode. The RPN helps prioritize the failure modes based on their overall risk level.

In Process FMEA (PFMEA), severity, occurrence, and detection are used as rating scales. The scales are typically defined using numerical values or descriptive categories to provide a standardized method for evaluation.

1. Severity: The severity scale assesses the potential consequences or impact of a failure mode on the process or product. It considers factors such as safety risks, customer satisfaction, compliance, and financial implications. Typically ranges from 1 to 10, with **1 representing a minor impact and 10 representing a catastrophic impact.**
2. Occurrence: The occurrence scale evaluates the likelihood of a failure mode occurring within the process. It takes into account historical data, process knowledge, and other relevant factors that contribute to the likelihood of the failure mode happening. Typically ranges from 1 to 10, with **1 representing a very low likelihood and 10 representing a very high likelihood.**
3. Detection: The detection scale assesses the ability of the process controls, inspections, or monitoring systems to detect or prevent the occurrence of a failure mode. It considers the effectiveness of existing detection methods, quality checks, and error-proofing mechanisms. Typically ranges from 1 to 10, with **1 representing a high detectability and 10 representing a low detectability.**

Maxware Logistics, a leading logistics company specializing in warehousing operations, has identified certain challenges in its warehouse process that are impacting customer satisfaction. The company aims to address these challenges to improve the overall efficiency and quality of its services.

The following is the high-level as-is process flowchart for warehouse operations:



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1. One of the key challenges identified in Maxware Logistics warehouse operations is inefficient storage space utilization. Warehouse staff are not optimizing the available storage space, leading to underutilized or congested areas. This results in longer search times, operational delays, and potential inefficiencies in the overall process. While this may not directly impact customer satisfaction, it affects the operational effectiveness of the warehouse. Inefficient storage space utilization occurs in approximately 1 out of 60 warehouse operations. The inefficiency in storage space utilization is detected in 20% of the cases during infrequent warehouse inspections and process audits.
2. Another challenge faced by Maxware Logistics is delayed order processing. Bottlenecks, inefficient order validation, and a lack of order prioritization contribute to orders not being processed in a timely manner. This results in delayed order fulfillment, which leads to customer dissatisfaction and potential loss of business. Addressing this challenge is crucial to meet customer expectations and ensure timely deliveries. Delayed order processing is observed in around 1 out of 20 orders processed by the warehouse team. The delayed order processing is detected in 100% of the cases through regular monitoring of order processing metrics, such as order fulfillment time and comparison with set targets.

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3. The third challenge identified in Maxware Logistics warehouse operations is the occurrence of incorrect item picking. Warehouse staff sometimes pick the wrong items from the storage locations, resulting in incorrect products being shipped to customers. This can lead to customer dissatisfaction, returns, and potential loss of business. Incorrect item picking occurs due to human errors, lack of proper checks, or ineffective communication within the warehouse team. Incorrect item picking occurs in approximately 1 out of 300 items picked by warehouse staff. The incorrect item picking is detected in 90% of the cases during manual checks performed by the packaging agents before packing for shipment.

Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severity (1-10)	Potential Cause(s)	Occurrence (1-10)	Current Process Controls	Detection (1-10)	RPN
Storage	Inefficient Storage Space Utilization	Inefficiencies, longer search times, and potential operational delays	2	No optimization of storage space	6	Infrequent warehouse inspections and process audits	8	96
Order Processing	Delay in Order Processing	Delayed order fulfillment, customer dissatisfaction	3	Bottlenecks, inefficient order validation, and a lack of order prioritization	7	Order tracking system	1	21
Order Picking	Incorrect Item Picking	Extreme customer dissatisfaction, returns, and potential loss of business.	4	Human error, lack of proper checks, or ineffective communication	5	Manual checks before packing	2	40

1. **Prioritize:** Identify the failure modes with the highest RPN values. These are the ones that pose the greatest risks and require immediate attention.
2. **Investigate Root Causes:** Analyze the failure modes to understand their underlying causes. Use tools such as the 5 Whys or Cause and Effect Diagrams to dig deeper into the root causes.
3. **Implement Corrective Actions:** Develop and implement appropriate corrective actions to address the root causes of the failure modes. These actions should be aimed at preventing or mitigating the identified risks.
4. **Reassess RPN:** After implementing corrective actions, reassess the RPN for the failure modes to determine if the risks have been adequately reduced. The RPN may decrease as a result of effective corrective actions.
5. **Verify Effectiveness:** Monitor the performance of the implemented corrective actions to ensure they are effective in reducing the risks associated with the failure modes. This may involve conducting tests, collecting data, and performing inspections to verify the effectiveness of the actions.
6. **Communicate and Document:** Communicate the findings, actions taken, and results to relevant stakeholders. Document the process changes, lessons learned, and any updated FMEA documentation.
7. **Continual Improvement:** Use the insights gained from addressing high RPN failure modes to drive continuous improvement. Apply the lessons learned to future designs, processes, or products to prevent similar risks from occurring.

Value-Added Activities are characterized by meeting three essential criteria:

1. **Customer pays for it:** Value-added activities are tasks or processes that the customer is willing to pay for. These activities directly contribute to meeting customer requirements, needs, or preferences.
2. **Physically transforms the product (or document/information):** Value-added activities involve actions that physically alter or transform the product, document, or information being worked on. These activities bring about a change in the form, shape, composition, or quality of the output.
3. **Right the first time:** Value-added activities are performed correctly and effectively from the outset. They are executed without errors, rework, or the need for additional corrective actions. Value is added by getting it right the first time, thereby reducing waste and optimizing efficiency.

Non-Value-Added (NVA) refers to activities or tasks within a process that do not contribute to meeting customer requirements, needs, or preferences. These activities are considered wasteful or unnecessary in terms of delivering value to the customer

There are two types of NVA activities commonly identified in process analysis:

1. **Pure Waste:** These are activities that add no value to the product or service and can be completely eliminated without impacting customer satisfaction. Examples include unnecessary movement or transportation, overproduction, waiting, excess inventory, or defects.
2. **Necessary NVA/Business VA:** Sometimes, certain activities are necessary but do not directly add value. Examples include process steps required for legal, regulatory, safety requirements. Inspections, audits, accounting, financial reporting and certain administrative tasks are necessary for the business to operate effectively but do not directly add value from the customer's perspective.

Pure waste, also known as the "Seven Wastes" or "Muda" in lean manufacturing and process improvement, refers to activities that add no value to the product or service from the customer's perspective. These wastes should be minimized or eliminated to improve overall efficiency.

1. **Transportation :** Unnecessary movement of materials or products.
2. **Inventory :** Excess stock or inventory that is not required.
3. **Motion :** Unnecessary movement or motion of people.
4. **Waiting :** Idle time or delays in the workflow.
5. **Overproduction :** Producing more than what is required or ahead of demand.
6. **Overprocessing :** Adding more value than what the customer requires.
7. **Defects :** Errors, mistakes, or defects that require rework or correction.

In addition to the traditional 7 Wastes, some practitioners and organizations have identified an eighth waste. This eighth waste is known as "Skillsunderutilization"

It refers to the unnecessary movement or transfer of materials, information, or tasks between individuals, departments, or systems.

Manufacturing examples	Non-manufacturing examples
1. Moving components or products between different workstations or production areas excessively or without a clear purpose. 2. Transporting raw materials or finished goods over long distances or between multiple storage locations unnecessarily. 3. Transferring materials or parts between suppliers and manufacturers inefficiently, resulting in delays and increased costs.	1. Handing off documents or files between departments or teams without a clear and efficient process, leading to delays and potential errors. 2. Handing off customer requests or inquiries between different customer service representatives, resulting in duplicated efforts or miscommunication. 3. Transferring tasks or responsibilities between team members without clear instructions or documentation, leading to confusion and errors.

The waste of inventory refers to excess or unnecessary stock or materials that are held within a process or system.

Manufacturing examples	Non-manufacturing examples
1. Excessive raw material inventory that surpasses the actual production requirements. 2. Finished goods (FG) inventory that exceeds customer demand or market needs. 3. Work-in-progress inventory that accumulates due to bottlenecks, delays, or inefficiencies in the production process. 4. Obsolete or expired inventory that cannot be used or sold.	1. In a service transaction, the equivalent of raw material could be the unused information, data, or resources required to deliver the service. This could include customer input, documentation, forms, or any other necessary inputs to initiate and execute the service. 2. WIP can be considered as the ongoing or partially completed tasks, activities, or processes involved in delivering the service. 3. FG can be finalized reports, recommendations, contracts yet to be closed.

The waste of motion refers to unnecessary or excessive movement of people or equipment within a process or workspace.

Manufacturing examples	Non-manufacturing examples
1. Workers frequently walking long distances or moving between workstations to retrieve tools, materials, or equipment. 2. Lack of ergonomic design in workstations, leading to unnecessary bending, reaching, or stretching by workers.	1. Unstructured or disorganized filing systems that result in employees spending excessive time searching for information or documents. 2. Employees having to move between different office spaces or departments frequently for meetings, document retrieval, or communication. 3. Excessive steps or movements required by customer service representatives to access information or complete tasks due to complex software systems, multiple screens, or outdated technology.

It refers to the idle time or delays experienced by individuals, equipment, or processes within a workflow or operation. It occurs when there is a lack of synchronization, coordination, or inefficiencies in the workflow, leading to unproductive waiting periods.

Manufacturing examples	Non-manufacturing examples
1. Machine downtime due to equipment breakdown, maintenance, or changeovers, causing delays in production. 2. Waiting for the availability of raw materials or components required for production. 3. Waiting for quality inspections or approvals before proceeding to the next production stage. 4. Waiting for scheduled maintenance or repairs on machinery or equipment, leading to production interruptions.	1. Customers waiting in line or queueing for service, such as in retail stores, banks, or government offices. 2. Waiting for service providers, such as waiting for a table at a restaurant, waiting for a doctor's appointment, or waiting for a response from customer support. 3. Waiting for information, documentation, or approvals from other departments or individuals within the organization, leading to delays in service delivery.

The waste of Overproduction refers to producing more goods, services, or outputs than what is required or demanded by customers or the subsequent processes.

Manufacturing examples	Non-manufacturing examples
1. Manufacturing finished goods in anticipation of future demand without proper forecasting, leading to the risk of product obsolescence or increased holding costs. 2. Overproducing components or sub-assemblies before they are needed in the production line, causing unnecessary work-in-progress inventory.	1. Printing or generating excessive paperwork or documentation that goes unused or becomes outdated, resulting in waste of resources and storage space. 2. Offering services or features that are not requested or valued by customers, leading to inefficiencies in resource allocation.

The waste of Overprocessing refers to performing unnecessary or excessive activities, steps, or processes that do not add value to the final product or service. It involves investing more time, effort, or resources than what is actually required to meet customer needs or specifications.

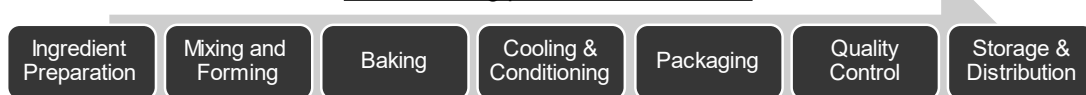
Manufacturing examples	Non-manufacturing examples
1. Performing additional machining, finishing, or polishing on a product beyond the required quality standards. 2. Using excessive packaging materials or elaborate packaging designs that go beyond what is necessary for product protection and presentation. 3. Conducting redundant inspections or quality checks at multiple stages of the production process instead of focusing on critical checkpoints	1. Collecting excessive customer information or documentation that is not necessary for delivering the service. 2. Implementing complex or convoluted administrative processes that involve multiple approvals or unnecessary paperwork. 3. Communications, reports, emails, contracts, etc. that contain more than the necessary points.

It refers to any errors, mistakes, or non-conformities in the products or services delivered to customers. It involves producing items that do not meet quality standards or customer expectations, resulting in rework, repairs, or customer dissatisfaction.

Manufacturing examples	Non-manufacturing examples
1. Manufacturing products that do not meet specifications or fail to function as intended. 2. Producing goods that have physical defects such as scratches, dents, or misalignments. 3. Packaging errors, such as incorrect labelling or missing components.	1. Errors in documentation or paperwork, such as incorrect invoices or contracts. 2. Service failures, including delays, disruptions, or inadequate resolution of customer issues. 3. Sending incorrect or incomplete communications, leading to misunderstandings or confusion

GEMBA WALK	DATA ANALYSIS	EMPLOYEE INPUTS
• Involves physically observing the process in action • Observe the flow of materials, the sequence of steps, and interactions between people and equipment. • Look for signs of waste, inefficiency, bottlenecks, or unnecessary activities.	• Analyze relevant data and metrics associated with the process. • Look for variations, delays, excessive cycle times, defects, or any other indicators of waste.	• Involve employees who work directly with the process. Seek their input and feedback on areas of waste they have observed or ideas for improvement. • Employees often have valuable insights as they are closest to the process and can provide practical suggestions.

ACTIVITY 14: PRACTICE CASE STUDY 8

Manufacturing process of snack bars

The following NVA activities were identified via Gemba walks, data analysis and interview with employees. Identify the waste:

1. There are delays in receiving the necessary ingredients causing idle time for the production team.
2. The production team prepares an excessive quantity of ingredients, resulting in unused material that need to be discarded.
3. The production line layout is not optimized, requiring workers to frequently move back and forth between different workstations to access tools or ingredients.
4. The mixing process involves unnecessary long mixing time.
5. Some snack bars come out of the oven with irregular shapes, burnt edges, or inconsistent texture, resulting in rejected products.

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PURPOSE

- Evaluate understanding of wastes in a process.

TASK

- Identify waste

OUTPUT

- Name of the waste

TIME ALLOWED

Preparation: 2-3 min

Feedback: 2-3 min

STEP 9: DETERMINE AND VALIDATE THE
ROOT CAUSE

Once the critical causes are identified, the Analyze phase aims to determine the root cause(s) that are responsible for the observed problem or opportunity. In-depth analysis, data investigation, hypothesis testing, and statistical techniques are utilized to validate the root cause(s) and establish a cause-and-effect relationship.

1. **Hypothesis testing** is a statistical approach to identify and confirm the underlying factors contributing to a problem or an outcome. It involves formulating hypotheses, collecting data, and performing statistical tests to evaluate the significance of potential root causes.
2. **5 Whys** technique is a simple yet powerful method to explore the underlying reasons behind a problem or an issue. It involves asking "why" multiple times, typically five times, to delve deeper into the chain of causation and uncover the fundamental cause of a problem. The 5 Whys technique helps identify the root causes by progressively peeling back the layers of symptoms and surface-level causes.

HYPOTHESIS TESTING

- Inferential statistics involves making inferences or drawing conclusions about a population based on a sample of data. Hypothesis testing is a fundamental concept in inferential statistics.
- Hypothesis testing is a systematic method for testing a claim or hypothesis about a parameter of a population, through a sample. A hypothesis test specifies whether to accept or reject this claim depending on the evidence provided by the sample.
- When we estimate the properties of a population from a sample, the sample statistics are unlikely to be exactly equal to the actual population values. The difference between the sample statistic and the population value is the **sampling error**.
- If sampling error causes the observed difference, the next time someone performs the sampling again, the results might be different. Hypothesis testing incorporates estimates of the sampling error to help us make the correct decision.



Delamination is a critical problem observed in the windshields produced by Sparkling Glass Ltd., the vendor supplying critical glass components to Marvel Motors. Delamination refers to the separation or peeling of the layers within the windshield, which compromises its structural integrity and safety. This defect has been causing concerns for both Sparkling Glass Ltd. and Marvel Motors, leading to increased rejection rates, customer complaints, and financial losses. The hypothesis aims to investigate whether the type of adhesive used (Securebond vs Sealstrong) in the manufacturing process plays a significant role in the occurrence of delamination.

Data File: Lamination

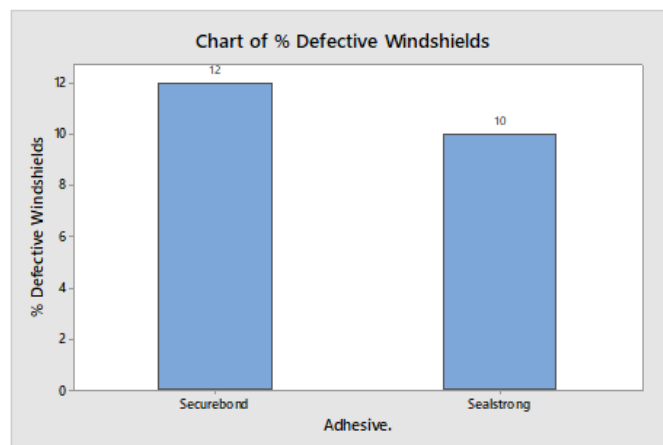
Adhesive	Securebond	Sealstrong
Defective Windshields	24	20
Total Samples	200	200

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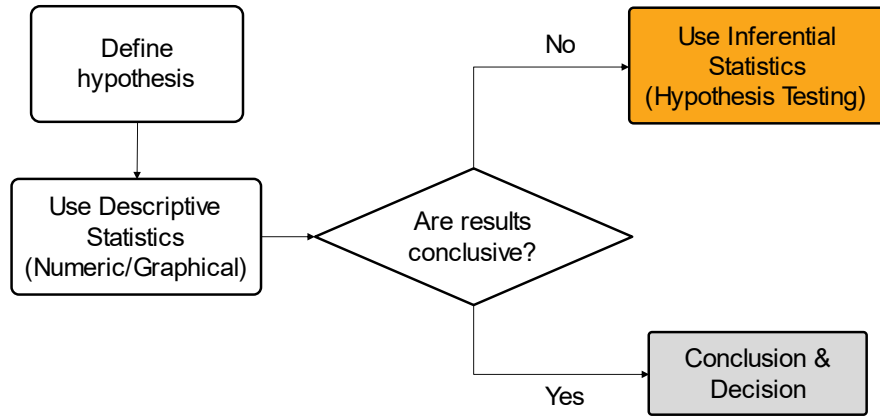
Adhesive	Securebond	Sealstrong
Defective Windshields	24	20
Total Samples	200	200
% Defectives	12	10

Is this difference in proportions of defectives just due to chance?



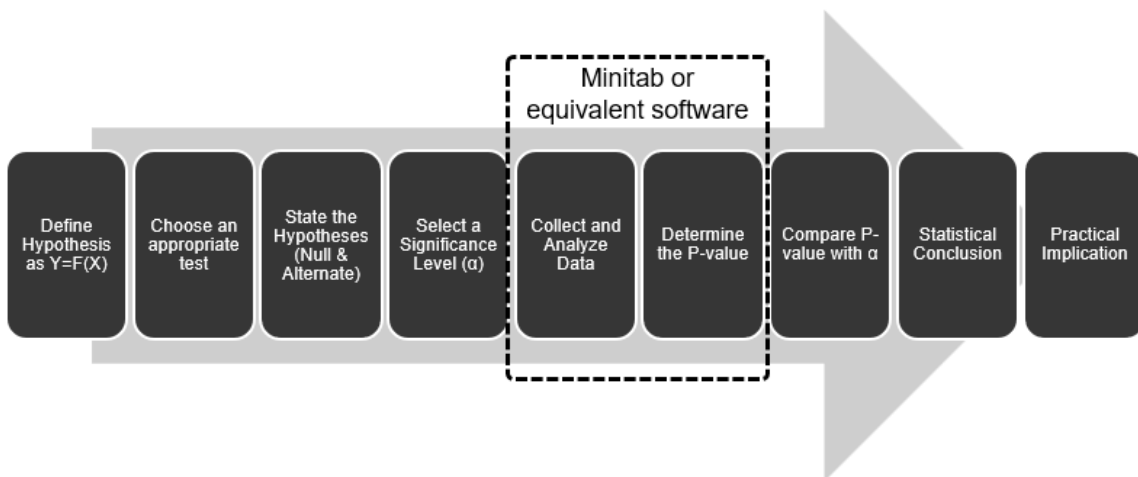
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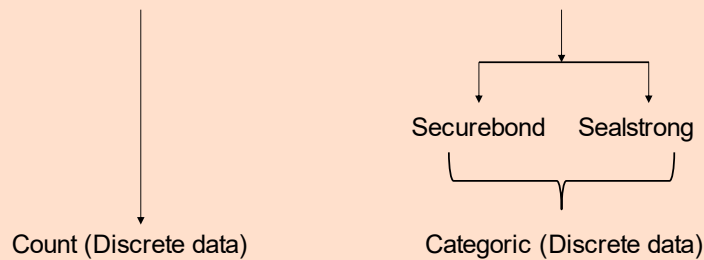
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- Y = Defective windshields due to lamination (Count/%)
- X = Type of Adhesive (Securebond & Sealstrong)

% Defection windshields due to lamination = f (Type of Adhesive)



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Output "Y" is...	Input "X" is ...	Appropriate Test	Purpose
Discrete	Discrete in 2 categories	2-proportion test	Compare 2 proportions/percentages
Discrete	Discrete multiple categories	Chi-square test	Compare >2 proportions/percentages
Continuous	Discrete in 2 categories	2- sample t -test	Compare 2 means (averages)
Continuous	Discrete more than 2 categories	One-Way ANOVA	Compare >2 means (averages)
Continuous	Continuous	Scatter Plot, Correlation Linear Regression	Understand relationship between two continuous variables

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In hypothesis testing, the null hypothesis (H_0) and the alternative hypothesis (H_A) are two opposing statements about a population parameter or the relationship between variables.

- **Null hypothesis (H_0):** It represents the default or initial assumption, suggesting that there is no significant difference, effect, or relationship in the population. It assumes that any observed differences or relationships in the sample data are due to random chance or sampling variability. The null hypothesis is often denoted as "no difference" or "no effect."
- **Alternative hypothesis (H_A):** Also known as research hypothesis, it contradicts the null hypothesis. It states that there is a significant difference, effect, or relationship in the population beyond what can be attributed to chance. The alternative hypothesis represents the hypothesis that researchers are trying to support or provide evidence for through their study.

Decision based on sample	Truth about the population	
	H_0 is True	H_A is True
Accept H_0	Correct Decision (probability = $1 - \alpha$)	Type II Error - fail to reject H_0 when it is false (probability = β)
Accept H_A	Type I Error - rejecting H_0 when it is true (probability = α)	Correct Decision (probability = $1 - \beta$)

- **Type 1 Error (α error):** Type 1 error, also known as an α error or false positive, occurs when the null hypothesis is incorrectly rejected, even though it is true.
- **Significance Level (α):** The significance level, denoted as α , is the predetermined probability of committing a Type 1 error. Commonly used significance level is 0.05 (5%).
- **Confidence Level:** The confidence level is the complement of the significance level.
- **Type 2 Error (β error):** Type 2 error, also known as a β error, occurs when the null hypothesis is incorrectly failed to be rejected, even though it is false.
- **Beta (β):** Beta is the probability of committing a Type 2 error in hypothesis testing.
- **Power ($1 - \beta$):** Power is the probability of correctly rejecting the null hypothesis when it is false, or in other words, the ability of a statistical test to detect a true effect or difference.

- The p-value, or probability value, is a measure used in hypothesis testing to determine the strength of evidence against the null hypothesis.
- P-value is the probability that we would obtain the effect observed in our sample, or larger, if the null hypothesis were true. In simpler terms, p-values tell us how strongly our sample data contradict the null.
- If the p-value is less than or equal to the significance level (α), typically 0.05, the null hypothesis is rejected in favour of the alternative hypothesis.
- If the p-value is greater than the significance level, there is insufficient evidence to reject the null hypothesis. This means that the observed data is reasonably likely to have occurred by chance even if the null hypothesis is true.
- A smaller p-value suggests stronger evidence against the null hypothesis.

$H_0 = \% \text{ Defective with Securebond} = \% \text{ Defective with Sealstrong}$
 $H_A = \% \text{ Defective with Securebond} \neq \% \text{ Defective with Sealstrong}$

Stat > Basic Statistics > 2 Proportions...

Two-Sample Proportions

Difference = (sample 1 proportion) - (sample 2 proportion)

Confidence level: 95.0

Hypothesized difference: 0.0

Alternative hypothesis: Difference $\neq$ hypothesized difference

Test method: Estimate the proportions separately

Significance level = 0.05

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Descriptive Statistics

Sample	N	Event	Sample p
Sample 1	200	24	0.120000
Sample 2	200	20	0.100000

Estimation for Difference

Difference	95% CI for Difference
0.02	(-0.041294, 0.081294)

CI based on normal approximation

Test

Method	Z-Value	P-Value
Normal approximation	0.64	0.522
Fisher's exact		0.632

We can be 95% confident that the true difference is between -4.13% and 8.13%.

P-value > α (0.05). There is not enough evidence to conclude that the % defectives differ at the 0.05 level of significance.

The % Defectives with Securebond is similar to the Sealstrong. Type of adhesive is not a critical X impacting lamination of windshields

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A bank wants to determine whether having a female co-applicant in the loan structure reduces the likelihood of default compared to single male applicants. The bank has historical data on loan applicants, including information on their gender and whether they defaulted on their loans. The bank selects a random sample of loan applications from their database. The sample includes both loan applications with female co-applicants and loan applications from single male applicants.

Data File: Default Rate

	Single Male Applicant	Female Coapplicant
Loans Defaulted	90	27
Total Loans	1313	700

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H_0 = Default rate with single male applicant = Default rate with female co-applicant
 H_A = Default rate with single male applicant $\neq$ Default rate with female co-applicant

Stat > Basic Statistics > 2 Proportions...

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Descriptive Statistics

Sample	N	Event	Sample p
Sample 1	1313	90	0.068545
Sample 2	700	27	0.038571

Estimation for Difference

Difference	95% CI for Difference
0.0299739	(0.010218, 0.049730)

CI based on normal approximation

Test

Null hypothesis	$H_0: p_1 - p_2 = 0$	
Alternative hypothesis	$H_1: p_1 - p_2 \neq 0$	
Method	Z-Value	P-Value
Normal approximation	2.97	0.003
Fisher's exact		0.007

We can be 95% confident that the true difference is between 1.02% and 4.97%.

P-value < α (0.05). We can conclude that the default rate differs at the 0.05 level of significance.

The default rate of the two loan structures is significantly different. The default rate of single male applicant is higher than female co-applicant loan structure.

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Soltech Corporation, an IT solutions provider, outsources its data entry and processing operations to three different vendors. The performance of these vendors is evaluated based on the accuracy of processing customer data. Soltech suspects that one of the vendors may have a different performance level compared to the others. Soltech randomly selects a sample of customer data processed by each vendor over a specific time period. For each vendor, Soltech records the total number of customer data entries and the number of accurate entries.

Data File: Soltech

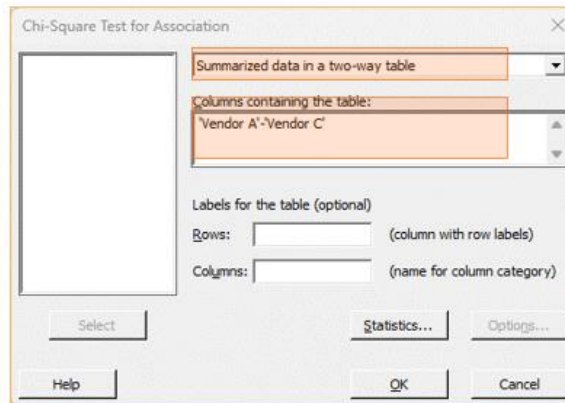
	Vendor A	Vendor B	Vendor C
Accurate Entries	3896	3580	3840
Inaccurate Entries	343	358	360

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H_0 = There is no difference in performance among the three vendors in terms of data entry accuracy.
 H_A = At least one of the vendors has a different data entry accuracy compared to the others.

Stat > Tables > Chi-Square Test for Association...



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Chi-Square Test

	Chi-Square	DF	P-Value
Pearson	2.602	2	0.272
Likelihood Ratio	2.599	2	0.273

P-value > α (0.05). There is not enough evidence to conclude that the % accuracy differs at the 0.05 level of significance.

The performance on all three vendors is similar. Vendor is not a critical cause impacting data accuracy.

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- While 2 proportion test is used to compare proportions or percentages between two groups, 2-sample t-test is used to compare means between two groups. E.g. comparing mileage of two fuels with different additives.
- Similarly, while chi-square test is used for categorical data to assess association (compare >2 proportions), one-way ANOVA is used for continuous data to compare means between three or more groups. E.g. Comparing TAT for three different vendors.
- The Anderson-Darling test is a hypothesis test used to assess whether a given dataset follows normal distribution. The null and alternative hypotheses for the Anderson-Darling test are as follows :

Null Hypothesis (H_0): The data follows a normal distribution

Alternative Hypothesis (H_A): The data does not normal distribution

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Scatter Plot

Graphical representation of the relationship between two variables



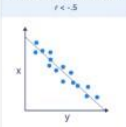
Correlation Analysis

Measures the strength and direction of the relationship between two continuous variables.

Strong positive correlation $r > .5$



Strong negative correlation $r < -.5$



Linear Regression

Establishes a mathematical model to describe and predict the relationship between variables.

$$Y_i = \beta_0 + \beta_1 X_i$$

Intercept (Constant) Regression Coefficient
 Y-axis Variable X-axis Variable

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Lamination typically involves the bonding of two or more layers of glass with an interlayer (usually polyvinyl butyral or PVB) using heat and pressure. Sparkling Glass Ltd. wants to analyse the relationship between the curing time (X) and the adhesion strength (Y). Study the relationship using appropriate statistical tools. The minimum adhesion strength of 250 Newtons is required to ensure adequate resistance to delamination .

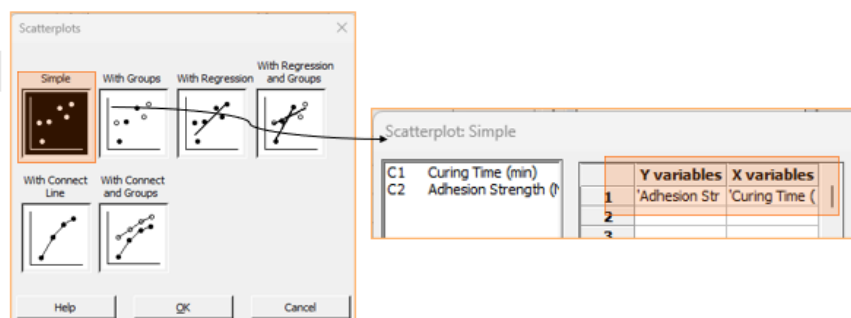
Data File: Curing Time

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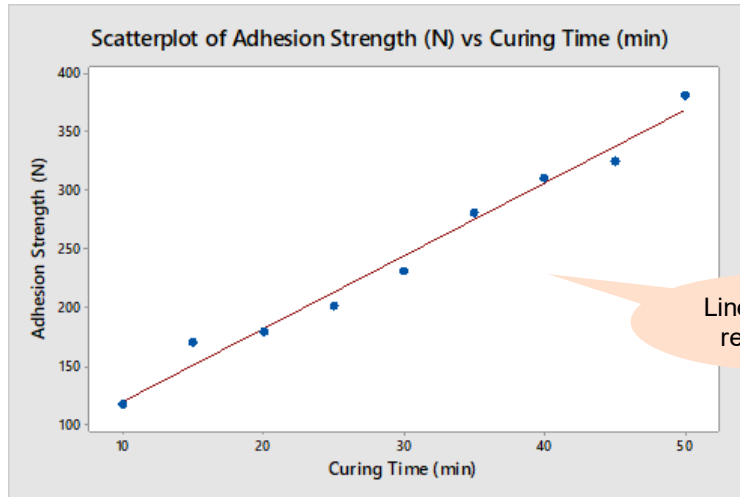
A scatter plot is a graphical representation of the relationship between two variables. It displays data points as individual dots on a graph, with one variable plotted on the x-axis and the other variable plotted on the y-axis. The scatter plot provides a visual depiction of how the two variables are related and helps identify any patterns, trends, or potential relationships between them.

Graph> Scatterplot...



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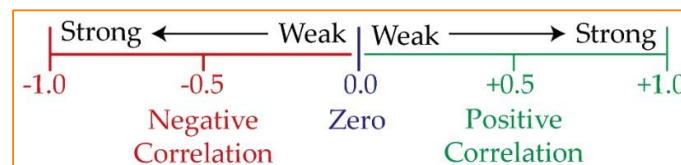
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Linear positive relationship

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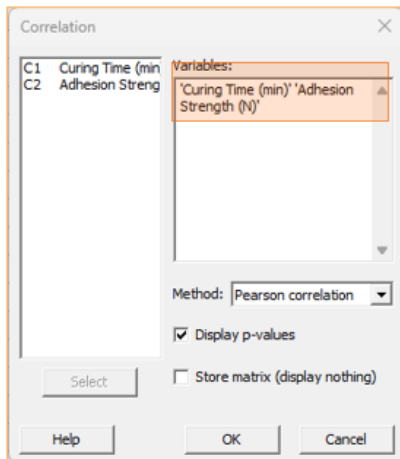


- Correlation analysis measures the strength and direction of the relationship between two continuous variables.
- It quantifies the degree of association between variables using a correlation coefficient, typically represented by the symbol "r".
- The correlation coefficient ranges from -1 to +1, where -1 indicates a perfect negative correlation, +1 indicates a perfect positive correlation, and 0 indicates no correlation.
- Values closer to -1 or +1 indicating a strong correlation, while values closer to 0 indicate a weak correlation.

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Stat > Basic Statistics > Correlation...



Correlations

Pearson correlation 0.991
P-value 0.000

The strong positive correlation ($r = 0.99$) indicates that when Curing time increases, Adhesion strength also tends to increase.

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- Linear regression is a statistical analysis technique used to model and understand the relationship between a dependent variable and one or more independent variables.
- It estimates the best-fit line that represents the linear relationship between the variables.
- The key output of linear regression is the equation of the line ($Y = c + mX$), where "Y" is the dependent variable, "X" is the independent variable, "c" is the y-intercept, and "m" is the slope.
- Linear regression helps determine the direction and magnitude of the relationship between variables, make predictions based on the model, and assess the significance of the relationship through statistical tests.
- R-squared (R^2) is a statistical measure that represents the percentage of the dependent variable's variance that can be explained by the independent variables. For example, an R-squared value of 0.75 means that 75% of the variance in the dependent variable can be explained by the independent variables in the regression model.

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Stat > Regression > Regression > Fit Regression Model...

Regression

C1	Curing Time (min)	Responses:
C2	Adhesion Strength	'Adhesion Strength (N)'
		Continuous predictors:
		'Curing Time (min)'

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Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	57536	57536.1	366.31	0.000
Curing Time (min)	1	57536	57536.1	366.31	0.000
Error	7	1099	157.1		
Total	8	58636			

The relationship between the two variables is statistically significant

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
12.5328	98.12%	97.86%	96.70%

98% of the variation in Adhesion strength can be explained by curing time

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	57.4	10.6	5.43	0.001	
Curing Time (min)	6.193	0.324	19.14	0.000	1.00

Since the model fits the data well, this equation can be used to predict Adhesion Strength for a particular value of Curing time or vice versa

Regression Equation

Adhesion Strength (N) = 57.4 + 6.193 Curing Time (min)

What is the minimum curing time (min) to achieve adhesion strength of at least 250N?

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ACTIVITY 15: PRACTICE CASE STUDY 9

Coel Retail is a leading online retailer specializing in consumer electronics. The company wants to understand the relationship between product ratings and sales volume to predict future sales based on customer reviews. They believe that higher product ratings will lead to increased sales. The retailer collects data on product ratings and corresponding sales volume for a wide range of consumer electronics products over a specific time period. Each data point represents a specific product. If there is a significant relationship, then create a model the retailer can use to forecast future sales based on product ratings.

Data File: Coel Retail

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PURPOSE

- Evaluate understanding of scatter plot, correlation, and regression.

TASK

- Create regression model

OUTPUT

- Final regression equation, Interpretation of R-squared, etc

TIME ALLOWED

Preparation: 5 min

Feedback: 2-3 min

The 5 Whys analysis is a problem-solving technique that aims to get to the root cause of a problem by repeatedly asking "Why?" It helps uncover the underlying causes behind an issue rather than focusing on its symptoms. The process involves asking "Why?" five times or more/less if necessary, to dig deeper into the problem's origin.

Problem: Your car won't start.

1. Why won't the car start? Because the battery is dead.
2. Why is the battery dead? Because the alternator is not charging it.
3. Why is the alternator not charging the battery? Because the alternator belt is broken.
4. Why is the alternator belt broken? Because it was not replaced during the last scheduled maintenance.
5. Why was the alternator belt not replaced during maintenance? Because the maintenance checklist did not include checking and replacing the alternator belt.

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Problem: Delayed Shipment



- | | |
|---|---|
| <ol style="list-style-type: none"> 1. Why was the shipment delayed? Because the shipment was not picked up by the carrier on the scheduled date. 2. Why was the shipment not picked up on the scheduled date? Because the shipping documents were incomplete and required additional information. 3. Why were the shipping documents incomplete? Because the person responsible for preparing the documents did not have access to all the necessary information. 4. Why didn't the person have access to the necessary information? Because the information was not communicated effectively between the departments involved. 5. Why was the information not communicated effectively between the departments involved? Because there is no established communication process or system in place for sharing important shipment details. | <ol style="list-style-type: none"> 1. Why was the shipment delayed? Because the shipping department didn't send it on time. 2. Why didn't the shipping department send it on time? Because they were too busy with other tasks. 3. Why were they too busy with other tasks? Because the warehouse didn't provide them with the necessary items on time. 4. Why didn't the warehouse provide the items on time? Because the procurement team didn't place the order in time. 5. Why didn't the procurement team place the order in time? Because the purchasing manager didn't prioritize it. |
|---|---|

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ANALYZE PHASE TOLLGATE CHECKLIST

- Have potential causes (Xs) for the problem or opportunity been identified and listed?
- Have critical causes (Xs) been identified and prioritized based on their impact and occurrence?
- Has a root cause analysis been performed to determine the underlying factors contributing to the problem or opportunity?
- Have appropriate statistical tools and techniques been applied to analyze the data and validate the findings?
- Have process map-based approaches, such as Process Failure Mode and Effects Analysis (PFMEA) and Value-Added/Non-Value-Added (VA/NVA) analysis, been conducted to gain insights into process flow, failure modes, and areas of waste or inefficiency?
- Have potential root causes been validated through hypothesis testing, correlation analysis, or other appropriate methods?
- Has the impact of identified root causes on the problem or opportunity been assessed?
- Have opportunities for process optimization and waste reduction been identified through VA/NVA analysis and other techniques?
- Has the analysis been documented and communicated effectively to the project team and stakeholders?
- what is the final list of prioritized root causes?

- The Analyze phase is a critical step in the DMAIC methodology, following the Measure phase, where potential causes (Xs) are listed to identify the root cause(s) of the problem.
- The main objective of the Analyze phase is to thoroughly analyze the data collected during the Measure phase to gain insights into the underlying factors contributing to the problem.
- During this phase, critical causes (Xs) are identified and prioritized based on their impact on the problem and their potential for improvement.
- Various analytical tools and techniques, such as hypothesis testing, correlation analysis, and regression analysis, are used to determine the relationships between the potential causes and the problem.
- Root cause(s) are determined and validated through techniques like 5 Whys analysis.
- It is essential to ensure that the identified root cause(s) are supported by evidence and data, and not solely based on assumptions or opinions.
- The results and findings of the Analyze phase serve as the basis for developing effective solutions and improvement strategies in the subsequent Improve and Control phases.

IMPROVE PHASE



Aim of the session

- Gain a clear understanding of the Improve phase in the DMAIC methodology .
- Define the objectives, focus, and key deliverables of the Improve phase.
- Learn how to generate and evaluate potential solutions to address the root causes identified in the Analyze phase.
- Understand the process of selecting and optimizing the most suitable solutions .
- Learn about pilot implementation and validate the selected solutions before full-scale implementation .
- Gain knowledge on how to measure the effectiveness of the implemented solutions and gather feedback for further improvements .

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By the end of this session, you will be able to:

- Understand the significance of the Improve phase in the DMAIC methodology and its role in driving process improvements .
- Conduct a comprehensive analysis of potential solutions and evaluate their effectiveness in addressing the root causes identified in the Analyze phase.
- Select and optimize the most suitable solutions that have the greatest potential for impact and feasibility.
- Pilot the implementation of the selected solutions and validate their effectiveness through data-driven evaluation .
- Apply various statistical techniques to measure process capability and assess the extent to which the process meets customer requirements .

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The Improve phase is the fourth stage of the DMAIC methodology . It focuses on generating and implementing solutions to address the root causes identified in the Analyze phase, with the goal of improving the process performance.

Objective: The primary objective of the Improve phase is to implement effective solutions that will lead to measurable improvements in the process.

Steps:

10. Generate and evaluate solutions
11. Select and optimize solutions
12. Pilot implementation and validation

Main Deliverable : Implemented solutions with measurable improvements

Other Deliverables : Updated process documentation, validation results

Generation	Evaluation	Selection
"Go Broad"	"Go Narrow"	"Select the best"
Brainstorming		
De Bono Six Thinking Hats		
Nominal Group Technique		
	Pugh Matrix	
	FMEA	
	Multi-voting	
Benchmarking		

Optimize: Regression, Design of Experiments (DoE)

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- Brainstorming is a collaborative approach that involves generating a large number of ideas and potential solutions to address the identified problems or opportunities.
- In the Improve phase, brainstorming is used to facilitate the generation of innovative and diverse ideas that can lead to process improvements. The aim is to encourage free thinking and open discussion among team members, allowing them to explore different perspectives and possibilities.
- Brainstorming sessions can be conducted with cross-functional teams or subject matter experts who have a deep understanding of the process. During the session, participants are encouraged to share their ideas without judgment or criticism. The focus is on quantity rather than quality in the initial stages, as this allows for a broader range of ideas to be considered.
- The generated ideas are then evaluated and refined to identify the most viable and effective solutions for implementation.

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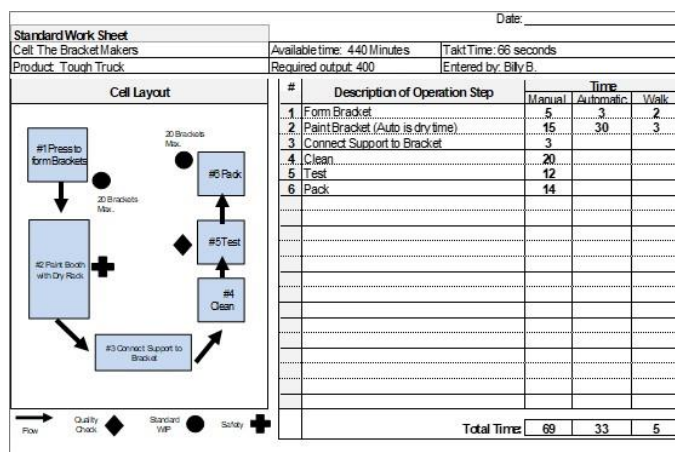
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- Benchmarking generate solutions by providing insights into best practices and performance standards of similar processes or organizations.
- By studying practices of industry leaders or top-performing organizations., one can identify innovative techniques, strategies, or technologies that have proven successful in achieving superior results. These best practices can serve as a source of inspiration and guide the development of solutions
- Benchmarking also helps in establishing realistic and ambitious performance targets. These targets serve as a reference point for the solutions implemented in the improve phase.
- Through benchmarking, one can gain insights into the challenges faced by other organizations and the lessons they have learned along the way. This knowledge can help one avoid common pitfalls and roadblocks, allowing design and implementation of solutions that are more likely to succeed.



A few established solutions discussed below can be considered during brainstorming for solutions depending on the problem/root cause.

1. **Standard work:** If the RCA reveals inconsistency or variation in how a particular process is performed or lack of clarity among employees regarding how a task or process should be executed, then standard work can be an effective solution.
2. **5S:** 5S can be a solution when the root causes relate to workplace organization, inefficiency, waste, safety hazards, or lack of visual management . It can help optimize workplace layout, eliminate waste, and enhance productivity.
3. **Mistake Proofing (Poka Yoke) :** Highly effective against human error, miscommunication, process complexity, equipment/tool issues, etc.



Standard work refers to the documented and agreed-upon set of instructions, procedures, and best practices that define the most effective and efficient way to perform a specific task or process. It provides a standardized framework for work execution, ensuring consistency, quality, and productivity.

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5S is a systematic approach for organizing and maintaining a clean, efficient, and safe work environment. In the Improve phase of a process improvement initiative, 5S can be utilized to enhance workplace efficiency, productivity, and overall performance.

Sort (Seiri)	➡	Identify and remove unnecessary items
Set in order (Seiton)	➡	Arrange necessary items in a logical & organized manner. A place for everything and everything in its place
Shine (Seiso)	➡	Regularly clean and maintain the workplace
Standardise (Seiketsu)	➡	Develop and implement standardized practices and procedures
Sustain (Shitsuke)	➡	Foster a culture of continuous improvement and sustain the 5S practices over the long term.

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5S principles can be applied in a nursing station at a hospital

Sort	- Remove outdated or unnecessary paperwork, forms, and supplies from the nursing station. - Discard expired medications and dispose of medical waste properly. - Sort and organize patient files and documents, ensuring they are to update and easily accessible.
Set in Order	- Designate specific areas for different types of supplies such as gloves, syringes, and dressings. - Arrange medical equipment, such as blood pressure monitors and thermometers, in a consistent and easily accessible manner. - Label shelves, drawers, and cabinets to indicate their contents and ensure everything has a designated place.
Shine	- Clean and disinfect the nursing station regularly, including work surfaces, computer keyboards, and phones. - Ensure hand hygiene products, such as hand sanitizers and soap dispensers, are readily available and maintained. - Keep the nursing station free from clutter and remove any potential hazards or obstacles.
Standardize	- Develop standard operating procedures (SOPs) for tasks performed at the nursing station, such as patient admission or discharge processes. - Implement standardized checklists for medication administration, patient assessments, and shift handovers. - Establish protocols for maintaining confidentiality and data security when handling patient information.
Sustain	- Conduct regular audits and inspections to ensure adherence to 5S principles. - Provide ongoing training to nursing staff on the importance of 5S and its benefits. - Foster a culture of cleanliness, organization, and continuous improvement within the nursing station.

5S principles can be applied to digital data management:

Sort	- Review and delete unnecessary files and folders to declutter digital storage spaces. - Archive or remove outdated or irrelevant digital documents. - Identify and categorize files based on their relevance and purpose.
Set in Order	- Organize files and folders using a logical and consistent naming convention. - Create a hierarchical folder structure that reflects the organization's workflow and file relationships. - Use metadata or tags to further categorize and classify files for easy searching and filtering.
Shine	- Regularly back up important data and ensure data security measures are in place, such as encryption and access controls. - Clean up and optimize computer systems to improve performance and reliability. - Conduct regular maintenance and updates of software and applications used for data management.
Standardize	- Establish guidelines and protocols for file naming conventions, folder structure, and version control. - Develop a centralized system for document sharing and collaboration, such as a closed platform or document management software. - Define roles and permissions for accessing and modifying files to ensure data integrity and security.
Sustain	- Provide training and resources to employees on best practices for digital data management. - Regularly review and audit the digital data management system to ensure compliance and identify areas for improvement. - Encourage a culture of responsibility and ownership in maintaining organized and secure digital data.

Mistake Proofing or Poka -Yoke is a method for avoiding errors in a process.

- It is a technique or a mechanism for either eliminating errors by making it impossible to make mistakes in the process – Prevention
- or by making the errors obvious at a glance - Detection



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Technique	Purpose
Shutdown	Stop the process / machine either just before the error or as soon as the error occurs
Control	Error is impossible
Warning*	Alert that something is about to or has gone wrong in the process

* Warning can help in reducing errors but not eliminate the chances of committing one. It is generally used in combination with shutdown poka -yoke techniques.

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1. No need for formal training programs
2. Reduction/elimination of inspection operations
3. Opportunity for value -adding tasks
4. Achievement of defect -free work
5. Reduction/elimination of rework

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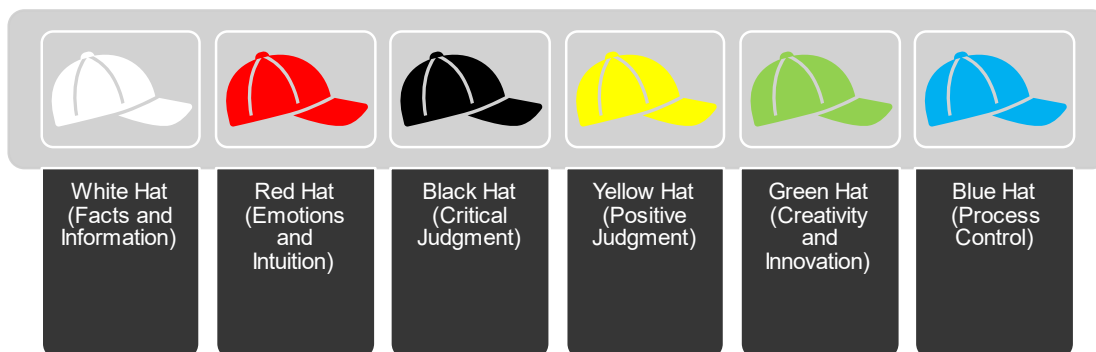
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Theme	Example
Combine Steps	A financial company cross-trains their file review team to make calls to customers in case of deficiencies instead of passing it another team trained only in calling.
Rearrange Steps	A hotel providing complementary airport pick up service provides room key and welcome document in the car to avoid waiting at the reception.
Eliminate Steps	Using view glass to check certain issues rather than opening a machine/lid
Segregate	Express counters at malls and retail stores
Magnify	A FMCG Company convinces their vendors to provide a larger diameter packaging material roll size to reduce stoppages during production for refilling.
Make the customer/user do it	CLTI activities done by production (shifting some activities from maintenance)

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De Bono's Six Thinking Hats is a creative thinking technique developed by Edward de Bono. It involves using six different "thinking hats" as metaphors to explore and analyze a problem or situation from different perspectives. Each hat represents a different mode of thinking, allowing individuals or groups to approach a problem systematically and thoroughly.



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- Multivoting helps streamline the decision-making process by narrowing down a large number of options into a smaller set of choices.
- In voting, participants typically choose only one option among several alternatives. In multivoting, participants have multiple voting points that they can distribute among the options.
- It helps prioritize options and focus the group's attention on the most promising or preferred choices.
- It encourages active participation from all members, prevents dominance by a few individuals, and promotes collaborative decision-making.
- By consolidating votes, multivoting provides a clearer picture of the collective preferences of the group, facilitating more efficient decision-making and consensus building.

- Steps in Multivoting**
1. Generate a List of ideas (via brainstorming)
 2. Assign voting points (most commonly 3)
 3. Participants vote for preferred options
 4. Tally
 5. Identify top choices
 6. Subject top ideas to detailed evaluation

Idea	Votes
Solution 1	● ● ●
Solution 2	●
Solution 3	● ● ● ● ●
Solution 4	● ● ●
Solution 5	●
Solution 6	●
Solution 7	●

Different colours represent multiple votes from different individuals

Further evaluation

The Nominal Group Technique (NGT) is a structured group discussion and decision-making process that encourages equal participation and generates a prioritized list of ideas or solutions. It is particularly useful when seeking input from a diverse group and ensuring everyone's ideas are considered.

Steps in NGT

1. State the problem
2. Silent Idea generation
3. Round robin sharing
4. Clarification and discussion
5. Voting or ranking
6. Consolidation and summary

When to use NGT?

1. When there is a diverse group of **participants with different backgrounds**, expertise, and perspectives.
2. When equal participation and **engagement of all** group members is desired.
3. When generating a prioritized list of ideas or solutions is important.
4. When avoiding the **influence of dominant individuals** and encouraging individual thinking and contribution.

- The Pugh Matrix is named after its creator Stuart Pugh, who developed the method in the late 1960s as part of his work on design methodology and decision-making in engineering design processes.
- It is a decision-making tool used to evaluate and compare multiple alternatives or concepts based on a set of criteria. It helps in selecting the most promising or optimal solution from a range of options.

Steps in creating a Pugh Matrix

1. Determine the datum
2. List the alternatives
3. Define evaluation criteria
4. Compare each alternative with datum. Begin filling in '1', '0' or '-1' ratings in the main area of the matrix, based on whether that alternative is better, equivalent, or worse than the datum for that criteria.
5. Calculate sum as well as weighted sum
6. Select the one with the highest score

Expectation	Importance	Datum (Status Quo)	Alternative 1	Alternative 2	Alternative 3	Alternative 4	Alternative 5
Criteria 1							
Criteria 2							
Criteria 3							
Criteria 4							
Criteria 5							
Sum of Same							
Sum of Positives							
Sum of Negatives							
Weighted Sum of Positives							
Weighted Sum of Negatives							

A pilot is a test of a proposed solution. It has the following properties:

- Performed on a small scale
- Used to evaluate both the solution and the implementation of the solution
- Purpose is to make the full scale implementation more effective
- Gives data about expected results and exposes issues in the implementation plan



Improve the solution



Understand risks



Validate expected results



Smooth implementation



Facilitate buy-in

Upon the completion of a pilot, a team should ask the following questions to help guide them further.

1. Did the pilot achieve the desired outcomes?
2. What were the key learnings from the pilot?
3. Were any unexpected issues or challenges encountered during the pilot?
4. Are there any gaps or areas for improvement that need to be addressed before moving forward?
5. How does the pilot data compare to the baseline data?
6. Are there any additional data or analysis needed before implementing the solution?
7. What are the next steps for solution implementation?
8. What are the potential risks and mitigation strategies for solution implementation?

- Based on the pilot results and analysis, the solutions can be refined and optimized further if necessary.
- A **comprehensive implementation plan needs to be created** that outlines the necessary steps, timeline, resources, and responsibilities for rolling out the solution on a larger scale. Factors such as change management, training, and communication strategies should be considered to ensure a smooth transition.
- The next step involves **execution of the implementation plan** and deployment of the solutions across the process or organization. Any challenges or issues that may arise, are to be addressed to ensure that the solution is **integrated smoothly into the existing operations**.
- Validating improvement against baseline involves comparing the performance of the process after implementing the improvement to the baseline data collected in the Measure phase.

Metric used in measure phase	Desired movement in improve phase
Sigma Level Overall (Z_{overall})	⬆
Capability Indices (P_p and P_{pk})	$P_p \geq 1.33$ and $P_{pk} = P_p$
DPMO	⬇
DPU/DPH/DPM	⬇
Yield (%) /FTY/RTY	⬆

Hypothesis testing can be used as a statistical tool to validate the success of the solution. It helps in assessing whether there is a significant difference between the pre-implementation and post-implementation data, indicating the impact of the solution on the process performance.

IMPROVE PHASE TOLLGATE CHECKLIST

- Have potential solutions been generated through brainstorming sessions or other creative problem-solving techniques?
- Have the potential solutions been evaluated and prioritized based on their feasibility, impact, and alignment with project goals?
- Has a pilot plan been developed to test the selected solutions on a small scale?
- Have the necessary resources and support been secured for conducting the pilot project?
- Has the pilot project been successfully executed, and has data been collected to evaluate the effectiveness of the solutions?
- Have the results of the pilot project been analyzed and validated to determine the impact of the solutions on process performance?
- Have any necessary adjustments or modifications to the solutions been identified based on the pilot results?
- Have the selected solutions been optimized to address the root causes and maximize their impact on process improvement?
- Is there a clear implementation plan in place to roll out the solutions on a larger scale?
- Have performance metrics and indicators been established to monitor the progress and effectiveness of the implemented solutions?
- Have potential risks and challenges associated with solution implementation been identified, and mitigation strategies developed?
- Has the team conducted a final evaluation of the solutions to ensure they meet the desired objectives and deliver the expected improvements?
- Have the necessary training and communication activities been conducted to ensure smooth implementation and adoption of the solutions?
- Is there a process in place to capture and address feedback from stakeholders and make necessary adjustments to the implemented solutions?

- The Improve phase in the DMAIC methodology focuses on generating and implementing solutions to address the identified root causes.
- Key steps include generating and evaluating solutions, selecting and optimizing the most appropriate solution(s), conducting pilot implementation and validation, analyzing pilot data, developing an implementation plan, mitigating risks, and ensuring organizational acceptance.
- The team can use various techniques such as brainstorming, benchmarking, Poka-Yoke, and De Bono's six thinking hats to generate a wide range of ideas. These ideas can be evaluated and prioritized based on their feasibility, potential impact, and alignment with the project goals.
- The pilot data is collected and analyzed to assess the effectiveness of the solutions and verify their impact on process performance.
- After successful piloting and optimization, the team proceeds with full-scale implementation of the chosen solutions. The success of the implemented solutions is continuously evaluated and monitored using appropriate metrics and indicators.

CONTROL PHASE



Aim of the session

- Provide a comprehensive understanding of the Control phase in the DMAIC methodology .
- Define the objectives, focus, and key deliverables of the Control phase.
- Highlight the significance of implementing control systems to sustain process improvements .
- Learn how to validate control measures using Statistical Process Control (SPC) techniques .
- Understand the importance of documentation and closure in ensuring the long-term success of process improvements .
- Empower participants with the knowledge and skills to implement effective control systems for critical process variables .

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By the end of this session, you will be able to:

- Recognize the importance of the Control phase in the DMAIC methodology and its role in sustaining process improvements .
- Understand the significance of implementing control systems to maintain process stability and prevent the recurrence of defects or deviations .
- Learn how to develop a comprehensive control plan that outlines the control measures for critical process variables .
- Apply Statistical Process Control (SPC) techniques to validate control measures and ensure process performance is within acceptable limits .
- Gain the skills to effectively monitor and maintain the control systems to sustain the improved process performance.
- Appreciate the importance of continuous improvement and ongoing monitoring to ensure long-term success and customer satisfaction .

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The Control phase is the final stage in the DMAIC methodology . It focuses on implementing control measures to sustain the improvements made and ensure the process remains stable and within the desired performance standards .

Objective: The primary objective of the Control phase is to establish control systems for critical process variables and monitor the process to prevent the process from reverting back to its previous state or performance level .

Steps:

13. Implement Control Systems for Critical Xs
14. Validate Controls
15. Documentation & Closure

Main Deliverable : Control plan

Other Deliverables : Implemented control systems, Updated process documentation, Closure documentation

A Control System refers to the comprehensive approach adopted to:

- Sustain improved process performance over time by implementing effective strategies .
- Identify the necessary actions and tools required to maintain and sustain the achieved process improvements or gains .\

The primary objectives of a Control System are:

1. To ensure that the process remains in control and operates within desired parameters even after implementing the solution or improvement measures .
2. To promptly detect any out-of-control state in the process and identify the specific causes associated with the deviation . This enables timely corrective actions to be taken to address the problem before non-conformances or undesirable outcomes are produced .

Implementing control systems involves developing and implementing a control plan that outlines the specific measures and procedures to be followed for each critical X.

A control plan is a detailed document that outlines the specific measures and procedures to be followed to maintain control over critical process variables. It serves as a guide for operators and personnel to ensure consistent and reliable process performance. The control plan includes various elements :

1. **Process Ownership:** Process ownership refers to assigning responsibility and accountability for the control and management of the process. It involves designating individuals or teams who are responsible for overseeing the process, ensuring its compliance with standards, and driving continuous improvement.
2. **Data Collection Plan:** A data collection plan outlines the specific data to be collected, the methods of data collection, and the frequency of data collection. It ensures that relevant data is consistently gathered to monitor and assess the performance of the process.
3. **Process Documentation:** This element includes all the relevant documents, instructions, procedures, and specifications that describe the process and its requirements.

4. **Risk Mitigation Plan:** The risk mitigation plan identifies potential risks or failure modes that may impact the process and outlines the strategies or actions to mitigate those risks. It helps in proactively addressing and minimizing the impact of potential issues or failures.
5. **Statistical Process Control (SPC):** Statistical Process Control involves using statistical techniques and control charts to monitor and control the process over time. SPC helps in identifying trends, patterns, and abnormalities in process data and provides insights into process stability and capability.
6. **Response/Reaction Plans:** Response/reaction plans outline the actions to be taken in case of deviations, non-conformities, or unexpected events in the process. These plans help in defining the appropriate response and corrective actions to address issues and minimize their impact on process performance and quality.
7. **Audit/Inspection Plans:** Audit/inspection plans define the frequency and methods of conducting audits or inspections to assess process compliance with standards.

Visual control is a method used to communicate and monitor the status of a process or system through visual aids. It involves using visual cues, such as signs, labels, color coding, charts, graphs, or displays, to provide quick and easy-to-understand information about the current state of the process.

Visual management includes a wide range of visual controls that help making all workplace elements and processes more visually apparent. Visual controls can be:

1. **Informative** to show identity, directions, strategic goals, customer expectations and compliance requirements.
2. **Instructional** to communicate SOPs, work-related information, and workplace organization and maintenance activities.
3. **Result-oriented** to display the status of processes, projects, production, productivity and performance.

Information

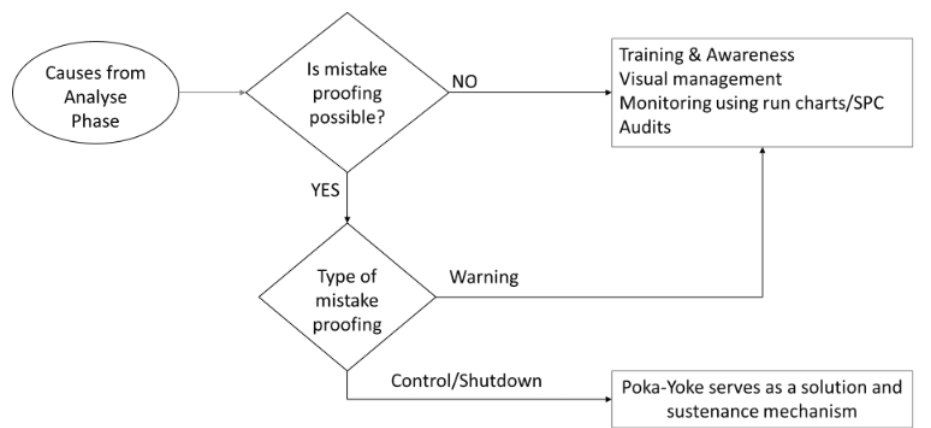
- Safety Visuals
- Signage
- Posters & Banners
- Markings

Instruction

- Standard Work
- 5s

Status

- Area Information Boards/Huddle boards
- Dashboards
- Performance metrics



- Validating control refers to the process of ensuring that the implemented control measures are effective in maintaining process stability and meeting the desired performance standards (process capability). One important aspect of validating control is the use of Statistical Process Control (SPC).
- SPC is a statistical method used to monitor and control a process by analyzing data collected over time.
- It involves the use of control charts to visualize the process data and identify any patterns, trends, or deviations that may occur.
- Control charts provide a graphical representation of process variation and help in distinguishing between common cause variation and special cause variation.
- In the Measure phase, SPC techniques were used to understand the current state of the process and assess its stability and capability. In the Control phase, SPC is used to validate the effectiveness of the implemented control measures and monitor the ongoing stability and performance of the process post implantation of solutions.

This step focuses on finalizing the project documentation, ensuring process ownership, and closing out the project.

1. **Documentation:** It is important to update and finalize all project documentation to capture the changes made during the improvement process. This includes updating process maps, standard operating procedures (SOPs), work instructions, and any other relevant documents. The documentation should reflect the improved process, including the implemented control measures and any changes in roles or responsibilities.
2. **Process Ownership:** Establishing process ownership is crucial for the long-term sustainability of the improvements. It involves assigning a responsible individual or team who will be accountable for the ongoing management and performance of the improved process.
3. **Closure:** The closure of the project involves formally wrapping up the improvement initiative. This may include conducting a project review or lessons learned session to reflect on the project's successes, challenges, and key takeaways. The project team should evaluate the achieved results, compare them to the project objectives, and ensure that the desired improvements have been realized.

By completing the Documentation & Closure step, the improvement project is formally concluded, and the improved process is handed over to the process owner for ongoing management.

CONTROL PHASE TOLLGATE CHECKLIST

- Have control plans been implemented for critical process variables identified during the Improve phase?
- Are the control plans aligned with the desired performance standards and customer requirements?
- Have control measures been validated using Statistical Process Control (SPC) techniques?
- Are there documented procedures and instructions for operating the controls?
- Have the operators and relevant personnel been trained on the new control measures?
- Is there a process in place to monitor and track key process metrics and performance indicators?
- Are there established procedures for addressing any deviations or out-of-control situations?
- Has a schedule been established for regular review and evaluation of process performance?
- Has the ownership of the improved process been transitioned to the appropriate process owner or team?
- Are there mechanisms in place to communicate and share the knowledge gained during the project with the wider organization?
- Is there a plan for ongoing continuous improvement and periodic reassessment of the process?
- Have any required resources, tools, or equipment for sustaining the control phase been identified and secured?
- Are there mechanisms in place to regularly communicate the progress and results of the control phase to relevant stakeholders?
- Has a closure plan been developed, including final documentation, reporting, and project evaluation?
- Has a feedback loop been established to capture lessons learned and apply them to future improvement initiatives?
- Are there mechanisms in place to celebrate and recognize the achievements and successes of the control phase?

- The Control phase is a critical stage in the DMAIC methodology, focusing on sustaining process improvements and ensuring long-term success.
- Its objective is to implement control systems, validate their effectiveness, and establish measures to maintain process stability.
- Key activities include developing a comprehensive control plan, conducting Statistical Process Control (SPC) analysis, and documenting the implemented controls
- The Control phase ensures that the process remains stable, defects are prevented, and performance meets or exceeds customer requirements.
- Ongoing monitoring and continuous improvement are essential to sustain the gains achieved during the Improve phase.
- The Control phase marks the transition from project-based improvements to the integration of improved processes into everyday operations.

APPENDIX

APPENDIX 1: PROJECT GOVERNANCE

Lean Six Sigma project governance is a structured framework that ensures effective management and oversight of Six Sigma projects. It encompasses various elements to support successful project execution and drive organizational improvement. Key aspects of project governance include:

1. Project Selection and Prioritization
2. Stakeholder Engagement
3. Leadership and Sponsorship
4. Change Management
5. Tollgate Reviews

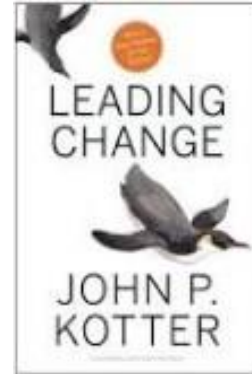
- It involves the systematic process of identifying, evaluating, and selecting projects that align with the organization's strategic objectives and have the potential to deliver significant improvements.
- The goal is to ensure that limited resources are allocated to projects with the highest impact and value. This process considers factors:
 1. Project feasibility
 2. Expected benefits
 3. Risks
 4. Resource availability
 5. Alignment with customer and business needs.

Effective stakeholder engagement is essential for the success of Lean Six Sigma projects as it helps gather diverse perspectives, build support, and ensure the project's objectives align with stakeholder needs and expectations. Some key points about stakeholder engagement:

1. **Identification:** The first step in stakeholder engagement is identifying all the individuals or groups who have an interest or can be affected by the project.
2. **Communication:** Project managers and team members need to establish effective channels of communication to keep stakeholders informed, engaged, and involved.
3. **Needs Assessment:** Understanding the needs, concerns, and expectations of stakeholders is crucial.
4. **Collaboration and Collaboration:** Involving stakeholders in key project activities, such as defining requirements, identifying improvement opportunities, and validating solutions, enhances their commitment and helps build consensus.
5. **Continuous Engagement:** Stakeholder engagement should be an ongoing process throughout the project lifecycle.

- Leadership and sponsorship are critical aspects of Lean Six Sigma project governance.
- Leadership provides direction, guidance, and support to the project team, driving a culture of improvement and ensuring resource allocation.
- Sponsors are senior executives or influential stakeholders who actively support the project, provide necessary resources, and champion its objectives. Their involvement fosters commitment, removes barriers, and enables successful implementation of project recommendations.
- While leaders provide overall guidance and direction, sponsors offer the necessary support and influence to overcome resistance, allocate resources, and facilitate change. Together, they create an environment conducive to project success, empower the project team, and ensure that the project aligns with organizational goals.

- Change management is crucial in Lean Six Sigma projects as it helps overcome resistance, manage uncertainties, and facilitate the smooth integration of new processes, systems, or ways of working.
- Change management is a structured approach to transitioning individuals, teams, and organizations from a current state to a desired future state. It involves planning, implementing, and reinforcing changes to ensure successful adoption and sustained benefits.
- Overall, change management provides a structured framework to address the human and organizational aspects of change. By effectively managing the people side of change, organizations can increase the likelihood of successful Lean Six Sigma implementations, enhance employee engagement, and maximize the benefits of process improvements.



- Tollgate reviews are structured checkpoints or milestones that occur at specific stages throughout the lifecycle of a Lean Six Sigma project.
- These reviews serve as decision points where project progress and alignment with objectives are assessed before proceeding to the next phase. The purpose of tollgate reviews is to ensure that the project is on track, deliverables are met, and appropriate actions are taken to address any issues or risks.
- During a tollgate review, the project team presents their findings, analysis, and progress to key stakeholders, sponsors, or a review panel. The review may include a formal presentation, documentation of project activities, data analysis, and demonstration of the project's impact on key metrics.
- The stakeholders then evaluate the project's readiness to move forward and make decisions regarding project continuation, modification, or closure.

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